

# BRAIN BEFORE THE SWIPE: CAN CONSUMER-GRADE EEG PREDICT VIEWER ENGAGEMENT DURING NATURALISTIC TIKTOK BROWSING?

DAS GEHIRN VOR DEM SWIPE: KANN CONSUMER-EEG  
DAS ENGAGEMENT DER ZUSCHAUENDEN BEIM  
NATURALISTISCHEN TIKTOK-BROWSING  
VORHERSAGEN?

Master's Thesis by

Gregor Lederer

Submission date 01. Oct. 2026

Chair Digital Global Public Health

Supervisor Dr. Irina-Catrinel Crăciun

# Acknowledgements

I am deeply grateful to my three supervisors, each of whom shaped this work in a distinct and necessary way.

Dr. Irina-Catrinel Craciun has been part of this thesis from long before it had a topic. Through the Digital Story Lab, which we founded together, she shaped my development as a researcher well beyond the scope of this document, introduced me to teaching workshops on this subject, gave me the opportunity to present at conferences and in classes at HPI, and involved me in academic publication for the first time. Her critical eye consistently sharpened this work when it needed sharpening, and I could not have asked for a more generous or more exacting introduction to scientific practice.

Prof. Dr. Bert Arnrich gave me a humble and realistic view of what consumer-grade EEG hardware can and cannot do, connected me with the right people in his group to understand the technological opportunities and limits of this space, and consistently pushed me to include perspectives I had not yet considered myself. His encouragement to look beyond my own assumptions is reflected throughout this thesis.

Prof. Dr. Dr. h.c. mult. Lothar H. Wieler helped me stay focused and on topic, and to maintain a rigorously scientific approach to reporting these results. He kindly took the time to answer questions at every stage, and his guidance kept this thesis grounded when its scope threatened to grow beyond what could be responsibly claimed.

I am also sincerely grateful to HPI PhD Student Sidratul Moontaha, whose extensive expertise in EEG-based engagement and brain-activity measurement was invaluable to this work. She generously gave her time to help me understand the field, and lent me two EEG devices for the study, which I returned considerably later than promised. I hope the results below go some way toward justifying the delay.

My thanks go to every participant in this study: a genuinely curious and academically engaged group of people whose interest in the research question, and whose thoughtful input during and after their sessions, motivated me throughout the data collection phase and improved this thesis in ways a

## ACKNOWLEDGEMENTS

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dataset alone could not have.

I also thank the Ethics Committee of the University of Potsdam for their efficient, constructive, and collegial handling of the ethics application that made this study possible in the first place.

Finally, and most personally, I thank my partner and my family, who supported and pushed me to keep writing, keep researching, and keep going, again and again, with new energy each time it was needed.

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# Abstract

Short-form video platforms such as TikTok structure user attention as a rapid sequence of binary micro-decisions: continue watching, or swipe away. Billions of such decisions are made every day, shaping what a population sees, knows, and acts on, though the data behind them rarely leaves the companies that collect it. Platform metrics and much of the surrounding public discourse treat continuous, uninterrupted viewing time as a proxy for user engagement, yet recent behavioral evidence suggests that prolonged, uninterrupted scrolling can reflect passive, dissociative consumption as well as goal-directed attention. This thesis investigates whether electroencephalography (EEG), recorded unobtrusively during naturalistic short-form video browsing, can discriminate the neural state preceding a voluntary skip decision from the neural state during sustained, non-skip viewing, and whether any such discrimination exceeds what is achieved by the conventional Engagement Index ( $\beta/(\alpha + \theta)$ ), a composite band-power ratio drawn from human-factors neuroscience.

Twenty-five participants ( $n = 25$ ; 13 female, 12 male; age  $M = 24.6$ ,  $SD = 3.0$ ) were recorded with a consumer-grade, four-channel Muse S headband (AF7, AF8, TP9, TP10) while freely browsing a TikTok “For You” feed, reset before each participant, for 25 minutes in a private, distraction-free setting, following a two-minute resting baseline. Manual swipe events were synchronized to the EEG stream via keypress markers. A literature-motivated labeling scheme operationalized two classes: **SKIP**, drawn from a 3.0-second SKIP period ending 2.0 seconds before each swipe (a pre-swipe gap exceeding both the typical reported onset of the readiness potential (Bereitschaftspotential) and directly measured pre-movement EMG onset in a self-paced reaching task), and **STAY**, comprising all remaining continuous, non-boundary viewing. From this labeling, 236 univariate features spanning seven frequency bands, four electrodes, and the raw broadband signal were engineered, including statistics of band strength, Hjorth parameters, and two features defined in the present thesis that count peaks of band strength per second (Peak Frequency, Macro Frequency).

Class separability, quantified via Cohen’s  $d$ , was higher for every frequency band, every electrode and most statistical descriptors than for the Engagement Index; the single strongest feature, raw-signal standard deviation at TP10, exceeded the Engagement Index’s separability by about ninefold. In logistic regression benchmarking against a coin-flip baseline, the Engagement Index model scored near chance within participants and was one-sided across participants: it labelled about 90% of all windows **STAY**, which raised **STAY** recall without separating **STAY** from **SKIP**. Three candidate features identified through the Cohen’s  $d$  ranking, namely TP10 raw standard deviation, AF7 high-gamma mean and AF8 delta peak frequency, each yielded statistically significant improvements over the Engagement Index in **SKIP** detection at the inter-subject level, and, in **SKIP** precision only, for AF7 high-gamma mean and AF8 delta peak frequency at the intra-subject level, though absolute performance remained close to chance and, in the inter-subject setting, stayed below the chance level of about 50%. A secondary behavioral analysis further showed that swipes cluster into rapid, individually variable “burst” sequences rather than occurring as uniformly spaced discrete events.

These findings indicate that the conventional Engagement Index captures only a limited, asymmetric component of the neural signal associated with short-form video browsing, and that more granular, region- and frequency-specific EEG features carry additional, currently underexploited discriminative information about imminent disengagement. The thesis further contributes a fully reproducible, open-source recording and analysis pipeline for consumer-grade EEG in naturalistic digital-media settings, and discusses the neuroscientific, psychological, behavioral, and public-health implications of these results, while explicitly flagging electromyographic and powerline-noise contamination as unresolved confounds requiring artifact-controlled replication.

# Chapter 1

## Introduction

### 1.1 Motivation

On TikTok, each swipe ends a micro-decision: continue watching, or swipe away. TikTok also carries health messages [1]. A viewer swipes health messages away like any other video.

Social media designers are tasked with maximising time on site [2]. Yet long viewing can reflect absorbed attention or zoned-out scrolling [2, 3] (Section 2.6).

The present thesis therefore studies the seconds before a swipe. Can EEG tell the seconds before a swipe apart from sustained viewing? And how well does the Engagement Index  $\beta/(\alpha + \theta)$  [4], the standard EEG measure of engagement, separate the same seconds from sustained viewing?

In one fMRI study, brain activity predicted how much of each video viewers watched before skipping [5]. The viewers lay in a scanner, saw the same 32 videos and skipped with a button. A consumer EEG headband lets a viewer browse a TikTok feed on a phone. EEG gives less spatial detail and little access to deep brain regions.

### 1.2 Defining Engagement

Different research fields define engagement differently (Section 2.4). The present thesis therefore states a working definition before measuring anything.

**Working definition (this thesis).** *Engagement is a viewer’s attention to the content on screen, held against the always-available next video, and ending at the latest at the swipe.*

The word *attention* leaves out dissociated viewing, in which users use a feed without really paying attention to what they are doing [2]. The next video is always one thumb movement away. Each swipe has a time stamp. The swipe therefore dates the latest end of engagement. Engagement may end earlier without a visible sign.

The working definition holds only where a screen and a swipe exist. Section 4.6 proposes a general definition for other settings. Watch time, sustained viewing and the **STAY** and **SKIP** labels (Section 3.4.2) are indicators, meaning measurable signs of engagement. No indicator is engagement itself.

### 1.3 Engagement in Health Communication

Engagement is a precondition for the effect of a health intervention [6] (Section 2.5). Theories of how a health story persuades start from an audience that stays with the story [7, 8]. The present thesis looks at the step before persuasion: the seconds before a viewer swipes away.

### 1.4 Research Gap

To the author’s knowledge, four gaps lead to the research questions. Section 2.15 lists all gaps.

**Gap 1: Time spent is an unvalidated indicator.** Health and social media research measures engagement mostly by amount of use, such as time spent and likes [9, 10]. Yet “more engagement does not necessarily equate to more effective engagement” [6]. To help fill exactly that gap, the present thesis studies the swipe-event.

**Gap 2: Health communication research seldom uses EEG.** Perski et al. and Yardley et al. list body signals used to measure engagement, and name no EEG [9, 6]. To contribute to closing this gap, the present thesis records EEG.

**Gap 3: The Engagement Index comes from a different task.** Pope et al. chose the Engagement Index for a computer task modelled on flight-deck work [4]. To the authors knowledge, no study reports the Engagement Index before a self-paced action comparable to the short-form video swipe. The present thesis benchmarks the Engagement Index.

**Gap 4: Engagement lacks a general definition.** O’Brien and Toms found no agreed definition of engagement [11]. Yardley et al. call for “defining engagement more consistently and appropriately” [6]. The working definition holds only where a screen and a swipe exist. The present thesis proposes a

general definition as a starting point for other fields.

Gaps 1 to 3 lead to research questions 1 and 2. Gap 4 leads to research question 3 (Section 1.5).

## 1.5 Research Questions

The present thesis asks three research questions:

1. **Separability:** Can EEG from a consumer headband with four electrodes tell the seconds before a swipe (**SKIP**) apart from sustained viewing (**STAY**), while viewers browse TikTok freely?
2. **Benchmark comparison:** How well do single EEG features separate **SKIP** from **STAY**, compared with the Engagement Index? Two measures answer the question. Cohen’s  $d$  measures how far the **SKIP** average and the **STAY** average lie apart, free of any unit. Logistic regression models predict **SKIP** or **STAY** for each second of recording, within each participant and on new participants.
3. **Definition:** Which general definition of engagement, bound to no platform and no measuring instrument, do the reviewed definitions and the results of the present thesis suggest as a starting point for other fields?

Answering the three questions required a recording and analysis pipeline built for self-paced swipes. The pipeline is public and forms one contribution of the present thesis (Section 1.7).

## 1.6 Approach

Twenty-five participants ( $n = 25$ ) each wore a consumer Muse S headband with four electrodes. After a two-minute resting baseline, each participant browsed a TikTok feed alone for 25 minutes. At every swipe, the participant pressed a key on a laptop. The keypress marked the swipe in the EEG (Sections 3.1–3.3).

Four labeling parameters set the **SKIP** windows and the **STAY** windows around each keypress (Section 3.4). A sensitivity analysis varied three of the four labeling parameters (Section 3.4.5). Each 1.0 s window gave 236 features (Section 3.7). Cohen’s  $d$  ranked the 236 features and the Engagement Index. Logistic regression then tested the Engagement Index against a coin-flip baseline, and three features picked from the ranking against the Engagement Index (Section 3.8). Section 4.6 answers the third research question.

## 1.7 Contributions

Chapter 4 reports six contributions of the present thesis, one section each:

1. **The experiment** (Section 4.1). Twenty-five participants browsed TikTok freely. A consumer headband recorded the EEG, and a keypress marked every swipe.
2. **The reusable pipeline** (Section 4.2). The full pipeline is all code from recording to results. The code is public for reuse.
3. **Swipe bursts** (Section 4.3). Swipes often come in quick runs, and participants differ widely. The bursts are a secondary result.
4. **Engagement Index performance** (Section 4.4). Every frequency band, every electrode and the raw signal separated **SKIP** from **STAY** better than the Engagement Index. No model predicted **SKIP** or **STAY** clearly better than chance.
5. **Discussion** (Section 4.5). The Discussion asks what the results may mean. Muscle activity and powerline-noise may partly explain the EEG differences.
6. **A proposed definition of engagement** (Section 4.6). A general definition names no platform and no measuring instrument. Four reporting requirements and an untested hypothesis for cheaper field measures follow.

## 1.8 Thesis Structure

Chapter 2 reviews related studies and ends with seven gaps. Chapter 3 describes the experiment, the **STAY** and **SKIP** labels, the full pipeline and the statistical tests. Chapter 4 reports the six contributions of Section 1.7. Chapter 5 answers the three research questions, lists the limitations and names future research. Chapter 6 answers common questions in plain language. Appendix A gives a German summary.

## Chapter 2

# Literature and State of the Art

The following sections outline the core theoretical claims, methodologies, and prior research required to contextualize the present thesis.

### 2.1 Adjacent Study

One published study sits closest to the present thesis: Rahman et al. record EEG while people watch short-form video, and train a classifier to tell short-form video viewing apart from other activities [12]. Rahman et al. present the classifier as a detector of Short-form Video Addiction (SVA), although the paper reports no measure of addiction, such as an addiction questionnaire. The present thesis calls the study of Rahman et al. the *adjacent study*. The adjacent study records EEG with a consumer-grade Muse S headband (the paper names both Gen 1 and Gen 2) at four scalp electrodes, TP9, AF7, AF8 and TP10, sampled at 256 Hz, plus two auxiliary channels of the headband, which the paper does not describe further; the band-wise analysis runs on all six channels. In a laboratory, a small sample of student volunteers ( $n = 10$ , ages 20–35) watches short-form video on their own phones, in their own feeds, TikTok included, and may swipe on at any moment. The present thesis calls self-paced viewing of an algorithmic feed, with a swipe possible at any moment, *naturalistic use*, and the feed itself the *infinite feed*.

The adjacent study labels whole blocks of time. A 15-minute block of short-form video viewing is the positive class; a block of other daily activity without media consumption on a smartphone is the negative class, balanced 1:1. The classifier of the adjacent study therefore answers one question: is this person inside the infinite feed, or not? Swipes play no role in the viewing-versus-not-viewing label. The two blocks may also differ in posture, gaze and hand movement: a person watching short-form video looks down at a phone

and moves a thumb, while a person doing something else may not. Part of the reported accuracy may therefore come from body movement as well as from brain activity.

The adjacent study pairs EEG with electrodermal activity, a measure of skin conductance, from an Empatica E4 wristband. Neural networks learn one shared representation of the EEG and the skin-conductance signal. The purpose of the shared representation is a sensor swap: learn with the head-band, then use the less obtrusive wristband alone. Evaluation repeatedly trains on 80% of the data and tests on the remaining 20% ( $k$ -fold cross-validation). The paper does not state whether windows from one 15-minute block can land in both the training data and the test data. If windows from one block land in both, the classifier can partly recognise the block itself, much like a student who has seen the exam questions during practice, and the reported accuracy may overstate performance on a new block. The adjacent study also reports no test on participants left out of training. Averaged over ten runs, the classifier reaches 76.3% accuracy from EEG alone and 84.2% from EEG and wristband together. For the sensor swap, the wristband alone reaches 71.5%, and 80.2% after training with help from the EEG.

On frequency, the adjacent study applies a band-stop filter at 50 Hz and the multiples of 50 Hz, and the paper names a “110V 50Hz” power supply. The ethics approval of the adjacent study comes from the University of Massachusetts Lowell, in the United States, where power lines carry alternating current at 60 Hz. The adjacent study then splits the signal into five bands: Delta 1–4, Theta 4–8, Alpha 7.5–13, Beta 13–30, and Gamma 30–44 Hz. The analysed spectrum stops at 44 Hz, so the mismatch between the 50 Hz filter and the 60 Hz power lines likely leaves the reported band results largely unaffected. Beta and Gamma are reported as the most informative bands, each above 80% accuracy.

**Gap the present thesis fills:** The adjacent study labels whether a person is inside the infinite feed; the present thesis on the other hand labels two states inside the infinite feed, anchored to each self-paced swipe: **STAY**, sustained viewing without a swipe, and **SKIP**, the seconds just before the swipe. Because the labels differ, the accuracies of the present thesis and of the adjacent study measure different things and are not directly comparable. More specifically, the present thesis asks whether EEG features with a known meaning, such as the strength of one frequency band, separate **STAY** from **SKIP** better than the Engagement Index does (Chapter 4). The adjacent study analyses the spectrum up to 44 Hz, whereas the present thesis analyses the spectrum up to 100 Hz, including the high-gamma (40–60 Hz)

and very-high-gamma (60–100 Hz) bands.

## 2.2 Secondary Adjacent Studies

Six further studies record EEG in connection with short-form video, all with laboratory EEG systems of 32 to 64 electrodes.

**Short-video recommendation (Zhang et al.; He et al.)** [13, 14]. *Research goal:* use EEG to improve short-video recommendation (Zhang et al.) and to detect how immersed a viewer feels (He et al.). *Design:* two papers from one research group report the same lab study with 30 participants. Each participant gets a fresh account on a real platform, trained by ten hours of browsing; the researchers then compose 15-minute sessions of 20–30 personalised, non-personalised and random videos. Participants swipe freely while a 64-channel cap records at 1,000 Hz, filtered to 0.5–50 Hz. After each session, participants re-watch the first seconds of each video and rate the video from 1 to 5 on several scales; Zhang et al. call six of the scales, among them valence, interest and immersion, *affective engagement*, and He et al. predict the immersion rating. *Result:* adding EEG raises the AUC of the recommendation models, the chance that a model scores a randomly picked positive case above a randomly picked negative case ( $0.5 =$  guessing,  $1 =$  perfect), by at most about 0.04, and in a few models lowers the AUC slightly. He et al. predict high versus low immersion with 69.1% accuracy without EEG and 70.1% with EEG. *Limits:* the ratings come from memory after the session, and Zhang et al. and He et al. do not report keeping each participant on one side of the training/test split, so viewings of one participant may land on both sides, the leak described for the adjacent study (Section 2.1).

**Opinion polarisation (Du et al.)** [15, 16]. *Research goal:* show whether one-sided short videos shift a viewer’s opinion of a historical or fictional figure, and whether EEG reveals the shift. *Design:* two papers report one dataset of 23 participants. The design has three steps. In a first lab session with EEG, a web page built by the researchers plays, for each of ten such figures, one one-minute clip praising the figure and one clip criticising the figure, mixed with film-clip distractors; a “Next One” button skips a clip after 30 seconds. Over the following six days, participants watch clips on the same web page daily without EEG, and each figure appears to a participant only in praising clips or only in criticising clips, decided at random per figure. A second lab session repeats the first. A classifier then guesses, for each participant–figure pair, whether the six days praised or criticised the figure. Input: alpha-band power at one forehead electrode, FPz, from a 64-channel

cap filtered to 0.5–50 Hz, averaged over the praising and the criticising clip of the figure in both lab sessions. *Result:* opinion ratings of a figure rise after six praising days and fall after six criticising days; the classifier reaches 64.1% accuracy and an AUC of 0.672. *Limits:* the five cross-validation folds are drawn at random over participant–figure pairs, so pairs of one participant can land in training and test data; FPz was chosen because FPz scored best in the same cross-validation, which likely makes the 64.1% optimistic; and the six browsing days carry no EEG.

**Short-video advertising (Zhang and Zhang) [17].** *Research goal:* find which selling approach (transactional or relational) and which displaying style (high or low enthusiasm and humour) of an online celebrity evokes more “neural engagement”. *Design:* 25 participants watch the same 64 TikTok advertisements of 10–35 s each while a 64-channel cap records at 250 Hz. Neural engagement here means four EEG band-power ratios: emotion, relaxation, attention and cognitive load. The relaxation index is the Engagement Index,  $\beta/(\alpha + \theta)$ , turned into the negative logarithm, so a higher relaxation score means a lower Engagement Index. *Result:* transactional selling and the high displaying style evoke higher values on most of the four indices (analysis of variance). *Limits:* each advertisement plays to the end, so viewing is not self-paced, and the study compares group averages without predicting the state of a single viewer.

**Remaining studies.** Wade and Dumitru are unpublished student theses. *Wade asks:* does TikTok addiction impair response inhibition? EEG runs during a fixed 3-minute TikTok compilation and a following Go/NoGo task, in which a participant presses a button for one signal and holds back for another [18]. *Dumitru asks:* does TikTok-style content impair sustained attention? Participants scroll for 5 minutes through the own TikTok or Instagram feed; the EEG of the 5 minutes enters one left-versus-right alpha comparison, and a  $\beta/\alpha$  engagement index is computed during attention tasks at baseline and after viewing [19]. *Yan et al. ask:* does a tendency toward short-video addiction go with weaker attention control? Yan et al. correlate an addiction questionnaire score with EEG recorded during an attention task, and record no viewing [20].

**Gap the present thesis fills:** In the six studies, no label or comparison is anchored to the moment of a swipe: Zhang et al. and He et al. log every swipe but leave the moment of the swipe unused, keeping only the share of each video watched, Du et al. allow a skip only after 30 seconds, Dumitru does not record the swipes, and the remaining studies involve no swipe. The present thesis anchors every SKIP label to one time-stamped, self-paced swipe

in the TikTok “For You” infinite feed, and analyses the spectrum above 50 Hz, which the 0.5–50 Hz filters of Zhang et al., He et al. and Du et al. remove. The present thesis also records with a consumer headband, which, unlike the laboratory systems of the six studies, is affordable for a single consumer and can be worn during naturalistic use outside a laboratory.

## 2.3 Recording Software

One route from a Muse headband to a laptop is `muselsl`, an open-source tool created by Barachant et al. [21]. `muselsl` connects to the Muse EEG headband over Bluetooth, and the command `muselsl stream` passes the EEG on to the laptop in near real time; further `muselsl` commands display and record the EEG. `muselsl` runs as a command-line tool and as an importable Python library, and the current documentation lists the Muse S used in the present thesis as supported.

`muselsl stream` sends the EEG into the Lab Streaming Layer (LSL), a system built to unify the collection of time-series data in research experiments. LSL puts each data source on the network as a separate stream, stamps every sample with a time, keeps the streams synchronized in time, and gives near-real-time access to every stream; the `muselsl` documentation calls LSL standard in EEG-based brain-computer interface research. As far as the `muselsl` documentation shows, `muselsl` offers no way to mark a keypress in the EEG recording and no automatic reconnection after a Bluetooth dropout.

A second route is Mind Monitor, an app that records the Muse on a phone; the adjacent study used Mind Monitor on Android [12]. The adjacent study labels whole 15-minute blocks (Section 2.1), so the adjacent study needs no marker for a single event such as a swipe.

**Gap the present thesis fills:** Every SKIP label of the present thesis is anchored to one swipe, marked by a keypress on the laptop, and every STAY label covers the remaining viewing time minus a pre-swipe gap and a post-swipe gap around each keypress (Sections 3.2 and 3.4.1), so the EEG recording needs a time mark for every keypress and a record of every Bluetooth dropout. A recording pipeline by the author of the present thesis therefore starts `muselsl stream`, stamps each keypress with the laptop clock, converts the keypress time into LSL time, flags the EEG sample that falls within a defined tolerance window of the keypress, and restarts the stream in a new file after a Bluetooth dropout (Section 3.5). Preprocessing then measures

each Bluetooth dropout from the timestamps between consecutive files, and no analysis window crosses a Bluetooth dropout (Section 3.6).

## 2.4 Defining Engagement: Existing Definitions and Their Gaps

The fields that measure engagement do not define engagement consistently. O’Brien and Toms found no agreed definition when O’Brien and Toms set out to deconstruct the term for human–computer interaction [11]. More than a decade later, a systematic review of social media engagement describes engagement as a term with many meanings and many dimensions [10]. In digital behaviour-change research, Yardley et al. call for “defining engagement more consistently and appropriately” [6]. Five fields bear on the present thesis.

**Human–computer interaction.** O’Brien and Toms interviewed 17 users of web search, online shopping, webcasting, and gaming applications. O’Brien and Toms define engagement as “a quality of user experience characterized by attributes of challenge, positive affect, endurability, aesthetic and sensory appeal, attention, feedback, variety/novelty, interactivity, and perceived user control”. In the model of O’Brien and Toms, engagement unfolds as a four-stage process: point of engagement, sustained engagement, disengagement, and re-engagement [11]. A recent review of EEG-based engagement research adopts the attribute list of O’Brien and Toms [22]. The four stages rest on the interviews, in which participants recall a past experience, and O’Brien and Toms note that a recalled experience is a “selected interpretation” of the event [11].

**Marketing and platform analytics.** Trunfio and Rossi, following Hollebeek, describe customer engagement as the positive, brand-related cognitive, emotional, and behavioural activity of a consumer during interactions with a brand. Trunfio and Rossi review social media engagement and find the behavioural dimension to be the most common way to measure engagement: counts of likes, comments, shares, and views, and indices derived from the counts [10]. Salminen et al. measure engagement with short-video advertising as viewer churn, the share of viewers who leave during playback, in 12 TikTok advertisements of one company. TikTok reports churn per quarter of a video, counted over all viewers of an advertisement. Most viewers left within the first quarter of an advertisement [23]. Churn shows in which quarter of a video viewers leave, while the state of each viewer before leaving stays unmeasured.

**Human-factors neuroscience and later uses of the Engagement Index.** Pope et al. tested candidate EEG indices with six participants in a computer task modelled on flight-deck work. A feedback loop switched a tracking task between the participant and the computer whenever an index rose or fell over a time interval. Among the candidate indices, the ratio of beta power to the sum of alpha and theta power,  $\beta/(\alpha + \theta)$ , the Engagement Index, reflected task engagement best [4]. Pope et al. judged each index by the switching pattern of the loop, and report no separate measure of engagement. Nuamah et al., citing earlier work, define task engagement as “effortful striving towards task goals”. Nuamah et al. used the Engagement Index as the input for recognising which of five mental tasks a participant performed, trained one classifier per participant, and report an average accuracy of 88.67% across six participants, where guessing among five tasks gives 20%. The recordings come from an existing dataset of 10-second trials, with 25 to 75 trials per participant [24]. In a controlled learning task, Apicella et al. classified two levels of cognitive engagement from the Engagement Index alone with 51 to 55% accuracy, close to the 50% that guessing between two levels gives, while features from twelve narrow frequency bands, combined across electrodes, reached about 77%. Apicella et al. note that EEG signals change over time and differ between people, which makes classification across participants difficult [25], and a recent review names the differences in EEG between people as a pressing issue in EEG engagement research [22]. In a conference abstract, de Carvalho et al. treat the Engagement Index ratio as a vigilance index, that is, a marker of sustained attention. In the study of de Carvalho et al. with 15 students, three of four markers chosen for sustained attention, the Engagement Index ratio among them, separated a sustained-attention test from a cognitive-engagement test, while the markers chosen for engagement did not. De Carvalho et al. attribute the failure of one engagement marker, theta power at the front midline of the head, mainly to a resting baseline recorded after the effortful tests [26].

**Digital health and behaviour-change intervention science.** Perski et al. define engagement with two components: the extent of use, such as the amount, frequency, duration, and depth of use, and a subjective experience characterised by attention, interest, and affect. Perski et al. describe engagement as “a dynamic process” that is expected to vary within and across individuals over time [9]. Yardley et al. describe engagement as a process that starts with a trigger and first use, followed by sustained engagement or disengagement, and separate moment-to-moment engagement with the intervention from engagement with the wider goals of the intervention. Yardley et al. note that an observed end of use can mean disengagement, or can mean that the user has learned enough to need the intervention no longer.

Yardley et al. shift the target to *effective engagement*, and note that “more engagement does not necessarily equate to more effective engagement” and that effective engagement “may or may not require sustained engagement” [6].

**Learning analytics.** Karimah and Hasegawa review automatic engagement estimation and follow Fredricks et al., who divide engagement into behavioural, emotional, and cognitive engagement. Emotional engagement, measured from affective cues, is the type studied most often. The largest share of the reviewed studies (43%) rates engagement by external observation, followed by self-report (20%). Karimah and Hasegawa note that self-report carries bias, and that external observation limits the judgement of the actual effort of learners [27].

**Gap the present thesis fills:** O’Brien and Toms reconstruct disengagement from recalled accounts, and an observed end of use, such as viewer churn counted per quarter of an advertisement or the end of use of a health intervention, shows roughly when people stopped but leaves the state of each person before stopping unmeasured. In small studies, the Engagement Index recognised five mental tasks well (Nuamah et al.), classified two engagement levels close to guessing (Apicella et al.), and behaved as a marker of sustained attention (de Carvalho et al.); classification across participants remains an open problem, and, to the author’s knowledge, the behaviour of the Engagement Index in the seconds before a self-paced decision has not been reported. The present thesis states a working definition anchored to the swipe, an observed end of viewing (Section 1.2), tests whether the Engagement Index and other EEG features separate the seconds before each self-paced swipe from sustained viewing, within and across participants, and later proposes a general definition of engagement that names no platform and no measuring instrument (Section 4.6).

## 2.5 Measuring Engagement in the Health Communication Literature

The present section covers two bodies of work within health communication. The first body studies digital behaviour change interventions (DBCIs), such as health apps and health websites. The second body studies health messages on TikTok. Section 2.4 gave the definitions of engagement from the DBCI work. The present section asks how the two bodies of work *measure* engagement.

**Three routes.** In the DBCI work, engagement with a health intervention is a precondition for the effectiveness of the health intervention [6]. Perski et al. reviewed 117 articles on engagement with DBCIs and name three routes to measure engagement [9]. The present thesis calls the three routes the *ask route*, the *log route*, and the *body route*.

- The ask route: self-report questionnaires, interviews, or think-aloud studies.
- The log route: automated recording of use, such as logins and page views.
- The body route: recording of physiological or psychophysical correlates.

In the behavioural-science literature, the log route was the most common. Typical log-route measures were logins, time spent online, and the amount and type of content used. Body-route measures came from the computer-science and HCI literatures: cardiac activity, respiratory depth, electrodermal activity, and eye tracking. Perski et al. note two limits. Ask-route questionnaires typically measure engagement after a person has used the intervention, instead of during use. In the one study Perski et al. cite on the question, body-route measures and ask-route measures were only weakly associated [9].

**The log route in practice.** Yardley et al. judge that usage alone cannot be taken as a valid indicator of engagement. The evidence linking usage to health outcomes is mixed and largely correlational [6]. Evans et al. review digital media for behaviour change and name inconsistent, non-standardised measures, such as engagement and reach, as the research gap cited most often. Evans et al. count objective dosage and exposure data from platform analytics among the strengths of social-media interventions. Evans et al. recommend dose-response designs, which test whether more exposure goes with more behaviour change, and state that “what qualifies as a critical threshold for engagement in digital interventions has yet to be established” [28].

**The body route in practice.** Yardley et al. list body-route measures of task engagement: skin conductance, heart rate, facial expression, and fMRI. According to Yardley et al., body-route measures can show which parts of an intervention attract attention or evoke emotional arousal in the short term, and add an objective record of user reactions to ask-route measures. Yardley et al. also rate body-route measures as often intrusive, expensive, and hard

to scale. Body-route measures taken in a laboratory may lack ecological validity, meaning that laboratory results may not carry over to everyday use. Yardley et al. further warn that differences in attention “may not always translate into differences in intervention use” [6].

**Health messages on TikTok.** McCashin and Murphy systematically reviewed 24 studies of TikTok for public or mental health. In 20 of the 24 studies, the primary method was content analysis, which codes what the videos show [1]. Three studies describe health messages on TikTok through content analysis and the counts TikTok displays, such as views, likes, and comments.

Basch et al. coded 100 videos under #Coronavirus and all 17 posts of the World Health Organization (WHO) in April 2020. The 117 videos had 1,194,081,700 views in total. Fewer than 10% of the videos mentioned transmission, symptoms, or prevention. WHO videos covered transmission and symptoms more often than consumer videos, yet still in fewer than 10% of videos [29]. Very high reach therefore came with little health content. Basch et al. coded content and counts only, so the study does not show what viewers took in.

McCashin and Murphy also analysed the content of 29 Irish TikTok accounts: 2 public-health accounts, 5 charity accounts, and 22 personal creators. Both public-health videos came from one account, the Irish Health Service Executive. Public-health videos used humour in 0 of 2 videos and a popular sound in 0 of 2 videos. Personal creators used humour in 16 of 22 videos and a popular sound in at least 17 of 22 videos. Videos with a popular sound averaged 132,419 likes. Videos without a popular sound averaged 51,161 likes. McCashin and Murphy conclude that institutional accounts, meaning the public-health and charity accounts, used fewer TikTok features and received fewer likes, comments, and shares. The comparison uses no statistical test and rests on 2 public-health videos. McCashin and Murphy also ask whether TikTok analytics reflect in-depth user engagement that may lead to positive outcomes, and call for controlled studies that relate TikTok engagement to behaviour change [1].

Chen et al. analysed 355 COVID-19 videos from “Healthy China”, the TikTok account of the National Health Commission of China, from January to April 2020. Chen et al. measure engagement as the counts of likes, shares, and comments. Regression models for count data linked longer video length to fewer likes (incidence rate ratio,  $IRR = 0.19$ ,  $p < 0.001$ ) and fewer comments ( $IRR = 0.39$ ,  $p < 0.001$ ). An IRR below 1 means fewer counts, and an IRR above 1 means more counts. The emotion of the title changed the

link: longer videos with a positive title received more likes (IRR = 21.72,  $p = 0.04$ ) and more comments (IRR = 10.14,  $p = 0.047$ ) [30]. Chen et al. record the length of each video and do not record how long viewers watched.

**The research agenda.** Zenone et al. wrote when TikTok had passed one billion monthly users, and videos tagged #MentalHealth had 17 billion views. Zenone et al. propose six public-health research priorities for TikTok. Most of the six priorities concern the content on TikTok or TikTok as a company. One priority, on medical advice, also asks how users take up the health information received. Zenone et al. ask that research runs independently of TikTok, with conflicts of interest declared when TikTok funds or joins the research [31]. For DBCI research, Yardley et al. list a related priority: to “investigate and validate complementary and non-intrusive measures of effective engagement” [6].

**Gap the present thesis fills:** The health communication literature reviewed here measures engagement mostly through the log route, as counts of use and the counts TikTok displays, and the body-route examples of Perski et al. and Yardley et al. do not include EEG. To the author’s knowledge, no study in the health communication literature reviewed here records a body-route signal in the seconds before a viewer swipes a health message away, a moment that counts of use do not capture. The present thesis tests consumer EEG, recorded independently of TikTok and without data from TikTok itself (Section 4.2), as one candidate for the complementary measures Yardley et al. call for, keeping in mind that a headband remains a body-route measure that Yardley et al. rate as often intrusive and hard to scale.

## 2.6 Viewing Time as an Indicator of Engagement

Social media designers are tasked with maximising time on site [2], and Knierim et al. advise platforms to measure the quality of the experience rather than time on the platform alone [3]. To the author’s knowledge, no study tests directly whether long viewing of short-form video means engagement. Four studies bear on the question from two sides: two *daily-life studies* of social media use, and two *lab EEG studies* of attention within a task. None of the four studies records EEG during short-form video viewing.

**Knierim et al.** [3]. *Research goal:* test whether social media produces flow, a state of deep and effortless absorption. 40 university students took

part in a five-day field study. Tracking software logged the use of a preset list of phone apps. Three times a day, participants reconstructed the preceding hours and listed activities with flow. The survey did not ask about social media directly. Participants attributed 14 of 673 flow reports (2%) to social media. Knierim et al. sort apps, not behaviours, into two categories. Passive apps are built around feeds and videos, for example Instagram, TikTok, and YouTube. Active apps serve direct communication. When a participant used more social media than usual in the hours before a report, the participant reported fewer flow episodes. The drop in flow was significant for passive apps and not significant for active apps. More use than usual also went with lower mental readiness, a mix of mood, sleepiness, and motivation, for both app categories. Knierim et al. suggest physiological signals such as EEG to tell genuine flow apart from absorption that only resembles flow. *Limits:* the tracking software counts minutes per app, and participants recall flow hours later. No physiological signal is recorded.

**Baughan et al.** [2]. *Research goal:* learn how feed design shapes normative dissociation. Normative dissociation is full absorption with reduced self-awareness and a reduced sense of control. Baughan et al. count both mindless scrolling and deep absorption in the content itself as normative dissociation. 43 U.S. participants used a custom Twitter app for four weeks, and the Twitter app changed its design features every week. One in-app question measured dissociation, “I am currently using Chirp without really paying attention to what I am doing”, averaged per day. 11 participants were interviewed. Interviewees described getting “lost” in the feed and losing track of time. Custom lists and reading-history labels lowered dissociation. On days with more views of usage statistics, participants reported less dissociation. On days when participants closed the Twitter app through a time-limit prompt, participants reported more dissociation, and Baughan et al. read the time-limit prompt as a cue to notice the dissociation. Time spent reading, entered as a control variable, was not significantly associated with dissociation ( $p = 0.52$ ). *Limits:* the feed shows text posts, dissociation is one self-reported item, and daily averages cannot show whether dissociation came before or after an action. No physiological signal is recorded.

**deBettencourt et al.** [32]. *Research goal:* separate two kinds of attention in memory. *Sustained attention* is attention held steady over time. *Spatial attention* is attention aimed at one place on the screen. In three lab experiments, a cue first points to one place on the screen. Images of objects then appear. Later, participants mark where each object appeared, or which colour each object had. In the EEG experiment ( $n = 30$ ), a classifier reads the EEG after the cue and before the objects appear, and predicts whether an object will be remembered better or worse than the participant’s median,

at 57.2% accuracy against a 50% chance level. The classifier is trained and tested on averages over groups of trials, not on single trials. deBettencourt et al. read the 57.2% prediction as sustained attention rising and falling from trial to trial. During trials with low sustained attention, the EEG still showed spatial attention at the cued place. deBettencourt et al. therefore conclude that low sustained attention is not the same as dropping out of the task. *Limits:* each trial starts at a cue set by the experimenter, and the task offers no way to leave. The study does not test how sustained attention changes with time on task.

**De Carvalho et al.** [26]. *Research goal:* find EEG frequency markers that separate sustained attention from cognitive engagement. After the two tests described in Section 2.4, the 15 students watched a 27-minute science lecture video while a 32-channel research cap recorded the EEG. During the lecture video, several markers changed along curves over time. The Engagement Index ratio, measured over central and parietal sites, rose and then fell. Theta power at the back of the head fell and then rose. Alpha power over parietal and occipital sites, a marker chosen for engagement, rose steadily. De Carvalho et al. conclude that a marker averaged over a whole task hides changes within the task. *Limits:* the curves explain only about 1–2% of the variation in the markers, and no marker correlated with test scores. The lecture video offers no way to skip.

**What the four studies show together.** In the two daily-life studies, time spent in a feed was not significantly associated with dissociation, and more passive use than usual went with fewer flow reports. In the two lab EEG studies, attention changed within a task on two different timescales: from trial to trial in deBettencourt et al., and along slow curves over 27 minutes in de Carvalho et al. In deBettencourt et al., a drop in sustained attention did not mean leaving the task.

**Gap the present thesis fills:** To the author’s knowledge, no study has yet checked whether long viewing without a skip reflects engagement, and none of the four studies records a signal during short-form video viewing that could separate attention held on the content from continued viewing alone, a separation for which Knierim et al. suggest EEG. The present thesis records EEG during naturalistic use and puts all continuous viewing without a skip into one class, **STAY**, treated as a partial operationalisation of engagement rather than as engagement itself. The EEG analysis (Sections 3.7–3.8) tests what separates **STAY** from **SKIP**, keeping in mind that, as in deBettencourt et al., **STAY** may contain stretches of low attention that do not end in a swipe.

## 2.7 Parameter Choice in EEG Classification

An EEG classifier does not read a recording in one piece. The recording is first cut into windows. A window is a short slice of the recording. Each window becomes one example for the classifier. Two timing choices shape every window. The window length sets how long each window is. The window placement sets where each window sits relative to an event, such as a cue on the screen. Five studies show how window length and window placement change classification results. A separate line of work describes how window length and window placement are usually tested.

**Moontaha et al.** [33]. *Research goal:* classify emotions from wearable EEG in real time. In the main experiment, participants watched 16 short emotional videos, 8 with each of two consumer EEG devices. One of the two devices was the Muse S. Moontaha et al. tested window lengths of 1, 2, 3, 4, and 5 s, on the recordings of the study and on a public dataset (AMIGOS). The 1-second window scored best for every emotion dimension, classifier, and device. In most cases, the score fell as the window grew longer. Moontaha et al. read the fall as a sign that the classifier needs more examples. A shorter window gives more examples per video. *Limits:* the windows were shuffled before cross-validation, so neighbouring windows from the same video likely appear in both the training data and the test data, which may favour short windows.

**Tzamourta et al.** [34]. *Research goal:* detect Alzheimer’s disease from clinical EEG. 24 people took part: 14 patients and 10 healthy people of similar age. Participants sat relaxed with eyes closed. Window lengths ran from 5 to 12 s, in steps of 1 s. Accuracy rose at every step, in all six classification problems. The 12-second window scored best. *Limits:* the target is a disease, and a disease stays the same across the whole recording.

**Window length: two directions.** Moontaha et al. found shorter windows better. Tzamourta et al. found longer windows better. The finding of Moontaha et al. covers 1–5 s. The finding of Tzamourta et al. covers 5–12 s. The targets differ as well. An emotion can change within one video. A disease stays the same across a recording, so a longer window may simply average more of the same state.

**Mi et al.** [35]. *Research goal:* improve motor-imagery classification by combining several window lengths. In motor imagery, a participant imagines moving a hand, a foot, or the tongue after a cue. On a public dataset of nine participants, Mi et al. tested single window lengths from 0.5 to 3 s. The best window length differed from participant to participant. For one participant, accuracy differed by about 13 percentage points between window lengths. A combination of several window lengths beat every single window length.

*Limits:* the window-length test used one dataset of nine participants.

**Rimbert et al.** [36]. *Research goal:* test how the placement of a rest window changes motor-imagery results. 71 participants imagined left-hand or right-hand movements. A classifier separated imagined movement from rest. The rest window lasted 3 s and sat at one of four positions around the next cue, from 5 s before the cue to 1 s after the cue. The rest window furthest from the next cue gave the highest accuracy. The furthest rest window also sat closest to the end of the previous trial. The furthest rest window scored 4 percentage points above the rest window ending at the cue ( $p < 0.05$ ), and 8 points above the rest window overlapping the cue ( $p < 0.001$ ). Rimbert et al. warn that part of the gain may come from the previous trial. After a movement ends, real or imagined, beta power over the motor cortex rises above the resting level for a while. The rise peaks about 1.5 s after the movement. A rest window close to the previous trial can fall inside the rise. Measured against the raised level, the power drop during imagined movement looks larger than the true drop. Rimbert et al. therefore recommend at least 3 s between the end of one movement and the start of a rest window. The best position also differed between people. The furthest position was best for 52–55% of participants, and the second-furthest position for 22–30%. *Limits:* every rest window lasted 3 s, and the study covers one motor-imagery task.

**How timing choices are tested.** A sensitivity analysis asks how much a result changes when a setting changes. The simplest design changes one setting and holds every other setting fixed, like turning one knob on a mixing desk while the other knobs stay put. The design is called one-at-a-time (OAT) [37]. OAT never changes two settings together. OAT therefore cannot show an interaction, meaning a case where the effect of one setting depends on the value of another setting [37]. OAT is described as the most popular sensitivity-analysis practice in published modelling work [38]. Saltelli et al. argue that OAT explores the possible settings inefficiently, and recommend alternatives from statistical theory, such as experimental design and regression analysis [38]. Moontaha et al., Tzimourta et al., and Mi et al. each vary window length one setting at a time. Rimbert et al. vary window placement one setting at a time. McCartney et al. vary two settings together.

**McCartney et al.** [39]. *Research goal:* find the best combination of time window and frequency band for motor-imagery classification. On a public dataset of 109 people, McCartney et al. built one model per person for every combination of 5 time windows and 23 frequency bands. The time windows differed in length and in start after the cue. A repeated-measures ANOVA, a standard test for comparing the same people across conditions, found an effect of time window ( $p < 0.001$ ) and an effect of frequency band ( $p < 0.001$ ). The interaction of time window and frequency band gave  $p = 0.044$ , close

to the usual 0.05 threshold, and McCartney et al. report the interaction as significant. The best combination for the whole group was 0–4 s after the cue at 4–12 Hz. The best time window differed between people: 0–2.25 s was best for 34 of the 109 participants, and 0–4 s for 31. *Limits:* each per-person model learns from about 90 trials, and the best time window averages 57% accuracy, close to the 50% of guessing between the left and the right hand. The study is a preprint.

**Gap the present thesis fills:** Every study above either places its windows around an event set by the experimenter, such as a cue or a video, or cuts a resting recording with no event at all. EEG studies of self-paced movement do classify windows before the movement [40], but, to the author’s knowledge, no study varies window length and window placement before a self-paced action during media viewing, such as a swipe. The present thesis labels windows before each swipe as **SKIP** and varies three labeling parameters: the window length, the pre-swipe gap between the **SKIP** windows and the swipe, and the **SKIP** period, meaning the stretch that holds the **SKIP** windows. The pre-swipe gap and the **SKIP** period are crossed on a grid; the window length is varied one at a time (Section 3.4.5).

## 2.8 Windows at Label Borders

A labelled EEG recording consists of stretches. Each stretch carries one label. In an emotion study, one stretch can be one video, labelled with the rating of the participant. In the present thesis, one stretch is either **STAY** or **SKIP**. A window that crosses the border of a stretch holds EEG from both sides of the border. Yet the window can carry only one label. One EEG study and one review handle windows at a border.

**Moontaha et al.** [33]. Section 2.7 introduced the emotion study of Moontaha et al. Moontaha et al. cut the EEG into windows without overlap. Every window inside a video gets the rating the participant gave that video. Where possible, Moontaha et al. align the windows with the end of the video. Otherwise, Moontaha et al. use only windows that lie completely inside the time range of the video, so windows that cross the start or the end of a video stay unused. *Limits:* Moontaha et al. give no reason for leaving out windows that cross a video border, and report no test of windows at a border. The version read for the present thesis is the preprint.

**Frénay and Verleysen** [41]. *Research goal:* review what wrong labels do to a classifier, and how methods deal with wrong labels. Frénay and Verleysen name disagreement between experts as one source of wrong labels. For

example, in electrocardiogram analysis, experts seldom agree on the exact borders of the waves in a heartbeat recording. One reviewed electrocardiogram study therefore removes the labels of examples close to a border between two classes, because experts label less reliably near such a border. The chance of a wrong label can differ between examples. An example that resembles the other class may get a wrong label more often. Wrong labels can lower the accuracy of a classifier and raise the number of training examples a classifier needs. Several reviewed works found that removing examples with wrong labels worked better than correcting the labels. Removing too few or too many examples can both lower performance. *Limits:* the review treats wrong labels in general and includes no EEG study.

**Gap the present thesis fills:** In Moontaha et al., the experimenter sets every border, so the time of every border is known, and a window that crosses a border can be found and left out. In the present thesis, the viewer ends each video with a self-paced swipe, and the moment the decision to swipe begins is unknown, so even a window that lies completely inside one **STAY** or **SKIP** stretch may carry a wrong label. To the author’s knowledge, no EEG study tests where to place a label border whose time is unknown; the present thesis moves the border between **STAY** and **SKIP** by varying the pre-swipe gap and the **SKIP** period, and reports how the separation between **STAY** and **SKIP** changes (Section 3.4.5).

## 2.9 Class Separation Metric

A study that compares features measured in different units, such as microvolts and hertz, cannot compare the raw differences between two class averages directly. Cohen’s  $d$  divides the difference between two class averages by the spread of the feature within the two classes, so the unit cancels out. One book defines Cohen’s  $d$ , and two EEG studies use Cohen’s  $d$  to compare two groups of people.

**Cohen** [42]. *Research goal:* give behavioural researchers tables that show how likely a study with a given number of participants is to detect an effect of a given size. Cohen defines  $d$  for two groups as the difference between the two group averages, divided by the standard deviation within the groups. Cohen calls  $d$  a “pure” number, free of the original unit, and argues that  $d$  helps especially where the raw unit is arbitrary, such as points on an attitude scale. For two groups that spread differently, Cohen divides by the square root of the average of the two variances. For two measurements taken from the same people, such as a sensory threshold measured on both sides of the

body of each patient, Cohen gives a second version of  $d$ , which first takes one difference per person and then divides the average of the differences by the standard deviation of the differences.

**Henao Isaza et al.** [43]. *Research goal:* develop a way to enlarge small EEG samples, tested on telling carriers of a gene mutation that causes Alzheimer’s disease (PSEN1-E280A), who show no cognitive symptoms yet, apart from healthy non-carriers, using EEG recorded at rest. Henao Isaza et al. pool four EEG datasets and, aiming to balance age and sex, form samples with 2, 5, and 10 non-carriers per carrier. A decision tree selects the 100 most important features, and Cohen’s  $d$  then describes how far carriers and non-carriers differ in each selected feature. *Limits:* to reach the higher ratios, Henao Isaza et al. remove carriers, from 79 carriers at a ratio of 2 to 15 carriers at a ratio of 10, and the remaining carriers are older: the average age of the carriers rises from about 35 to 42 years, while the non-carriers stay near 30 years. Over the same steps, the largest  $d$  of an EEG feature rises from 1.22 to 2.28 in absolute value, and  $d$  for age rises from 1.10 to 2.78. The rise of  $d$  for age suggests that  $d$  changes with the make-up of the sample as well as with the EEG.

**Kiarashi et al.** [44]. *Research goal:* tell older adults with mild cognitive impairment (24 people) apart from cognitively normal older adults (19 people), using sleep EEG recorded at home over several nights with a device worn in the ear. A classifier gives each person a probability of mild cognitive impairment, and Cohen’s  $d$  measures how far the probabilities of the two groups lie apart. Averaging over all recorded nights, instead of using the first night alone, raises  $d$  for all three neural networks tested, for example from 0.65 to 0.96. Kiarashi et al. attribute the rise to the averaging, which reduces the night-to-night variation within each person while keeping the differences between people. Kiarashi et al. also compute  $d$  for single EEG features between the two groups. The study is a preprint.

**Gap the present thesis fills:** In Henao Isaza et al. and Kiarashi et al., Cohen’s  $d$  compares two groups of different people, with one value per person, and  $d$  changes with the people in the sample and with the variation within each person. In the present thesis, STAY windows and SKIP windows come from the same participants, yet Cohen’s  $d$  is computed with the two-group version over the windows of all participants together, instead of with the version for two measurements from the same people; the baseline z-scoring of each participant reduces the differences between participants (Section 3.6), but part of the differences may remain in the standard deviation and keep  $d$  small. The raw features, such as TP10 raw standard deviation, are not

z-scored at all (Section 3.6.2), so for the raw features the differences between participants remain unreduced. The share of SKIP windows also differs between participants (Figure 3.19), so for the raw features the differences between participants may also shift the two class averages, and so may raise  $|d|$ . The present thesis uses Cohen’s  $d$  to compare the seconds before a self-paced swipe (SKIP) with sustained viewing (STAY), both to rank single features (Section 3.8.2) and to compare values of the labeling parameters (Section 3.4.5).

## 2.10 Estimating Frequency from Local Maxima

Peak Frequency, one feature of the present thesis, counts per second how often the strength of one frequency band peaks (Section 3.7.4). Each peak is a local maximum, a point higher than both neighbouring points. The count need not equal the frequency of the oscillation within the band. In EEG research, “peak frequency” often names the frequency with the largest value in the spectrum, meaning the strength of each frequency in the signal; in the present thesis, Peak Frequency names the count of local maxima per second. Three works estimate frequency, or information about frequency, from the signal over time.

**Hjorth** [45]. *Research goal:* describe the general character of an EEG trace with a few numbers, computed from the signal over time. One of the numbers is Mobility. Mobility is the square root of the ratio between the variance of the first derivative, meaning the slope of the signal, and the variance of the signal. Hjorth shows that Mobility equals the standard deviation of the spectrum along the frequency axis, and reads Mobility as a mean frequency. Computing Mobility from the signal over time needs no spectrum. Hjorth also notes that measuring the time between zero crossings, the moments the signal crosses zero, usually ends up as a frequency measure. *Limits:* Hjorth links Mobility to frequency by a mathematical derivation, and tests the link on no EEG recording. The only recording shown covers one night of sleep, and plots a mean frequency that Hjorth calls very similar to Mobility, and not Mobility itself.

**Khambampati et al.** [46]. *Research goal:* predict the recovery of patients in a coma after cardiac arrest from EEG, as an entry to a public competition, the PhysioNet Challenge 2023. Khambampati et al. compute three features from the spectrum, among them the dominant frequency, the frequency with the largest value in the spectrum. Khambampati et al. compute three features from the signal over time, among them the zero-crossing rate, the number of times the signal crosses zero. In a pure sine wave, each

cycle crosses zero twice. Khambampati et al. describe the zero-crossing rate as information about the frequency of rapid changes in the EEG. A classifier, XGBoost, combines all features with patient data such as age, and predicts the outcome with 81% accuracy on the hidden test data. *Limits:* the zero-crossing rate is one of many features. The paper reports no separate contribution of the zero-crossing rate, and does not compare the zero-crossing rate with the frequency of the signal.

**Niknazar et al.** [47]. *Research goal:* measure how similar two nonlinear signals are, such as signals of chaotic systems, from the local extrema of the signals. A local extremum is a local maximum or a local minimum. Niknazar et al. use the height of each local extremum and the time between consecutive local extrema. Niknazar et al. state that local extrema carry information about the amplitude and the “global frequency” of a signal, and treat local extrema as a down-sampled version of the signal. Niknazar et al. also state that finding local extrema is highly sensitive to noise, especially high-frequency noise. Added noise raises the number of local extrema. On a public EEG dataset, the method of Niknazar et al. classifies seizure EEG, seizure-free EEG of patients, and EEG of healthy people in five tasks, with 90% to 97% accuracy depending on the task, and 99% to 100% when a second classifier is added. *Limits:* Niknazar et al. test the whole similarity method, and do not test the claim about frequency on its own.

**Gap the present thesis fills:** The three works count maxima, extrema or zero crossings of the signal itself. Shin et al. count local maxima of beta power above a threshold, and report the count per trial as the rate of beta events [48]. To the author’s knowledge, no study counts every local maximum of the strength of an EEG band, without a threshold, so Peak Frequency is a feature the present thesis defines without prior validation. Because added noise raises the number of local extrema [47], the present thesis computes Peak Frequency within each frequency band, where filtering removes noise outside the band. The present thesis tests only how well Peak Frequency separates **STAY** from **SKIP**, with Cohen’s  $d$  (Section 4.4.1) and with logistic regression (Section 4.4.3).

## 2.11 EMG Artifacts in EEG

Muscles produce electrical activity of their own. A recording of muscle activity is called an electromyogram (EMG). A scalp electrode records the EMG of nearby muscles together with the EEG of the brain. EMG and EEG share many frequencies, much like two singers who sing the same notes: a filter that

keeps some frequencies and removes others cannot cleanly pull the two voices apart. Two head muscles are named among the main sources of EMG in the EEG [49, 50]: the frontalis, the forehead muscle that raises the eyebrows, and the temporalis, the muscle at the temple that closes the jaw. Four studies describe how EMG from head muscles enters the EEG, and how much EMG a cleaning method removes.

**Goncharova et al.** [49]. *Research goal:* describe which frequencies and which scalp sites the EMG of the frontalis and the temporalis reaches. 25 healthy adults wore a 64-electrode cap. Participants either relaxed or tensed one muscle at a set strength, for example at 15% of the strongest possible tension. *Result:* EMG covered all frequencies from 0 to more than 200 Hz. Near the tensed muscle, EMG reached every EEG band, delta included. EMG was strongest at the electrodes closest to the tensed muscle, and spread over the whole scalp at higher frequencies. For the temporalis, EMG at temporal and nearby electrodes was highest at 40–80 Hz. On electrodes placed over the muscles themselves, EMG often showed a sharp peak that resembles an EEG beta rhythm, at 16 to 38 Hz for the frontalis and at 13 to 34 Hz for the temporalis, depending on the person. For the frontalis, the strongest EMG lay at the front edge of the scalp in two thirds of participants, usually at Fp1 or AF8. *Limits:* participants tensed each muscle on purpose, guided by feedback on a screen.

**Whitham et al.** [51]. *Research goal:* measure how much of the scalp EEG comes from muscles. Two men were recorded with 115 EEG channels twice: once normally, and once under a drug that paralysed all skeletal muscles except one arm, kept free for answering. Paralysis removes EMG, so the difference between the normal and the paralysed recording shows the share of EMG. *Result:* under paralysis, power above 20 Hz fell, by about 100-fold at 40–100 Hz. The largest drop, above 200-fold, lay at the lower edge of the scalp, near the frontalis, the temporalis and the neck muscles. Below 20 Hz, the normal and the paralysed recording did not differ. Whitham et al. conclude that most scalp EEG above 20 Hz comes from muscles, even in a relaxed person. *Limits:* the study has two participants. Whitham et al. note that the participants were highly alert during the procedure, which may have raised muscle activity, so the reported share of EMG may be on the high side.

**Yilmaz et al.** [50]. *Research goal:* test whether head muscles send EMG into the EEG during rest, and whether independent component analysis (ICA) removes the EMG. ICA is a common cleaning method: ICA splits the EEG into parts, and the parts judged to come from muscles or eyes are removed. In 13 healthy adults, fine wires inside the frontalis recorded single motor units, the smallest units of a muscle that fire together, while

a 32-electrode cap recorded the EEG. For comparison, Yilmaz et al. reused recordings of the temporalis from an earlier study. *Result:* motor units of the frontalis fired continuously while participants rested. Averaged over many firings, many motor units left a small trace on nearby frontal electrodes, mostly Fp1, Fp2, F3 and F4. Traces from the temporalis spread wider, and were strongest at F7 or Fp1. At most electrodes, ICA made the traces of both muscles significantly smaller, yet did not remove the traces completely. *Limits:* the study covers two muscles, and ICA was tested only on the 19 of 58 frontalis motor units with a clear trace on more than one electrode.

**Li et al.** [52]. *Research goal:* remove more EMG by adding EMG recorded from head and neck muscles as extra input to ICA. *Result:* in 8 healthy adults clenching one hand, the method removed about 75% of the power at 40–100 Hz, a range dominated by EMG, where ICA alone removed about 49%. *Limits:* the method still cannot remove all EMG, partly because some head and neck muscles are hard to record from the skin.

**What the four studies show together.** Whitham et al. and Yilmaz et al. find EMG even at rest. Below 20 Hz, Goncharova et al. show EMG near a tensed muscle, while Whitham et al. found no difference below 20 Hz between the normal and the paralysed recording. ICA, with or without extra EMG input, reduces EMG without removing EMG completely.

The Muse S of the present thesis records at AF7 and AF8 on the forehead, near the frontalis, where Goncharova et al. and Yilmaz et al. found the strongest EMG, and at TP9 and TP10 behind the ears, at the lower edge of the scalp next to the small muscles around the ear [53], where Whitham et al. found the largest share of EMG. The present thesis applies no EMG correction, records no extra EMG, and removes no windows flagged as artifacts (Section 3.6.4). Facial muscle activity changes with mental effort and emotion [53], so if jaw or forehead tension changes before a swipe, the three features the present thesis tests further, TP10 raw standard deviation, AF7 high-gamma mean and AF8 delta peak frequency (Section 4.4.1), may reflect muscle activity, brain activity, or both. The present thesis cannot tell muscle activity and brain activity apart, and returns to the origin of the three features in Section 4.5.

## 2.12 Delta-Band Function in Awake Cognition

Delta is the slowest EEG band. The present thesis defines delta as 1–4 Hz (Section 3.6). Delta is best known from deep sleep. Two reviews describe what delta does in the awake brain, and one study relates delta to addiction to short-video apps.

**Güntekin and Başar [54].** *Research goal:* review delta responses in healthy people and in several disorders. A delta response is a change in delta right after a stimulus, such as a tone or a picture. *Result:* delta responses take part in cognitive processes, mainly in decision making and attention, and in emotional processes. Delta responses are smaller in Alzheimer’s disease, mild cognitive impairment, bipolar disorder, schizophrenia and alcoholism. Güntekin and Başar therefore propose a smaller delta response as a possible general marker of impaired cognition. *Limits:* the review mainly covers delta right after a stimulus.

**Knyazev [55].** *Research goal:* explain what delta does, from the view of evolution. *Result:* delta dominates the waking EEG of reptiles. In humans, delta is strong mainly early in development and in deep sleep. Delta rises with hunger, sexual arousal and pain, and in drug users, which links delta to motivation. In cognition, delta is linked to attention, to salience detection, meaning the noticing of what matters, and to perception below awareness. Knyazev concludes that delta reflects old, basic processes. In awake adults, newer processes tied to faster bands overshadow the old processes. The old processes rise when the newer processes lose priority, for example in sleep, early in development, or in disease. *Limits:* the conclusion is a hypothesis built on a review, and Knyazev notes that the evidence is scarce.

**Xu et al. [56].** *Research goal:* test whether addiction to short-video apps goes with weaker working memory and weaker cognitive control in adolescents. Cognitive control was tested as the switching between two tasks. *Design:* 84 Chinese adolescents (mean age 12.3) filled in an addiction questionnaire and did two cognitive tests. One week later, an 8-electrode cap recorded 1 minute of rest and then 3 minutes of free browsing on Douyin, the Chinese version of TikTok. Xu et al. compared the power of each band during browsing with the power at rest. *Result:* delta rose over the whole head during browsing, which Xu et al. read as attention to the videos. Of all bands, only the change in delta predicted the addiction score ( $F = 2.184$ ,  $p = 0.040$ , adjusted  $R^2 = 0.115$ ). The more addicted an adolescent, the less delta rose at left temporal and left occipital electrodes. The delta pattern tied to addiction predicted lower working memory ( $r = -0.226$ ,  $p = 0.043$ ) and lower cognitive control ( $r = -0.220$ ,  $p = 0.047$ ). *Limits:* the effects are small, browsing lasted 3 minutes, and the paper reports no times of single swipes.

**What the three works show together.** Güntekin and Başar link delta right after a stimulus to attention and decision making. Knyazev links delta to motivation, attention and salience detection, drawing also on delta at rest and in sleep. Xu et al. link the change in delta during free browsing to a questionnaire score of addiction.

**Gap the present thesis fills:** None of the three works relates delta to the seconds before a self-paced swipe: Güntekin and Başar review delta mainly right after a stimulus, and Xu et al. report no times of single swipes. The three works also study mainly the strength of delta, whereas AF8 delta peak frequency, one of the three features the present thesis tests further (Section 4.4.1), counts how often per second the strength of the delta band peaks (Section 3.7.4). The present thesis tests how well the delta features, AF8 delta peak frequency among them, separate the seconds before each swipe (SKIP) from sustained viewing (STAY), and compares the delta features with the features of the other bands and with the Engagement Index (Sections 4.4.1 and 4.4.3).

## 2.13 Notch Filtering and the High-Gamma Band

The recordings of the present thesis took place in Germany (Section 3.2). Power lines in Germany carry alternating current at 50 Hz. EEG electrodes pick up the electrical field of the power lines as powerline-noise, a sharp peak at 50 Hz. A notch filter removes a narrow band of frequencies around 50 Hz. The present thesis defines high gamma as 40–60 Hz (Section 3.6). The 50 Hz peak therefore lies in the middle of high gamma. A notch filter removes powerline-noise and any brain activity near 50 Hz together. One study compares a low-cost EEG amplifier with a research amplifier in the gamma range.

**Pattisapu and Ray [57].** *Research goal:* test whether a low-cost, open-source EEG amplifier (OpenBCI) records gamma as well as a research amplifier (Brain Products). *Design:* 11 adults looked at full-screen striped patterns. Full-screen striped patterns are known to raise gamma. Each participant was recorded in two sessions, once with each amplifier. Both amplifiers recorded through the same cap of gel electrodes. *Result:* OpenBCI recorded a gamma response, a rise in gamma power from before the patterns to during the patterns, in almost all participants who showed a gamma response with the research amplifier. The alpha responses and gamma responses of the two amplifiers were highly correlated. In a test with the electrodes in salt water and no person attached, OpenBCI showed more powerline-noise than the research amplifier. OpenBCI also showed three extra peaks, at 14, 36 and 64 Hz. 36 and 64 Hz lie 14 Hz below and above 50 Hz. Pattisapu and Ray trace the extra peaks to a distortion of the powerline-noise inside the OpenBCI amplifier. The research amplifier showed no extra peaks. In the recordings of people, the extra peaks remained in some participants. The gamma response was smaller with OpenBCI, especially in fast gamma

(35–66 Hz). Pattisapu and Ray name the extra peaks as one possible reason. *Limits:* the study used gel electrodes, unlike the dry electrodes of the Muse S.

In the bands of the present thesis, the three extra peaks would fall into beta (14 Hz), low gamma (36 Hz) and very high gamma (64 Hz). The present thesis did not test whether the Muse S shows the same extra peaks. To the author’s knowledge, no study measures for a consumer headband how much of high gamma is powerline-noise and how much is brain activity. The present thesis applies no notch filter in any reported analysis. A notch-filtered copy serves only the artifact check (Section 3.6.4). Two of the three features the present thesis tests further may therefore partly reflect powerline-noise: AF7 high-gamma mean, and TP10 raw standard deviation, computed on the raw signal, the signal before the split into bands, which keeps the 50 Hz peak. If the Muse S shows the same extra peaks, beta, low-gamma and very-high-gamma results may also partly reflect the extra peaks (Section 4.5). Very high gamma (60–100 Hz) also reaches 100 Hz, twice 50 Hz, where powerline-noise can show a second peak.

## 2.14 Pre-Motor Window

The brain prepares a voluntary movement before the movement starts, and the muscles activate shortly before the movement. EEG recorded shortly before a movement may therefore partly reflect the preparation of the movement. Three works give the timing of brain and muscle activity before a self-paced movement.

**Libet et al.** [58]. *Research goal:* compare the start of brain activity before a voluntary movement with the moment a person becomes aware of wanting to move. *Design:* 5 participants flexed the fingers or the wrist of the right hand whenever they felt like it. Libet et al. measure all times from the start of muscle activity in the forearm, which comes just before the movement. Participants watched a spot circling like a clock hand. After each movement, participants reported where the spot had been when the wish to move first appeared. *Result:* the brain activity before the movement is called the *readiness potential*, or *Bereitschaftspotential*. The readiness potential is a slow negative shift in the EEG. In series without planning ahead, the readiness potential began on average about 550 ms before the muscle activity. In series where participants sometimes planned ahead, the readiness potential began on average about 1,025 ms before the muscle activity. The reported wish to move came about 200 ms before the muscle activity, so the readiness potential began about 350 ms before the reported wish without planning

ahead, and about 800 ms before the reported wish with some planning ahead. Libet et al. conclude that the brain can start a voluntary movement before the person is aware of wanting to move. *Limits:* the study has 5 participants, the moment of awareness rests on what each participant recalled, and each EEG recording covered only 1.4 s before the muscle activity, so an earlier start of the readiness potential could not be seen.

**Shakeel et al.** [59]. *Research goal:* review methods that detect from EEG that a person is about to move. *Result:* the readiness potential begins about 1 to 1.5 s before the movement. The readiness potential has an early part, which rises slowly, and a late part, which begins about 400–500 ms before the movement and rises more steeply. The start of the readiness potential varies considerably between conditions and between people. The summary of the review gives a start of about 2 s before the movement. *Limits:* the timing values come from earlier studies, and the review measures none of them.

**Lew et al.** [40]. *Research goal:* detect from EEG, in single trials, that a person is about to start a self-paced reaching movement. *Design:* in the first experiment, 8 healthy adults rested one hand on a home button. A sound named one of four targets. Participants then reached for the target whenever they wished, but not earlier than 2 s after the sound. Release of the home button marked the start of the movement. Electrodes on the skin recorded the EMG of the biceps, the muscle at the front of the upper arm. *Result:* the biceps activated on average  $263 \pm 40$  ms before the start of the movement. A classifier detected the coming movement from the EEG above chance, on average about 460 ms before the start of the movement. *Limits:* the movement is an arm reach, and a sound set the target and the earliest start.

**What the three works show together.** The reported start of the readiness potential ranges from about 0.55 s before the movement in Libet et al. to about 2 s in the summary of Shakeel et al., and varies between people. All three works measure time from the movement, or from the muscle activity just before the movement.

The present thesis leaves the *pre-swipe gap*, the last 2.0 s before each keypress that marks a swipe, out of both **STAY** and **SKIP** (Section 3.4.1). The participant presses the key with the other hand at the moment of each swipe (Sections 3.2 and 3.3), so the brain likely prepares two movements, the swipe and the keypress. If the keypress comes slightly before or after the swipe, the true gap before the swipe is slightly shorter or longer than 2.0 s. The pre-swipe gap exceeds the start of about 1 to 1.5 s in Shakeel et al. by about 0.5 to 1 s, and leaves no margin to the start of about 2 s in the summary of Shakeel et al. The present thesis does not test whether the **SKIP** windows are

free of movement preparation. In bursts, some **SKIP** windows even lie inside the pre-swipe gap of a neighbouring swipe (Section 3.4.2).

## 2.15 Summary of Gaps

To the author's knowledge, the reviewed literature leaves seven gaps:

1. No EEG study anchors a label to the self-paced swipe (Sections 2.1 and 2.2).
2. Engagement lacks an agreed definition (Section 2.4).
3. Research measures engagement mostly by time spent and counts. Whether long viewing means engagement is untested (Sections 2.5 and 2.6).
4. No health communication study records a body-route signal before a viewer swipes away (Section 2.5).
5. No study reports the Engagement Index before a self-paced decision (Section 2.4).
6. No study varies window timing or label borders before a swipe (Sections 2.7 and 2.8).
7. For a consumer headband, the share of powerline-noise in high gamma is unmeasured (Section 2.13).

# Chapter 3

## Materials and Methods

### 3.1 Participants

Twenty-eight participants were recruited ( $N = 28$ ; 14 female, 14 male). Participants needed normal or corrected-to-normal vision and no history of neurological disorders. Much of the recruitment ran through university mailing lists. The sample therefore skews toward an academically oriented population (Sections 5.1 and 5.2).

Before the main recordings, three pilot sessions (P1–P3) tested the recording and analysis pipeline. P1 is a recording of the author and experiment lead. P2 and P3 are recordings of two other people. The pilot sessions are not part of the sample. The recruited participants therefore carry the IDs P4–P31.

Three recruited participants were excluded before analysis. P16 and P19 each reported more than 10 keypress errors: 45 and 40 swipes without a keypress. A swipe without a keypress leaves no time mark. The seconds before that swipe would then count as **STAY** instead of **SKIP**. P29 reported a diagnosed neurological disorder in the pre-session survey. Figure 3.1 shows the selection.

### 3 MATERIALS AND METHODS

#### 3.1 PARTICIPANTS

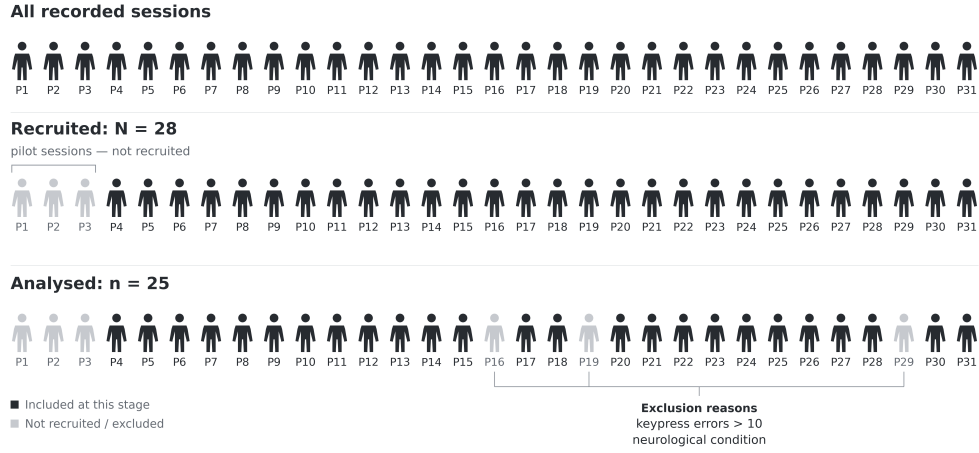


Figure 3.1: Participant selection from the recorded sessions to the analysed sample. Of 31 recorded sessions, P1–P3 were pilot sessions and not part of the sample. The  $N = 28$  recruited participants carry the IDs P4–P31. P16 and P19 were excluded for more than 10 self-reported keypress errors. P29 was excluded for a diagnosed neurological disorder. The analysed sample is  $n = 25$ . IDs stay unchanged throughout the present thesis, so IDs run up to P31.

The analysed sample has  $n = 25$  participants (13 female, 12 male; age 20–30 years,  $M = 24.6$ ,  $SD = 3.0$ ). In the analysed sample, participants reported about 1.5 keypress errors per session on average: 0.8 swipes without a keypress and 0.7 keypresses without a swipe. P8 did not count keypresses without a swipe. A session held about 134 keypresses on average. The reported errors therefore come to about 1% of keypresses, and to at most 4.5% (P7). P18, P21 and P25 reported an ADHD or attention-related diagnosis. P11 reported taking psychoactive medication. Neither ADHD nor psychoactive medication was an exclusion criterion. The missing criterion is an acknowledged limitation of the present thesis (Section 5.1).

Recording took five days. P4–P14 (11 participants) were recorded on the first two days, without payment. P15–P31 (17 participants) were recorded on the last three days and received a 20-euro voucher. In the analysed sample, 11 participants were unpaid and 14 were paid, about half each. Whether the voucher affected the data is beyond the scope of the present thesis and left to future research (Section 5.2).

All participants gave informed consent and signed data protection agreements before the study. The Ethics Committee of the University of Pots-

## 3.2 DATA ACQUISITION

dam approved the consent and data protection forms (No. 5/147, 08 Sep 2025; “Optimizing Large Scale Public Health Interventions on Social Media by Creating Neurological Markers of Engagement and Addiction in TikTok to Drive Disease Prevention”). Table 3.1 lists age, sex and pre-session survey answers per participant.

| ID  | Sex | Age | Daily SFV Use | Sleep (h) | Alertness      | Consumed   | Status   |
|-----|-----|-----|---------------|-----------|----------------|------------|----------|
| P4  | F   | 26  | 1-30m         | 6-7       | Neutral        | Caffeine   | Valid    |
| P5  | F   | 30  | 1-2h          | 7-8       | Alert          | None       | Valid    |
| P6  | F   | 24  | 1-30m         | 7-8       | Neutral        | None       | Valid    |
| P7  | F   | 23  | 30-60m        | 6-7       | Neutral        | None       | Valid    |
| P8  | F   | 23  | 30-60m        | 6-7       | Neutral        | None       | Valid    |
| P9  | M   | 27  | 30-60m        | 6-7       | Neutral        | Caffeine   | Valid    |
| P10 | F   | 24  | 1-30m         | >8        | Neutral        | Caffeine   | Valid    |
| P11 | F   | 26  | 30-60m        | 7-8       | Neutral        | Caffeine   | Valid    |
| P12 | F   | 20  | 30-60m        | 5-6       | Neutral        | Energy Dr. | Valid    |
| P13 | M   | 28  | 1-30m         | >8        | Neutral        | Caffeine   | Valid    |
| P14 | F   | 22  | 1-2h          | >8        | Tired          | None       | Valid    |
| P15 | M   | 28  | 1-30m         | 7-8, >8   | Neutral        | Caffeine   | Valid    |
| P16 | M   | 32  | 1-30m         | 7-8       | Alert          | None       | Excluded |
| P17 | F   | 30  | 30-60m        | 6-7       | Neutral        | Caffeine   | Valid    |
| P18 | M   | 27  | 1-2h          | 7-8       | Alert          | Caffeine   | Valid    |
| P19 | F   | 30  | 1-30m         | 7-8       | Neutral        | Caffeine   | Excluded |
| P20 | M   | 20  | 1-30m         | >8        | Alert          | Caffeine   | Valid    |
| P21 | M   | 21  | 30-60m        | 6-7       | Neutral        | None       | Valid    |
| P22 | F   | 21  | 1-30m         | >8        | Alert, Neutral | None       | Valid    |
| P23 | M   | 23  | 1-2h          | >8        | Neutral        | None       | Valid    |
| P24 | F   | 24  | 1-2h          | >8        | Neutral        | Caff+Nic   | Valid    |
| P25 | F   | 21  | >3h           | >8        | Neutral        | None       | Valid    |
| P26 | M   | 24  | 30-60m        | 6-7       | Neutral, Tired | None       | Valid    |
| P27 | M   | 23  | 2-3h          | 7-8       | Alert          | None       | Valid    |
| P28 | M   | 28  | 30-60m        | 7-8       | Very alert     | None       | Valid    |
| P29 | M   | 22  | 1-2h          | >8        | Very alert     | None       | Excluded |
| P30 | M   | 26  | >3h           | 5-6, 6-7  | Very alert     | None       | Valid    |
| P31 | M   | 27  | 1-2h          | 6-7       | Alert          | None       | Valid    |

Table 3.1: Participant demographics and pre-session survey responses. SFV = Short-Form Video. P16, P19, and P29 were excluded from the final analysis.

## 3.2 Data Acquisition

EEG was recorded with a consumer-grade Muse S headband (Generation 2, Figure 3.2). The Muse S has four dry electrodes, which touch the skin without gel. Two electrodes sit on the forehead (AF7, AF8), and two sit behind

the ears (TP9, TP10). The labels follow the international 10–20 system, a standard map of positions on the scalp. Each electrode measures voltage against a reference electrode at Fpz, the middle of the forehead [33]. The Muse S samples at 256 Hz [12]. Two units of the same model were used across participants. Differences between the two units therefore cannot be fully separated from differences between participants (Section 4.4.1). All recordings took place in Germany.

Earlier Muse models have been tested in research. Krigolson et al. tested the 2014 Muse, which uses the same electrode sites [60]. In two tasks, the 2014 Muse recorded three event-related potentials (ERPs): the N200, the P300 and the reward positivity. An ERP is the brain response to a stimulus, averaged over many repeats. The Muse ERPs closely matched those of a 64-electrode research system, once the research data were reduced to the Muse electrode sites and reference. Against the full research analysis, the N200 and the P300 had reversed sign. The test covers only frequencies below 15 Hz. Moontaha et al. used the Muse S Gen 1 to classify emotions during film clips [33]. A systematic review of EEG-based user engagement research lists the Muse in two of the reviewed studies [22].

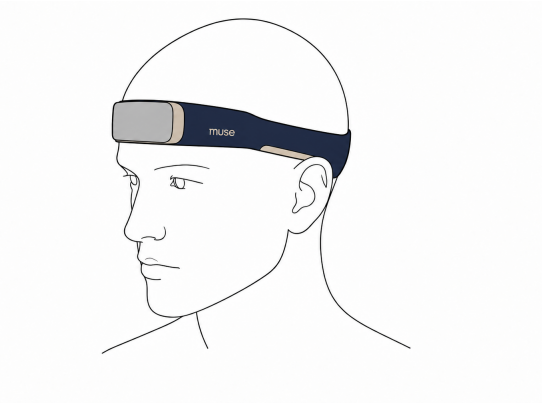


Figure 3.2: Sketch of the Muse S (Gen 2) headband as worn. The front sensors lie across the forehead, and the band runs over the ears.

The headband sent the EEG over Bluetooth to a laptop. The recording pipeline (Section 2.3) received the EEG through `muselsl` and the Lab Streaming Layer [21]. The recording pipeline displayed the sampling rate live during recording. After a Bluetooth dropout, the recording pipeline re-connected and continued in a new file, and the session went on. The EEG during a dropout was lost. Preprocessing logged each dropout and kept every analysis window clear of every dropout (Section 3.6). Section 3.5

describes the recording pipeline in full. The code is public at [https://github.com/coffeeandcoffee/eeg\\_tiktok\\_pipeline](https://github.com/coffeeandcoffee/eeg_tiktok_pipeline).

No video of the participant was recorded. Keypresses marked the events instead. A laptop stood next to the participant (Figures 3.3 and 3.4). The author pressed the “B” key to mark the start of the baseline, a two-minute resting recording (Section 3.3). The participant pressed the “A” key at every swipe, with the hand not holding the phone. The recording pipeline stamped each keypress into the EEG-CSV recording file.



Figure 3.3: The recording cabin, set up for privacy and few distractions.



Figure 3.4: Sketch of the recording setup. The participant wears the headband, browses TikTok on the author’s iPhone, and presses keys on the laptop.

### 3.3 Experimental Protocol

Each session had a 45-minute slot. The procedure itself took about 35 to 40 minutes. Figure 3.5 shows the order of the steps.

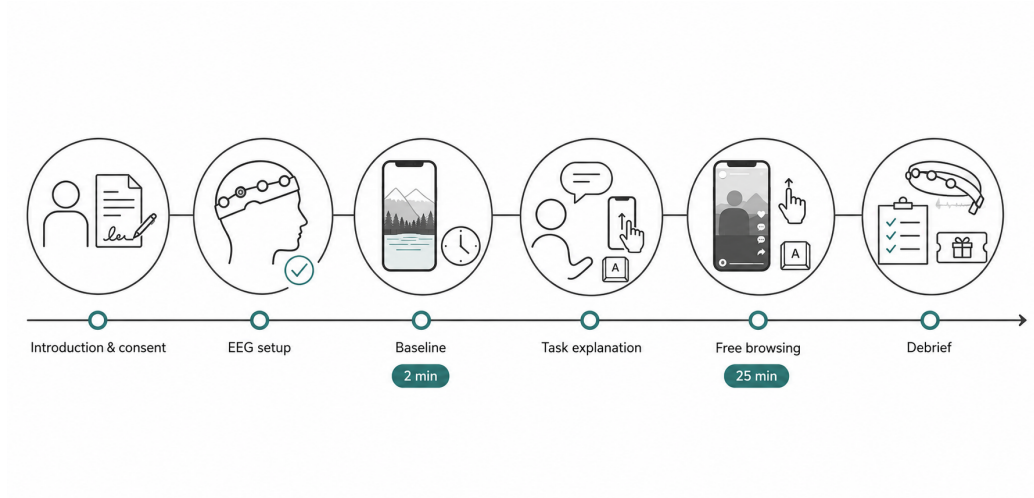


Figure 3.5: Session flow: introduction and consent, EEG setup, 2-minute baseline, task explanation, 25-minute free browsing, and debrief.

Participants first received an information sheet. Participants then signed the consent and data protection forms (Section 3.1) and filled in the pre-session survey. The pre-session survey took about five to ten minutes.

The author (experiment lead) then fitted the Muse S headband, adjusted the headband for good skin contact on the forehead and behind the ears, especially for participants with long hair, and checked the signal quality and the battery charge.

The session had two phases: the *baseline phase* and the *browsing phase*.

In the baseline phase, participants held the author’s iPhone, which played a two-minute nature video on YouTube. The author asked participants to relax and simply watch the video. Guided Access, an iPhone setting, kept the iPhone in the open app. The author started the recording pipeline, pressed the “B” key, and left the space, so that participants had privacy. After two minutes, Guided Access timed out on screen. The time-out ended the baseline phase.

The author then returned and explained the browsing phase. Before every participant, the author reset the TikTok “For You” feed with the “Reset feed” option in the content preferences. Participants were asked to use TikTok as usual: to watch videos they found interesting, and to swipe away videos they found boring. Participants were asked not to like, comment or visit profiles, so that the swipe stayed the only action. The browsing phase is naturalistic use (Section 2.1): self-paced viewing of an algorithmic feed, with a swipe possible at any moment.

Free viewing has been used in brain imaging before. In one fMRI study, five people freely watched 30 minutes of one film. Eye movements were free, and the film ran without breaks. Brain activity still correlated significantly across the five people, over large parts of the cortex [61]. Nobody in that study could skip.

In the present study’s experiment, participants were asked to press the “A” key at the same moment as every swipe (Section 3.2, Figure 3.6).

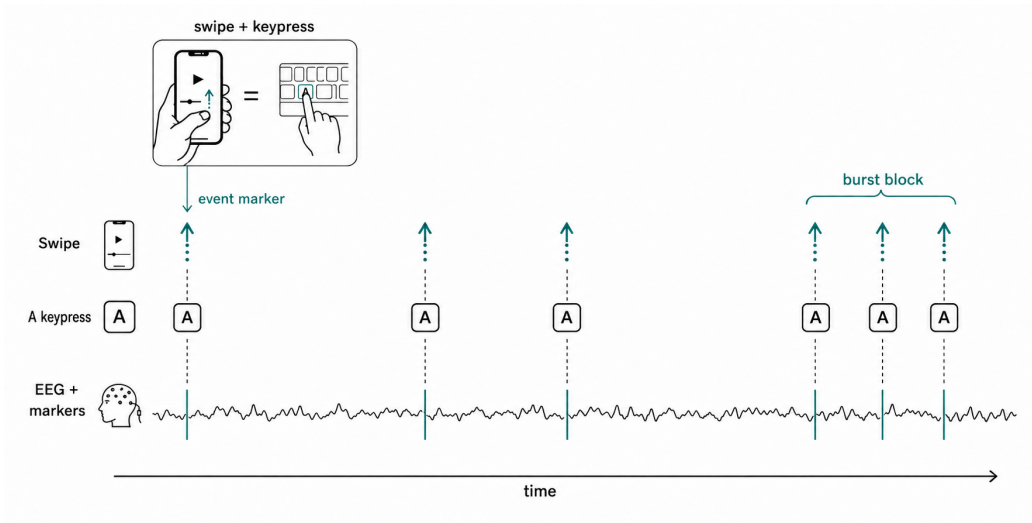


Figure 3.6: Swipe and keypress. The participant swipes to the next video and presses the “A” key at the same moment. Each keypress marks the swipe in the EEG recording. Several rapid swipes in a row form a burst block (Section 3.6.5).

The author set Guided Access to 25 minutes in the TikTok app. The author told participants that a Bluetooth dropout needs no action: participants should keep browsing, and the recording pipeline reconnects by itself (Section 3.2). Participants then browsed alone in the room, with the door closed, so that responses to the feed could stay natural. In case of problems, participants could knock on the glass door. No major interruption occurred.

After 25 minutes, the author entered the room, and the participant took off the headband. In a short post-session survey, participants reported their keypress errors (Section 3.1). Paid participants then signed the forms for the voucher (Section 3.1). The post-session survey, the forms and the debrief took about two to four minutes. The author thanked participants and, on request, explained the wider context of the study.

## 3.4 Labeling Strategy

Four labeling parameters define the **STAY** and **SKIP** labels: the pre-swipe gap, the **SKIP** period, the post-swipe gap and the window length. The present section sets the four labeling parameters, defines **STAY** and **SKIP**, and checks how much the separation between **STAY** and **SKIP** depends on three of the four labeling parameters.

### 3.4.1 Operational Definition of Labeling Parameters

Around each swipe, time runs in this order: **STAY** period, **SKIP** period, pre-swipe gap, “A” keypress, post-swipe gap, **STAY** period (Figure 3.7). The pre-swipe gap, the **SKIP** period and the post-swipe gap count from the “A” keypress that marks a swipe (Section 3.2).

- **Pre-swipe gap: 2.0 s.** The last 2.0 s before each keypress belong to neither label. The pre-swipe gap is meant to keep the preparation of the swipe out of **SKIP**. The readiness potential, a slow EEG shift before a voluntary movement, starts about 1 to 1.5 s before the movement [59]. Arm muscles activate on average  $263 \pm 40$  ms before a self-paced reach [40]. The 2.0 s are longer than the 1 to 1.5 s and longer than the 263 ms. The summary of the review by Shakeel et al. gives a start of about 2 s, and the start differs between people. The definition has a limit: in a person whose readiness potential starts earlier than 2 s before the keypress, the last **SKIP** window may already hold the start of the readiness potential (Section 2.14).
- **SKIP period: 3.0 s.** **SKIP** windows come from the 3.0 s just before the pre-swipe gap, from 5 s to 2 s before each keypress. To the author’s knowledge, no study gives a value for this period. The 3.0 s serve as a starting point, varied exploratively in Section 3.4.5.
- **Post-swipe gap: 0.5 s.** The first 0.5 s after each keypress belong to neither label. The post-swipe gap is meant to keep the end of the swipe movement, and the first moment of the new video, out of **STAY**. The post-swipe gap is not varied in Section 3.4.5.
- **Window length: 1.0 s.** The **SKIP** period and the **STAY** period are cut into analysis windows of 1.0 s. In a test of 1 to 5 s windows on wearable EEG, the shortest window, 1 s, scored best [33].

### 3.4.2 Operational Definition of STAY and SKIP

SKIP windows come from the SKIP period. STAY windows come from the STAY period. The STAY period is simply all remaining time of the browsing phase that does not belong to any other gap or period. In other words, the STAY period starts after a post-swipe gap and ends where the next SKIP period starts. With the default values, each SKIP period with default length 3.0 s holds three 1.0 s SKIP windows.

STAY captures engagement only in part. Long viewing without a swipe can also be viewing without attention (Section 2.6).

Swipes often come in bursts (Section 3.6.5). With the default values, when two swipes lie less than 5.5 s apart, the SKIP period of the later swipe reaches into the pre-swipe gap or the post-swipe gap of the earlier swipe. The overlapping time then counts as SKIP. Figure 3.8 shows an example near 15 s. Two keypresses lie only 1.2 s apart, so the SKIP periods of the two swipes merge into one SKIP period of about 4.2 s. The merged SKIP period holds four SKIP windows instead of three. Figure 3.8 uses 0.3 s windows instead of the default 1.0 s, so each SKIP window shows as a separate dash. In Figure 3.7, the same four SKIP windows join into one bar. Of all 9,307 SKIP windows, 2,179 (23%; 0 to 44% per participant) lie within the pre-swipe gap or the post-swipe gap of a neighbouring swipe. These SKIP windows may therefore contain preparation or movement of the neighbouring swipe. The present thesis keeps these SKIP windows (Section 5.1).

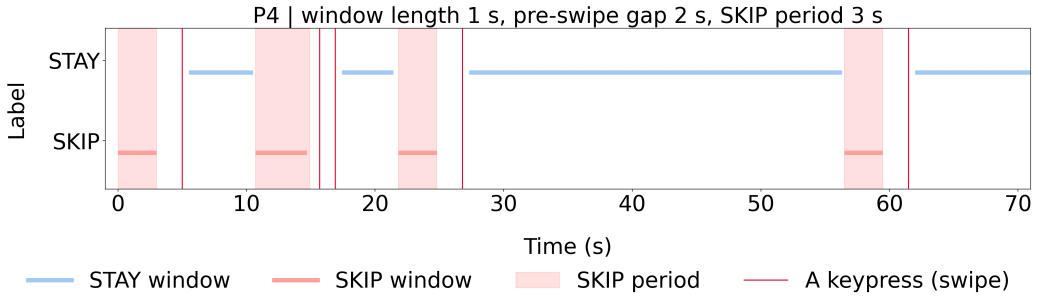


Figure 3.7: Labels under the default values, first 70 s of participant P4. Each vertical red line is an “A” keypress, which marks, approximately, the moment of a swipe to the next video. SKIP windows come from the SKIP periods. The white space before each keypress is the pre-swipe gap (2.0 s). The white space after each keypress is the post-swipe gap (0.5 s).

### 3.4.3 Baseline and Border Exclusion

The baseline phase gives no labels. The baseline phase serves only to normalise the signal of each participant (Section 3.6). The classifiers therefore separate **STAY** from **SKIP** within the browsing phase only.

The **STAY** period and the **SKIP** period meet where the **SKIP** period starts: with the default values, 5 s before the keypress. No window crosses this border (Section 3.6.3). Each window therefore carries one label only. Moontaha et al. likewise use only windows that lie completely inside one video [33].

Windows right next to the border stay in the analysis, although the moment the decision to swipe begins is unknown (Section 2.8). If the decision begins 7 s before the keypress, the **STAY** windows from 7 s to 5 s already hold the decision. If the decision begins 3 s before the keypress, the **SKIP** windows from 5 s to 3 s do not yet hold the decision. The sensitivity analysis moves the border by changing the pre-swipe gap and the **SKIP** period (Section 3.4.5).

### 3.4.4 Class Separation Metric

Cohen’s  $d$  measures how far the **STAY** windows and the **SKIP** windows lie apart in one feature, in units of the standard deviation of the feature (Section 2.9). A larger  $|d|$  means a stronger separation. It means the feature with a larger  $|d|$  distinguishes both the classes **STAY** and **SKIP** better from one another than a feature with a smaller  $|d|$ . The denominator is the square root of the average of the two class variances, the version Cohen gives for two groups that spread differently [42]:

$$d = \frac{\bar{x}_{\text{STAY}} - \bar{x}_{\text{SKIP}}}{\sqrt{\frac{s_{\text{STAY}}^2 + s_{\text{SKIP}}^2}{2}}} \quad (3.1)$$

where  $\bar{x}_{\text{STAY}}$  and  $\bar{x}_{\text{SKIP}}$  denote the means of the feature over all **STAY** windows and over all **SKIP** windows, and  $s_{\text{STAY}}^2$  and  $s_{\text{SKIP}}^2$  the corresponding variances.

Cohen’s  $d$  is computed once for every feature (e.g. TP10 raw standard deviation), over the windows of all participants together. Section 3.8.2 describes the two levels at which  $|d|$  is reported: exact values per feature, and means per category.

### 3.4.5 Parameter Sensitivity Analysis

The sensitivity analysis is exploratory and rough: the sensitivity analysis shows where the default values sit among nearby values, and points to future

research. The sensitivity analysis therefore belongs to the Methods chapter, and the start of Chapter 4 lists the result among the findings reported in the Methods chapter.

The sensitivity analysis asks how much the separation between **STAY** and **SKIP** depends on three labeling parameters: the pre-swipe gap, the **SKIP** period and the window length. The post-swipe gap stays at 0.5 s. Each tested setting reruns the pipeline from labeling to features. Each setting then yields one number: the largest  $|d|$  of any single feature out of 236, over the windows of all participants.

Crossing the three labeling parameters, with four values each, would need  $4 \times 4 \times 4 = 64$  pipeline runs. The main analysis needs the labeling parameters before the main analysis can start, so two less intensive designs were used, with 19 pipeline runs for basic exploration:

- **Pre-swipe gap and SKIP period, crossed.** Every pre-swipe gap (0.5, 1, 2 and 3 s) was combined with every **SKIP** period (2, 3, 4 and 6 s), with the window length fixed at 1.0 s: 16 runs.
- **Window length, alone.** The window length (0.3, 0.7, 1.0 and 2.0 s) was varied with the pre-swipe gap fixed at 2.0 s and the **SKIP** period at 3.0 s: 3 further runs, because the 1.0 s run is part of the grid. Changing one labeling parameter while the others stay fixed is the one-at-a-time (OAT) design of Section 2.7 [37, §2.4.2]. The stride, the step from one window start to the next, stayed at 1.0 s. The 2.0 s windows therefore overlap by half, and the 0.3 s windows leave 0.7 s of every second unused.

The sensitivity analysis has three limits. First, the OAT design cannot show an interaction (Section 2.7): whether the effect of the window length depends on the pre-swipe gap or on the **SKIP** period [37]. The window-length result therefore holds only near the fixed values. Second, the grid shows the joint effect of pre-swipe gap and **SKIP** period without a statistical test. Third, the sensitivity analysis pools all participants, so the sensitivity analysis cannot show a best value per person, as Mi et al. found for window length [35]. Section 5.1 lists the three limits.

Figures 3.7 to 3.9 show the same 70 s of participant P4 under three settings. Blue bars are **STAY** windows, red bars **SKIP** windows, shaded red areas **SKIP** periods, and vertical red lines keypresses. Figure 3.7 shows the default values. In Figure 3.8, the window length drops from 1.0 s to 0.3 s, so the bars turn into short dashes with spaces between them. In Figure 3.9, the pre-swipe gap shrinks from 2.0 s to 0.5 s, so the white space before each

keypress narrows, and the SKIP period grows from 3.0 s to 4.0 s, so the shaded areas widen.

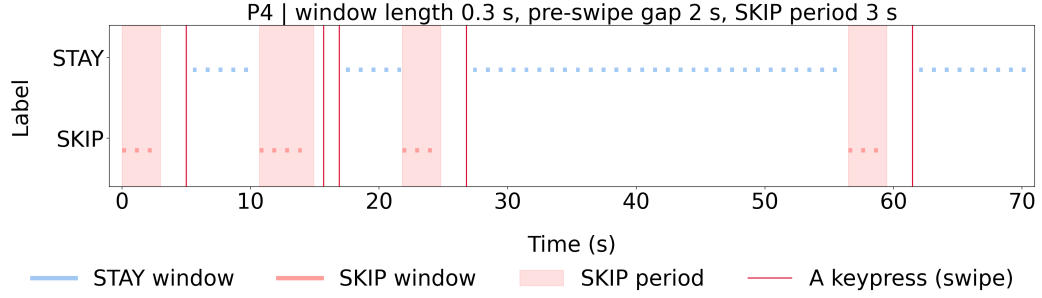


Figure 3.8: Window length 0.3 s; pre-swipe gap and SKIP period at the default values. The stride stays at 1.0 s, so the windows leave spaces between each other.

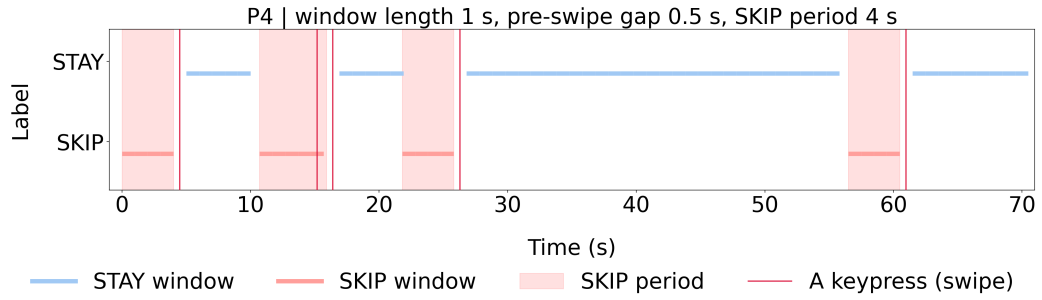


Figure 3.9: Pre-swipe gap 0.5 s and SKIP period 4.0 s; window length at the default value.

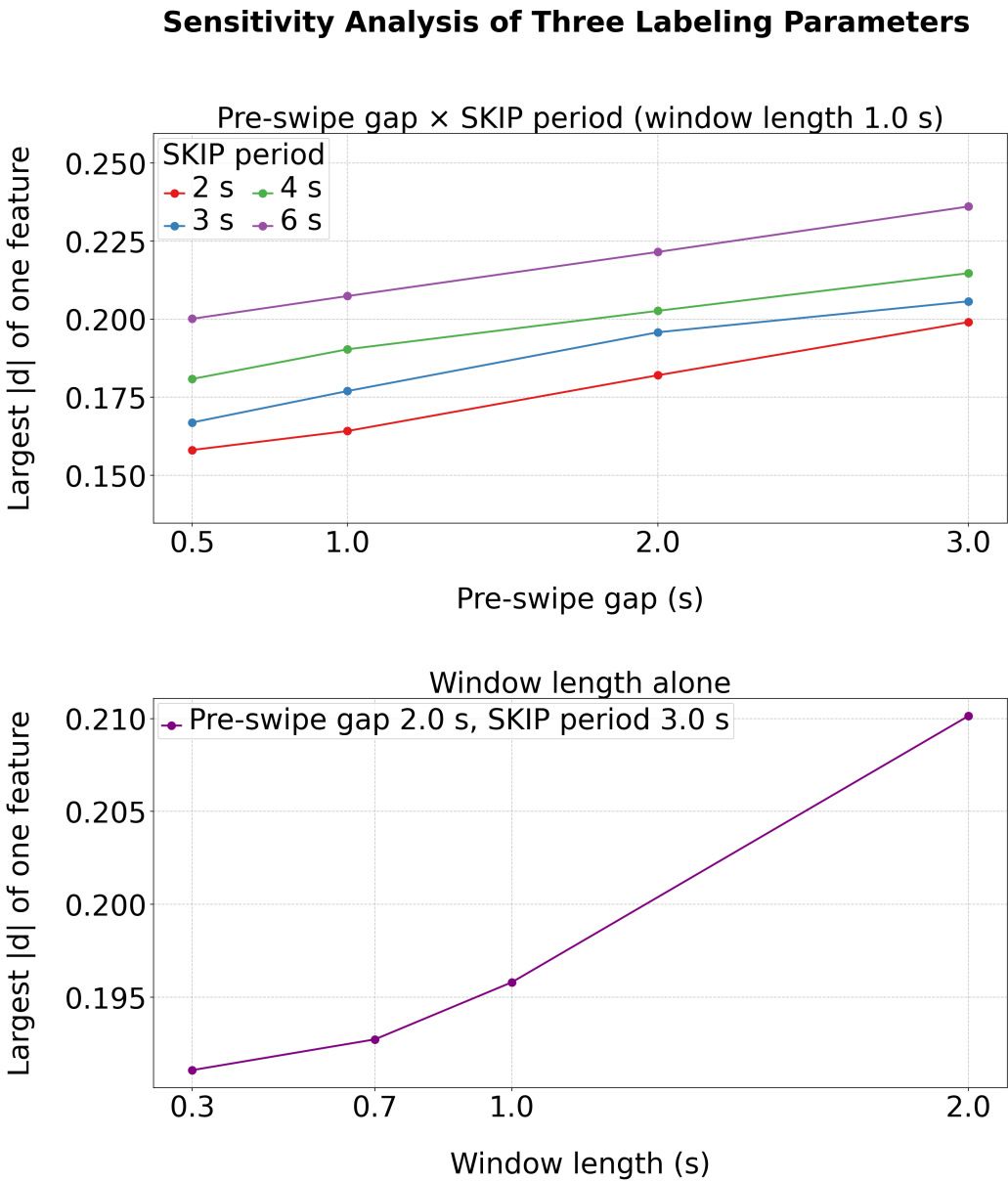


Figure 3.10: Sensitivity analysis of three labeling parameters. Each point is the largest  $|d|$  of any single feature, over the windows of all participants. Top: pre-swipe gap crossed with SKIP period, window length 1.0 s. Bottom: window length alone, pre-swipe gap 2.0 s, SKIP period 3.0 s. The sensitivity analysis covers single features only, and not the Engagement Index.

The largest  $|d|$  rose with every step of every varied labeling parameter

(Figure 3.10). Across the grid, the largest  $|d|$  ran from 0.158 (pre-swipe gap 0.5 s, SKIP period 2 s) to 0.236 (pre-swipe gap 3 s, SKIP period 6 s). Across window lengths, the largest  $|d|$  ran from 0.191 (0.3 s) to 0.210 (2.0 s). The default values give 0.196, the value of TP10 raw standard deviation in Section 4.4.1.

The window-length trend runs against Moontaha et al., the source of the 1.0 s default, where the score mostly fell from 1 to 5 s [33]. The comparison is loose: the windows here span 0.3 to 2.0 s, and the measure here is  $|d|$ , not classifier performance. The trend for the pre-swipe gap resembles Rimbart et al., where rest windows further from the next cue scored higher, a gain Rimbart et al. attribute partly to the rise in beta power after the previous movement [36]. Because swipes often come in bursts (Section 4.3), a longer pre-swipe gap or a longer SKIP period may likewise move SKIP windows toward the previous swipe.

**Conclusion:** The default values were fixed before the sensitivity analysis, and the sensitivity analysis changed none of the default values. The sensitivity analysis served only to understand the labeling parameters that the main analysis needs. A detailed study of the labeling parameters is beyond the scope of the present thesis (Section 5.2). All following analyses of the present thesis use the default values:

- **Pre-swipe gap:** 2.0 s
- **SKIP period:** 3.0 s
- **Post-swipe gap:** 0.5 s
- **Window length:** 1.0 s

## 3.5 Reproducible Recording Pipeline

The reproducible recording pipeline engineered by the author, records the EEG of the Muse S and the keypresses of each session. The recording pipeline is one Python file. The file is public at [https://github.com/coffeeandcoffee/eeg\\_tiktok\\_pipeline/blob/main/data\\_collection/recording\\_script.py](https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/blob/main/data_collection/recording_script.py). The recording pipeline can also record webcam video. The present thesis recorded no video (Section 3.2).

Each session started the recording pipeline with one command:

```
python recording_script.py --nocamera --duration 1800
```

The flag `--nocamera` switches the webcam off. The flag `--duration 1800` stops the recording after 1,800 s, that is, after 30 minutes.

The recording pipeline runs in five steps:

1. **Find the headband.** The recording pipeline scans for Muse headbands over Bluetooth and lists every Muse found. The author picks one Muse from the list, so no device address has to be written into the code.
2. **Start the stream.** The recording pipeline starts `muselsl stream` (Section 2.3). `muselsl` passes the EEG into the Lab Streaming Layer (LSL). The recording pipeline reads the EEG from LSL at 256 Hz, one sample at a time, with the Python library `pyls1`.
3. **Mark the keypresses.** A keypress listener, built on the Python library `pynput`, stamps every “A” keypress of the participant and every “B” keypress of the author with the laptop clock. At the first EEG sample of each connection, the recording pipeline measures the difference between the laptop clock and LSL time once. With this difference, the recording pipeline converts each keypress time into LSL time. The first EEG sample within 100 ms of the keypress gets a flag: 1 in the column for “A” or “B”, 0 otherwise. The flag sits in the same row as the EEG sample, in the EEG-CSV recording file.
4. **Reconnect after a Bluetooth dropout.** Bluetooth connections can drop. When no EEG sample arrives for 5 s, the recording pipeline stops `muselsl`, scans for the headband again, restarts the stream, and continues in a new file (`_2.csv`, `_3.csv`, and so on). The EEG during a Bluetooth dropout is lost. Earlier files stay intact.
5. **Keep reading while listening.** EEG reading and the keypress listener each run in a thread of their own. EEG reading does not wait for the keypress listener, and the keypress listener does not wait for EEG reading.

## 3.6 Reproducible Preprocessing Pipeline

The reproducible preprocessing pipeline engineered by the author, turns the recording files of the recording pipeline (Section 3.5) into labeled analysis windows. The code is public at [https://github.com/coffeeandcoffee/eeg\\_tiktok\\_pipeline](https://github.com/coffeeandcoffee/eeg_tiktok_pipeline). The preprocessing pipeline runs twice: once without and once with a 50 Hz notch filter. Only the artifact check uses the notch-filtered copy (Section 3.6.4).

### 3.6.1 Data Synchronization and Interpolation

Each Bluetooth dropout ends one recording file and starts the next recording file (Section 3.5). The preprocessing pipeline processes each recording file on its own and never joins two recording files (executed via [https://github.com/coffeeandcoffee/eeg\\_tiktok\\_pipeline/blob/main/data\\_analysis/step01\\_preprocess.py](https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/blob/main/data_analysis/step01_preprocess.py)). No analysis window can therefore cross a Bluetooth dropout.

The preprocessing pipeline keeps each recording file from 5.0 s before the first “A” keypress in the recording file to 0.5 s after the last “A” keypress. With the default values, the kept part starts with the first SKIP period and ends with the last post-swipe gap (Section 3.4.1).

The raw samples arrive at slightly uneven intervals. Over the whole recording file, the raw rate averages between 255.93 and 256.02 samples per second, across all 61 recording files of the 25 participants. The band filters of Section 3.6.2 assume evenly spaced samples. Linear interpolation therefore puts every recording file on an even grid of exactly 256 samples per second.

The preprocessing pipeline measures each Bluetooth dropout as the time between the last sample of one recording file and the first sample of the next recording file, and logs every Bluetooth dropout. Figure 3.11 shows the Bluetooth dropouts per participant.

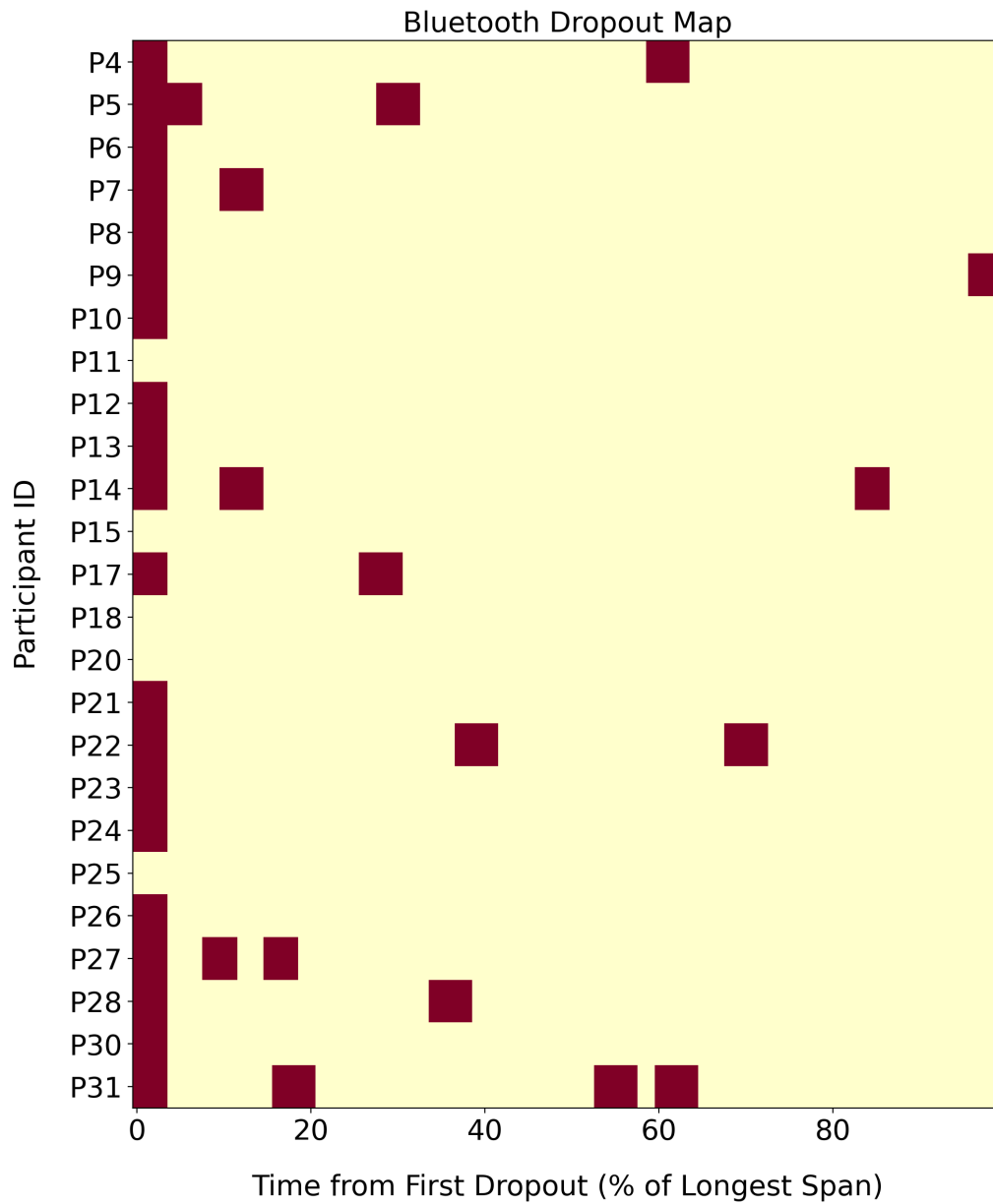


Figure 3.11: Bluetooth dropouts per participant. Each dark block marks a Bluetooth dropout. Two Bluetooth dropouts close together can show as one wider block, as for P5. The horizontal axis starts at the first Bluetooth dropout of each participant, not at the start of the session, and is scaled to the longest span from a first to a last Bluetooth dropout in the sample. The map therefore shows how far apart the Bluetooth dropouts of one participant lie, and not when in the session the Bluetooth dropouts happened. Rows without a block had no Bluetooth dropout. No analysis window crosses a Bluetooth dropout.

### 3.6.2 Baseline Extraction and Normalization

The preprocessing pipeline splits the signal of each electrode into seven frequency bands with a 4th-order Butterworth bandpass filter: Delta (1–4 Hz), Theta (4–8 Hz), Alpha (8–13 Hz), Beta (13–30 Hz), Low Gamma (30–40 Hz), High Gamma (40–60 Hz), and Very High Gamma (60–100 Hz). The undecomposed broadband signal is kept alongside the seven bands and is called the *raw* signal throughout. For each band, the Hilbert transform gives the *envelope*  $A$ : the strength of the band at every moment, like a line traced along the tops of the waves. The envelope is not squared to power.

The baseline gives every participant, electrode and band a separate zero point, so that values of different participants can be compared. The baseline is the 90 s from 10 s to 100 s after the first “B” keypress of the author, within the two-minute baseline phase (Section 3.3). The baseline comes from the recording files before the trimming of Section 3.6.1. For each band at each electrode,  $\mu_{base}$  and  $\sigma_{base}$  are the mean and the standard deviation of the envelope over the baseline. Every sample of the envelope is then z-scored:

$$z = \frac{A - \mu_{base}}{\sigma_{base}} \quad (3.2)$$

A value of  $z = 1$  means that the band is one baseline standard deviation stronger than during the baseline. The z-scoring reduces the differences between participants, but part of the differences may remain (Section 2.9).

The raw signal is not z-scored. For the raw signal, the preprocessing pipeline only subtracts the mean of each recording file. The raw features, such as TP10 raw standard deviation (Section 3.7), therefore stay in the unit of the recording and keep the differences between participants in signal strength.

### 3.6.3 Windowing and Class Labeling

After normalization, the preprocessing pipeline cuts the signal into **STAY** windows and **SKIP** windows around each “A” keypress, with the default values of Section 3.4.1: a 3.0 s SKIP period, a 2.0 s pre-swipe gap and a 0.5 s post-swipe gap (executed via [https://github.com/coffeeandcoffee/eeg\\_tiktok\\_pipeline/blob/main/data\\_analysis/step03\\_label\\_windows.py](https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/blob/main/data_analysis/step03_label_windows.py)). No window takes data from a pre-swipe gap or a post-swipe gap, except **SKIP** windows in bursts (Section 3.4.2).

The analysis windows last 1.0 s and do not overlap (stride = window length). Each window holds samples of one label only. At the end of a stretch, a window is kept if at least 0.8 s of the window lies inside the stretch, and

then holds only the samples inside the stretch. 531 of the 27,643 windows (1.9%) are such shortened windows at the end of a stretch.

### 3.6.4 Artifact Analysis

The artifact check flags likely blinks and likely muscle or jaw movement artifacts (executed via [https://github.com/coffeeandcoffee/eeg\\_tiktok\\_pipeline/blob/main/data\\_analysis/step02\\_artifact\\_flag.py](https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/blob/main/data_analysis/step02_artifact_flag.py)). The artifact check removes nothing. Every flagged sample stays in every analysis of the present thesis. The artifact check only shows that artifacts are present. The missing artifact removal is a limitation of the present thesis (Section 5.1) and is left to future research (Section 5.2).

The artifact check cuts the signal of each electrode into short segments that overlap by half:

- **Blinks:** AF7 and AF8, filtered to 1–40 Hz, segments of 0.5 s.
- **EMG:** TP9 and TP10, filtered to 30–100 Hz, segments of 0.25 s.

For each segment, the artifact check measures the peak-to-peak amplitude, the distance from the lowest point to the highest point. The artifact check flags a segment when the peak-to-peak amplitude lies above the 95th percentile of all segments of the same electrode in the same recording file.

The 95th percentile flags a fixed share of segments by design. The flagged share of samples in Figure 3.12, between 7.5% and 15% per participant and artifact type, therefore looks alike across participants. The flagged share cannot show which participant recorded cleaner data.

The artifact check runs on the notch-filtered copy, so that the 50 Hz powerline-noise does not raise the peak-to-peak amplitude. All other analyses use the signal without a notch filter, because a notch filter would also remove any brain activity near 50 Hz, inside high gamma (Section 2.13).

EMG is strongest at electrodes near head muscles [49, 51], such as TP9 and TP10 at the lower edge of the scalp (Section 2.11). AF7 and AF8 lie near the forehead muscle and pick up EMG as well (Section 2.11). At AF7 and AF8, however, the artifact check looks only for blinks.

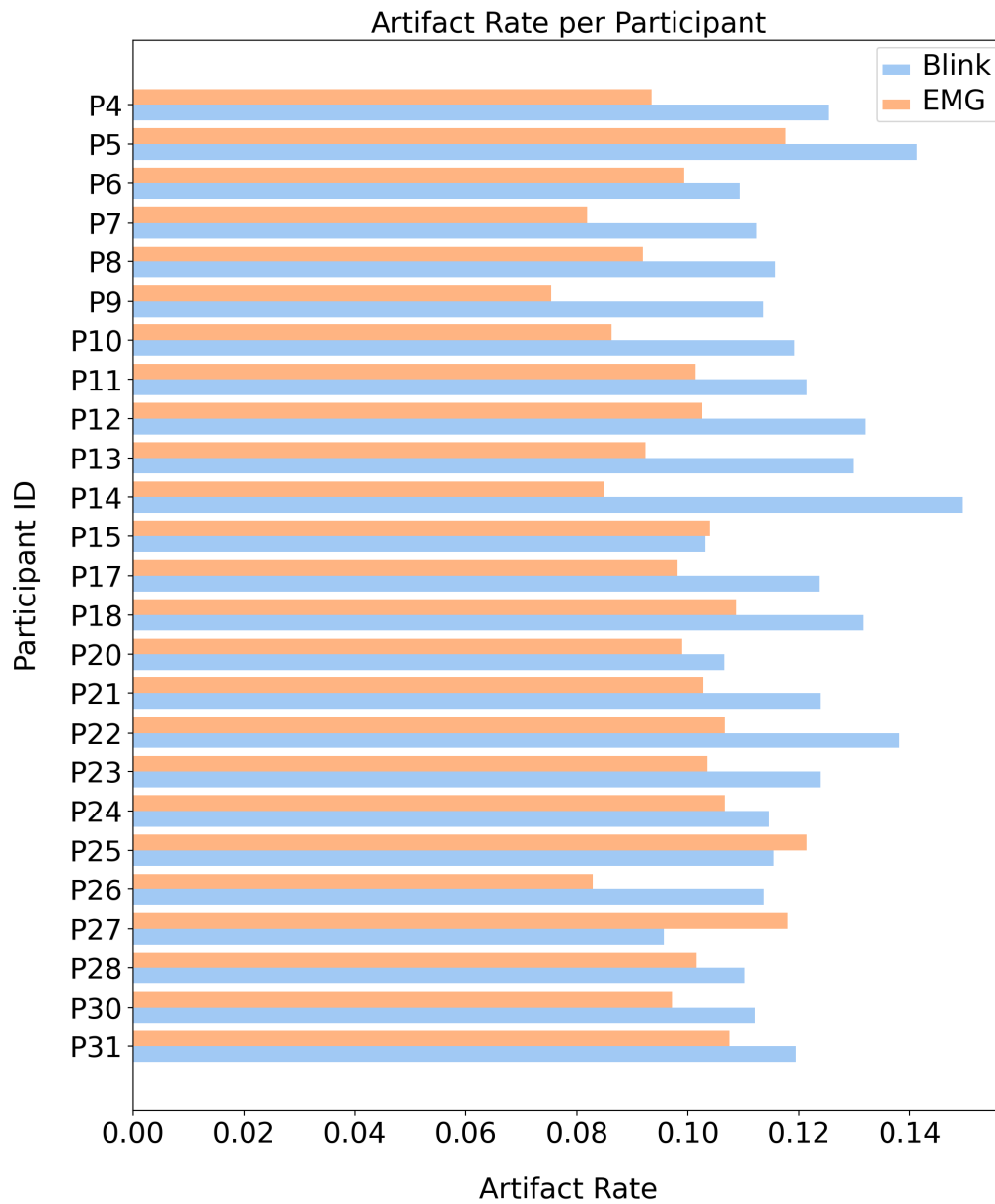


Figure 3.12: Share of samples flagged by the artifact check, per participant, for blinks (AF7, AF8) and EMG (TP9, TP10). The threshold is the 95th percentile of each electrode in each recording file, so the flagged share is similar across participants by design. No flagged sample was removed.

### 3.6.5 Behavioral Burst Analysis

Swipes often come in quick succession. The present thesis calls several rapid swipes in a row a *burst*. The *inter-swipe interval* is the time between two consecutive “A” keypresses, roughly the viewing time of one video. For the burst analysis, each swipe gets a *burst window* of  $\pm 3.0$  s. When two swipes lie less than 6.0 s apart, the burst windows of the two swipes overlap, and the two swipes merge into one *burst block*. Figure 3.13 shows the inter-swipe intervals of all participants and the 6.0 s merge threshold.

The burst window and the SKIP period are two separate definitions. The burst window only groups swipes for the burst analysis and changes no label. The SKIP period defines the SKIP windows (Section 3.4.1). Both definitions use the value 3.0 s, and the burst window and the SKIP period can overlap in time. Section 3.4.2 describes what bursts do to the SKIP period.

The author chose the 3.0 s of the burst window without a source and without looking at the inter-swipe intervals. Section 4.3 shows how the burst blocks change with a  $\pm 5.0$  s burst window. The burst analysis removes nothing: bursts stay in the data. Whether bursts should be removed is left to future research (Section 5.2).

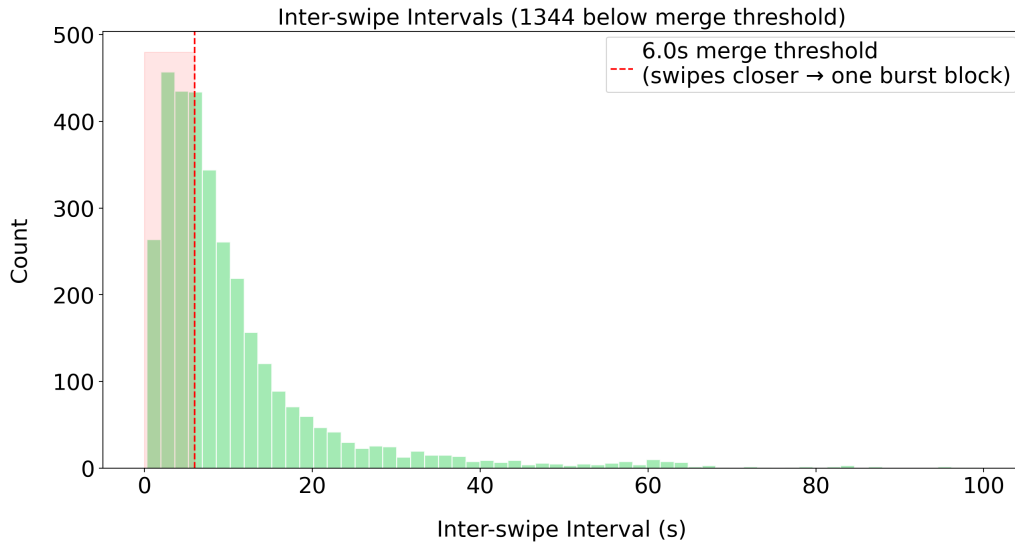


Figure 3.13: Inter-swipe intervals of all participants. The dashed line marks the 6.0 s merge threshold. The shaded area marks the inter-swipe intervals below the 6.0 s merge threshold, where swipes merge into burst blocks. The merge threshold comes from the  $\pm 3.0$  s burst window and is separate from the SKIP period.

## 3.7 Feature Engineering

Feature engineering turns each 1.0 s window (Section 3.6.3) into 236 numbers, called *features* (executed via [https://github.com/coffeeandcoffee/eeg\\_tiktok\\_pipeline/blob/main/data\\_analysis/step05\\_feature\\_engineering.py](https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/blob/main/data_analysis/step05_feature_engineering.py)). The 236 features form two groups:

- **Band features:** eight features for each of the seven bands at each of the four electrodes,  $7 \times 4 \times 8 = 224$  features. The eight band features are mean, standard deviation, Relative Power, the three Hjorth parameters, Peak Frequency and Macro Frequency.
- **Raw features:** three features of the raw signal at each of the four electrodes,  $4 \times 3 = 12$  features. The three raw features are mean, standard deviation and slope.

The Engagement Index (EI) is computed separately from the same windows and serves as the benchmark for the 236 features.

Every band feature is computed on  $y(t)$ , the z-scored envelope of one band at one electrode within one window (Equation 3.2). The window holds  $N$  samples and lasts  $T$  seconds. A full window has  $N = 256$  and  $T = 1.0$  s. The 531 shortened windows at the end of a stretch hold fewer samples (Section 3.6.3).

### 3.7.1 Mean, Standard Deviation and Relative Power

The mean of  $y(t)$  gives the level of the band strength within the window. The standard deviation of  $y(t)$  gives the spread of the band strength within the window. Figure 3.14 shows the mean and the standard deviation for one window. The 236 features do not include the minimum and the maximum of  $y(t)$ . The value of the minimum and the maximum as features is left to future research and lies beyond the scope of the present thesis.

Relative Power divides the mean of one band by the sum of the means of all seven bands at the same electrode. The means are z-scores and can be negative. Relative Power is therefore no share of total power, although the name suggests one. When the sum of the seven means is zero or negative, Relative Power is set to 0. Across all windows, Relative Power is 0 in 40% of the values and negative in 20%.

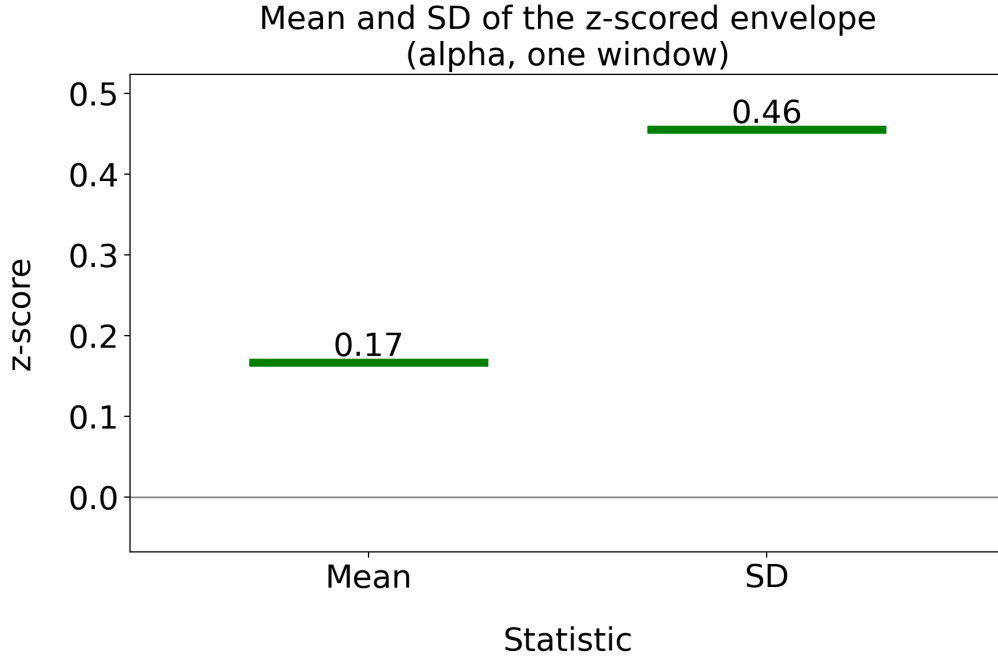


Figure 3.14: Mean and standard deviation of the z-scored envelope for one window of the alpha band, at one electrode of one participant.

### 3.7.2 Engagement Index

Pope et al. selected the EI to switch a computer task modelled on flight-deck work between manual and automated operation [4] (Section 2.4):

$$EI = \frac{\beta}{\alpha + \theta} \quad (3.3)$$

Pope et al. used band power, summed over four central and parietal electrodes (Cz, Pz, P3 and P4), with beta at 13–22 Hz [4].

The present thesis adapts the EI to the z-scored envelope. For each window, the pipeline computes the variance of the z-scored envelope of beta, alpha and theta at each of the four electrodes. The pipeline then averages each band over the four electrodes. The three averages enter Equation 3.3. Figure 3.15 shows the calculation for one window.

Band power, the input of Pope et al., is never negative. The mean of a z-scored envelope compares the band with the baseline, and can be negative or close to zero. A ratio of such means can flip its sign or become very large. The variance is never negative, like band power. The variance of the

z-scored envelope measures how much the band strength fluctuates within the window, relative to the baseline. The variance does not measure the level of the band strength, which band power measures. The present thesis did not test an EI built on the level of the band strength.

The EI of the present thesis also differs from the EI of Pope et al. in the electrodes (AF7, AF8, TP9 and TP10) and in the beta band (13–30 Hz). EI values of the present thesis are therefore not comparable with EI values of Pope et al. Section 3.8 describes how the EI serves as the benchmark.

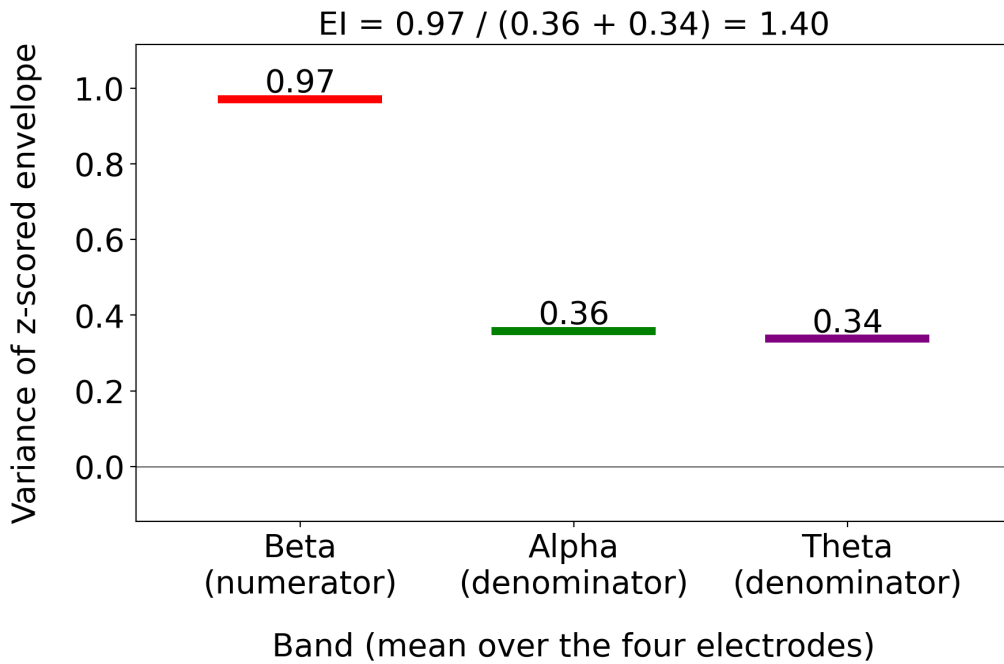


Figure 3.15: EI calculation for one window of one participant. Each line is the variance of the z-scored envelope of one band, averaged over the four electrodes. The EI divides the beta value by the sum of the alpha value and the theta value (Equation 3.3).

### 3.7.3 Hjorth Parameters

Mean and standard deviation ignore the order of the samples: shuffling the samples of a window leaves the mean and the standard deviation unchanged. Hjorth parameters depend on the order of the samples, because Hjorth parameters use the slope of the signal [45]. The pipeline takes the slope  $y'(t)$  as the change from one sample of  $y(t)$  to the next, and the change in slope

$y''(t)$  likewise.

**Activity.** Activity is the variance of  $y(t)$ , the square of the standard deviation:

$$\text{Activity} = \sigma_y^2 = \frac{1}{N} \sum_{t=1}^N (y(t) - \mu_y)^2 \quad (3.4)$$

where  $\mu_y$  is the mean of  $y(t)$ . The EI builds on the same variance of beta, alpha and theta.

**Mobility.** Mobility is the standard deviation of the slope, divided by the standard deviation of  $y(t)$ :

$$\text{Mobility} = \frac{\sigma_{y'}}{\sigma_y} = \sqrt{\frac{\text{var}(y'(t))}{\text{var}(y(t))}} \quad (3.5)$$

Hjorth reads Mobility as a mean frequency [45]. Here,  $y(t)$  is the envelope. Mobility therefore describes how fast the band strength rises and falls, rather than the frequency of the waves within the band. The slope is taken per sample, so Mobility carries the unit per sample.

**Complexity.** Complexity is the Mobility of the slope, divided by the Mobility of  $y(t)$ :

$$\text{Complexity} = \frac{\text{Mobility}(y')}{\text{Mobility}(y)} = \frac{\sigma_{y''}/\sigma_{y'}}{\sigma_{y'}/\sigma_y} \quad (3.6)$$

where  $\sigma_{y''}$  is the standard deviation of the change in slope. A pure sine wave gives a Complexity of 1. Complexity rises as the shape of  $y(t)$  departs from a sine wave [45].

### 3.7.4 Peak Frequency and Macro Frequency

The present thesis defines Peak Frequency and Macro Frequency without prior validation (Section 2.10). Both are purely experimental mathematical constructs created by the author after noticing repeated hill patterns in the EEG data, and therefore require further validation beyond the present thesis. Peak Frequency and Macro Frequency count per second how often  $y(t)$ , the band strength, peaks. Peak Frequency counts every bump. Macro Frequency counts only the larger hills. The counts need not equal the frequency of the waves within the band (Section 2.10).

**Peak Frequency.** A peak is a sample of  $y(t)$  higher than both neighbouring samples.  $P$  is the set of all peaks in the window. Peak Frequency divides the number of peaks by the window duration:

$$f_{peak} = \frac{|P|}{T} \quad (3.7)$$

The z-scoring shifts and scales the envelope, and leaves the number of peaks unchanged.

**Macro Frequency.** Added noise raises the number of local maxima and minima of a signal [47]. Peak Frequency may therefore also count small bumps from noise. Macro Frequency counts the larger hills in three steps:

1. **Threshold.** The threshold  $h_{peak}$  is the mean height of all peaks in the window:

$$h_{peak} = \frac{1}{|P|} \sum_{p \in P} y(p) \quad (3.8)$$

2. **Mark.** Every sample of  $y(t)$  at or above  $h_{peak}$  becomes 1. Every other sample becomes 0. The result is a sequence of ones and zeros,  $B(t)$ .
3. **Bridge.** A hill can hold a short dip below the threshold.  $\bar{G}$  is the mean length of all runs of zeros in  $B(t)$ . Every run of zeros between two ones that is no longer than  $\bar{G}$  becomes ones. The result is  $B_{smooth}(t)$ .

Each run of ones in  $B_{smooth}(t)$  is one hill. Macro Frequency divides the number of hills by the window duration:

$$f_{macro} = \frac{\text{number of hills in } B_{smooth}}{T} \quad (3.9)$$

A window without any peak gets a Peak Frequency and a Macro Frequency of 0.

### 3.7.5 Raw Features

The raw features describe the raw signal, the signal before the split into bands (Section 3.6.2). A split into bands can hide activity that spreads over all frequencies: band ratios, such as the ratio of theta to beta, can reflect such spread-out activity rather than the named bands [62]. Three raw features are computed at each electrode:

- **Mean:** the level of the raw signal within the window.

- **Standard deviation:** the spread of the raw signal within the window.
- **Slope:** the slope of a straight line fitted to the raw signal of the window, per second. The slope shows the drift of the raw signal within the window.

## 3.8 Result Synthesis Pipeline

### 3.8.1 Temporal Cross-Validation and Data Balancing

Two settings test the models. The *intra-subject* setting trains and tests one model per participant, as Nuamah et al. and McCartney et al. do (Sections 2.4 and 2.7). The *inter-subject* setting trains one model on 24 participants and tests the model on the 25th participant, once for every participant (Leave-One-Group-Out cross-validation, with one participant per group). A test on a new participant shows whether a result holds beyond the participants in the training data [63].

The intra-subject setting uses blocked temporal 5-fold cross-validation (executed via [https://github.com/coffeeandcoffee/eeg\\_tiktok\\_pipeline/blob/main/data\\_analysis/step04\\_balance\\_split.py](https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/blob/main/data_analysis/step04_balance_split.py)). Neighbouring windows of one recording look alike. A test window right next to a training window would let the model recognise the test window instead of predicting the class, much like a student who has seen the exam questions during practice (Section 2.1). Varoquaux et al. therefore call for a time separation between training data and test data, and for cross-validation on whole blocks of correlated data [63].

The pipeline cuts the windows of each participant, in time order, into five folds. The fold borders sit where each fold holds about one fifth of the SKIP windows of the participant (Figure 3.16). Each fold serves once as test data, while the other four folds serve as training data.

At each fold border, the pipeline drops every window that starts within 1.5 s of the border. The present thesis calls the dropped 3 s the *fold gap*. The start of every training window therefore lies more than 3 s from the start of every test window. A window lasts 1.0 s, so more than 2 s without any window separate a training window from a test window. The author set the 3 s; Varoquaux et al. give no value. The pipeline checks the fold gap in every fold and stops with an error when a training window and a test window start less than 2.9 s apart, 3 s minus a tolerance of 0.1 s. No fold triggered the check (Figure 3.17). Figure 3.18 shows the five folds and the four fold gaps of one participant.

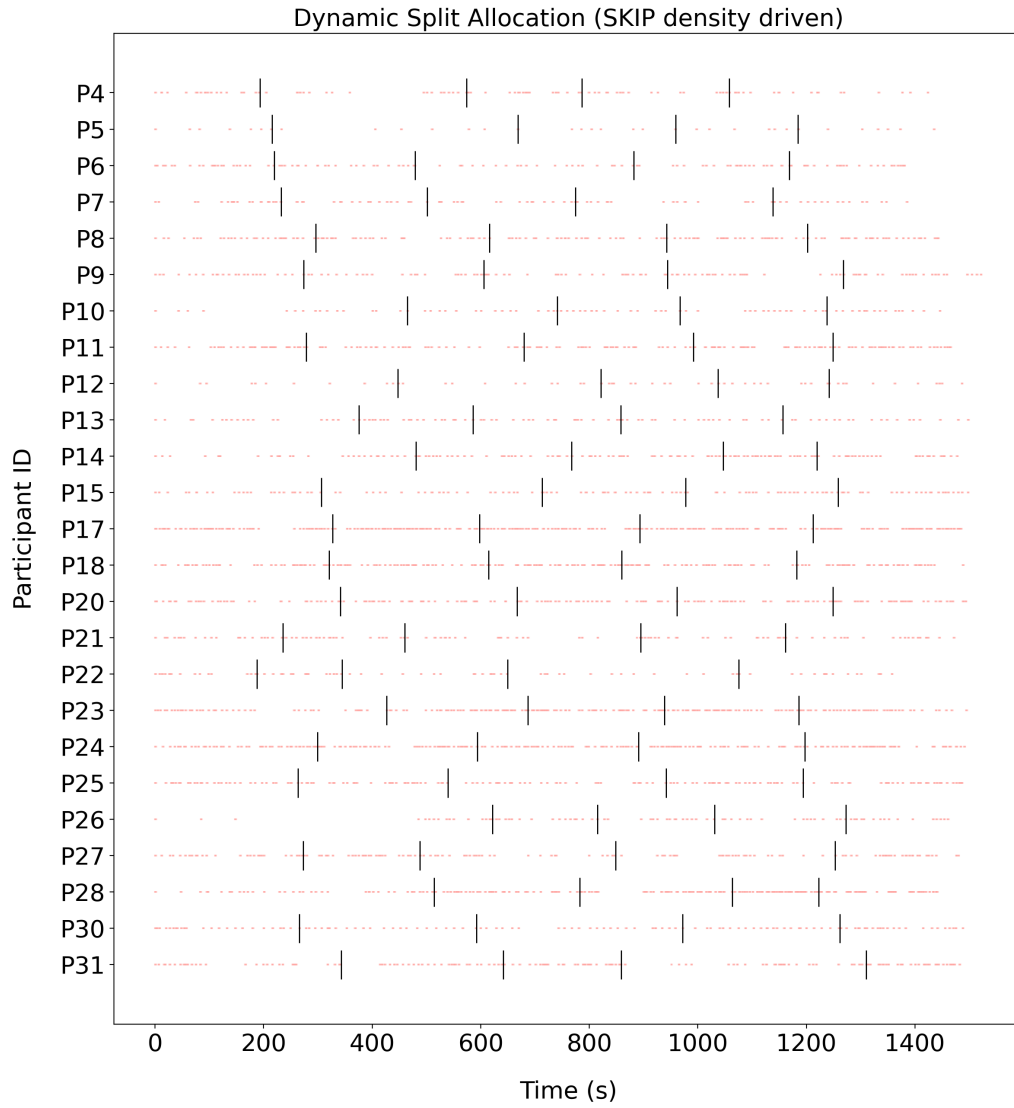


Figure 3.16: Fold borders per participant. Red marks are **SKIP** windows. Black lines are fold borders. Each fold holds about one fifth of the **SKIP** windows of the participant.

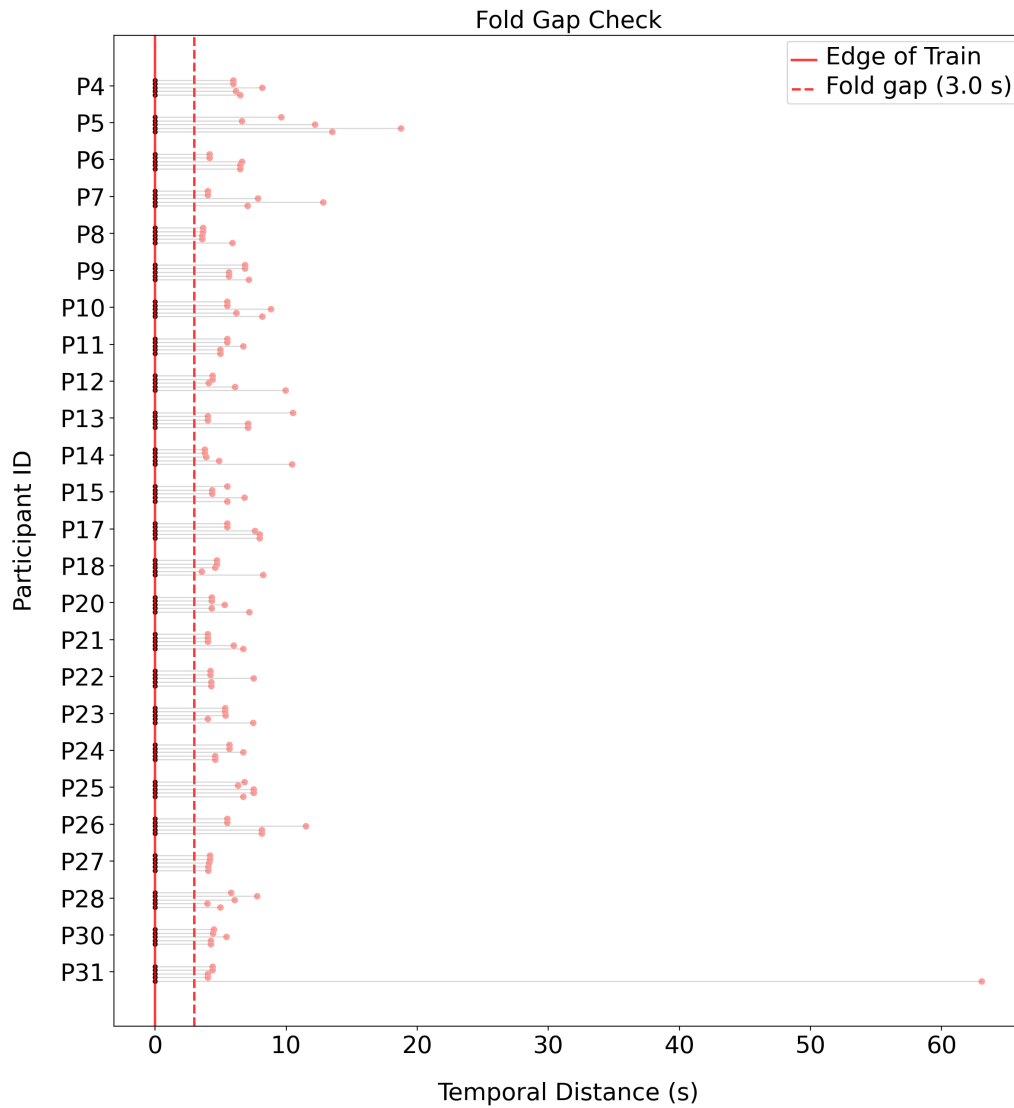


Figure 3.17: Check of the fold gap, first seed. Each red point is the smallest distance between the start of a training window and the start of a test window in one fold. The dashed red line marks 3 s. Every point lies beyond 3 s.

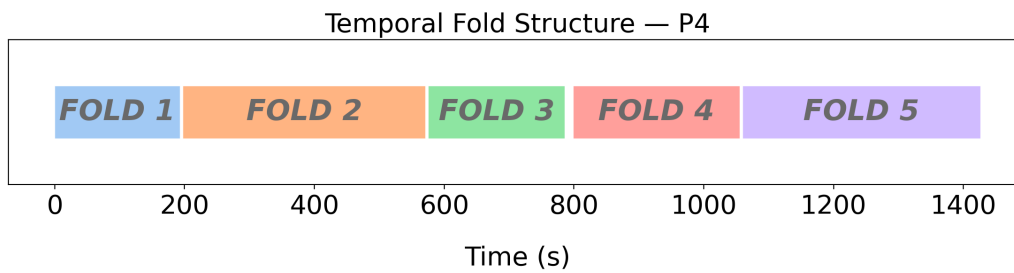


Figure 3.18: The five folds of participant P4 over time. Each coloured block is one fold. The narrow white spaces between the blocks are the fold gaps.

The two classes differ in size. **STAY** windows outnumber **SKIP** windows in 21 of the 25 participants. In P17, P23, P24 and P28, **SKIP** windows outnumber **STAY** windows (Figure 3.19). A model trained on unequal classes can score well by mostly predicting the larger class. After the fold split, the pipeline therefore drops randomly chosen windows of the larger class until the two classes hold the same number of windows (Figure 3.20). The pipeline balances every training fold and every test fold separately. On balanced test windows, random guessing scores about 50% (Section 3.8.4).

The random choice of dropped windows depends on a seed, the starting number of the random generator. The pipeline repeats the balancing with five seeds (0, 1, 7, 42 and 99). Each intra-subject result is the average over the five seeds. In the inter-subject setting, each participant contributes the balanced test windows of the five intra-subject folds under the first seed only. The fold gaps stay excluded in the two settings.

3.8 RESULT SYNTHESIS PIPELINE

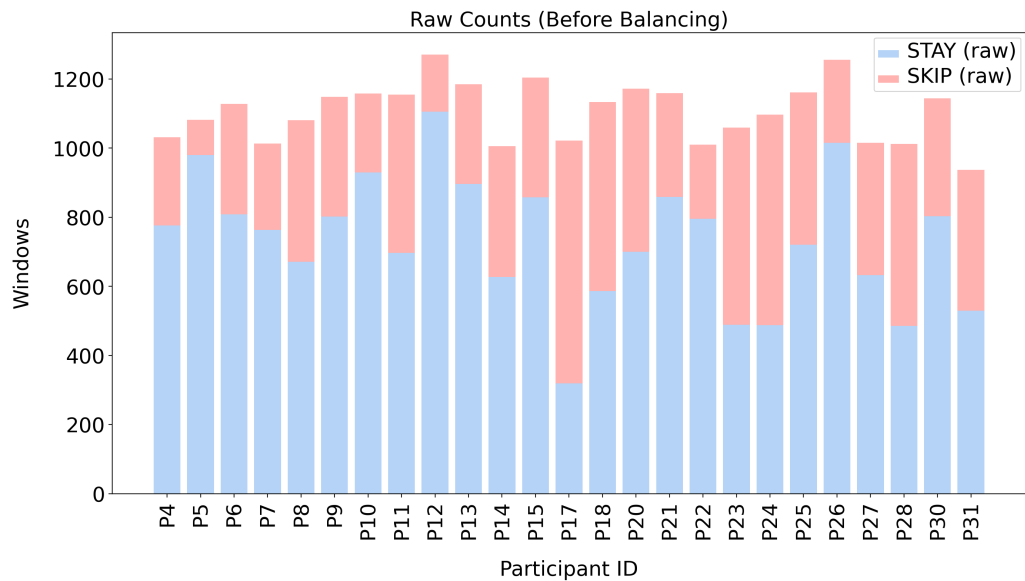


Figure 3.19: Window counts per participant before balancing. STAY windows (blue) outnumber SKIP windows (red) in 21 of 25 participants.

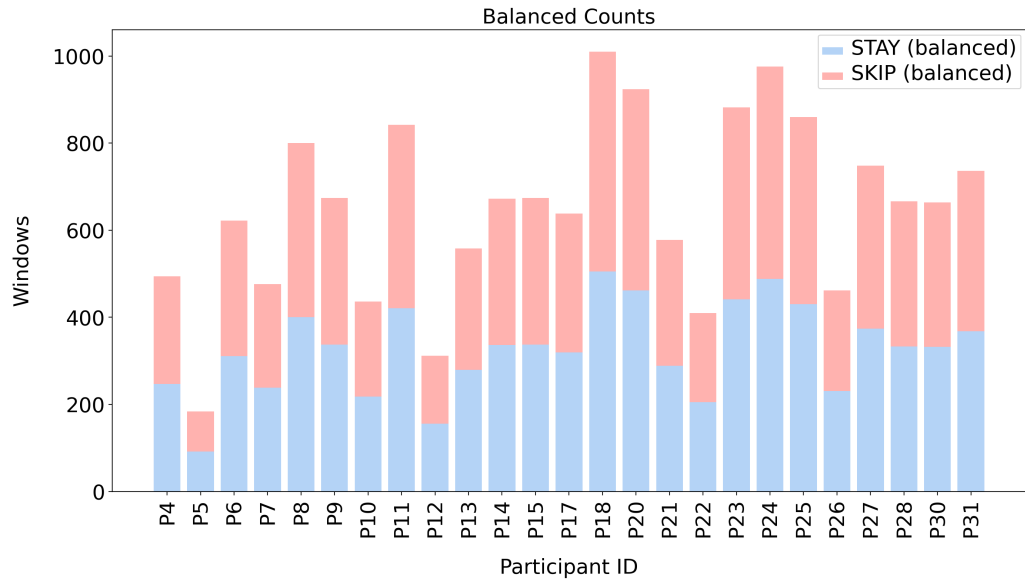


Figure 3.20: Window counts per participant after balancing: the test windows of the five folds, first seed. Each participant holds equal numbers of STAY windows and SKIP windows.

Varoquaux et al. report wide error bars for cross-validation of brain data, with typical confidence intervals of 10%. Many repeated random splits of blocks varied less than leaving out each block once, the scheme of the present thesis for folds and for participants [63]. Small differences between models therefore call for caution.

### 3.8.2 Feature Separation Analysis

Cohen’s  $d$  (Section 3.4.4, Equation 3.1) ranks the 236 features by how far the STAY windows and the SKIP windows lie apart (executed via [https://github.com/coffeeandcoffee/eeg\\_tiktok\\_pipeline/blob/main/data\\_analysis/viz17\\_feature\\_ranking.py](https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/blob/main/data_analysis/viz17_feature_ranking.py)). The EI gets a  $|d|$  the same way, as the reference line in the figures of Section 4.4.1. Cohen’s  $d$  carries no unit [42]. Features in different units can therefore share one ranking, for example TP10 raw standard deviation in microvolts and AF8 delta peak frequency in peaks per second.  $|d|$  serves only to pick candidate features. No statistical test is run on  $|d|$ ; the tests of Section 3.8.4 compare models instead.

Cohen’s  $d$  is reported at two levels:

- **Single features:** the exact  $|d|$  of one named feature, such as TP10 raw standard deviation (Table 4.1).
- **Categories:** the mean  $|d|$  over all features of one band, one electrode or one statistic, and over all features of one band at one electrode.

The categories differ in size. A band holds eight statistics per electrode, and the raw signal holds three (Section 3.7). A comparison between two category means is therefore rough. The spread within a category shows differences between features, rather than the uncertainty of one value. The category figures therefore show no confidence intervals. The one exception is the figure of each band at each electrode. The band-by-electrode figure shows bootstrapped 95% confidence intervals over the eight features, or three raw features, behind each point, so the intervals show the same spread between features.

Five picks lead to the models of Section 3.8.3: the two features with the largest  $|d|$  overall, and the feature with the largest  $|d|$  within the strongest band, within the strongest electrode and within the strongest statistic. Strongest means the largest mean  $|d|$  among the bands, among the electrodes or among the statistics. The five picks name three distinct features, the *top features*: TP10 raw standard deviation, AF7 high-gamma mean and AF8 delta peak frequency (Section 4.4.1).

### 3.8.3 Logistic Regression Modeling and Baselines

Three kinds of model predict **STAY** or **SKIP** for the same test windows (executed via [https://github.com/coffeeandcoffee/eeg\\_tiktok\\_pipeline/blob/main/data\\_analysis/step07\\_train\\_intra.py](https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/blob/main/data_analysis/step07_train_intra.py) and [https://github.com/coffeeandcoffee/eeg\\_tiktok\\_pipeline/blob/main/data\\_analysis/step16\\_parallel\\_universes.py](https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/blob/main/data_analysis/step16_parallel_universes.py)):

1. **Coin-flip baseline.** A classifier guesses **STAY** or **SKIP** for every window at random, with equal chance (`DummyClassifier(strategy='uniform')` of scikit-learn). The coin-flip baseline ignores the EEG and measures the chance level directly (Section 3.8.4). Random variation moves single values of the coin-flip baseline away from 50%, for example to 55.0% for intra-subject **STAY** recall (Table 4.2).
2. **EI model.** A logistic regression receives the EI of each window as the only input.
3. **Top-feature models.** A logistic regression receives one top feature of each window as the only input. Each of the three top features gets a model of its own.

Like the EI, every input is one number per window. A difference between a top-feature model and the EI model therefore reflects the input alone. Before training, the pipeline scales each input to a mean of 0 and a standard deviation of 1 over the training windows, and applies the same scaling to the test windows (`StandardScaler` of scikit-learn). The logistic regression runs with `class_weight='balanced'` (`LogisticRegression` of scikit-learn). The training windows are already balanced, so the setting leaves the two classes with equal weight.

### 3.8.4 Metrics and Significance Testing

Each class takes a turn as the positive class. With **SKIP** as the positive class, a true positive (TP) is a **SKIP** window predicted as **SKIP**, a true negative (TN) a **STAY** window predicted as **STAY**, a false positive (FP) a **STAY** window predicted as **SKIP**, and a false negative (FN) a **SKIP** window predicted as **STAY**. With **STAY** as the positive class, the two classes swap roles. A correct **SKIP** prediction finds a window from 5 s to 2 s before a keypress (Section 3.4.1).

$$\text{Precision} = \frac{TP}{TP + FP} \quad \text{Recall} = \frac{TP}{TP + FN} \quad F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Precision asks: of all windows the model calls **SKIP**, how many are **SKIP**? Recall asks: of all **SKIP** windows, how many does the model find? The  $F_1$ -score combines precision and recall and is the primary metric. A model that calls nearly every window **STAY** reaches a high **STAY** recall, yet a low **SKIP**  $F_1$ -score. Reporting the  $F_1$ -score for each class therefore reveals such one-sided models. On balanced test windows, random guessing gives a precision, a recall and an  $F_1$ -score of about 50% for each class.

For each participant, the pipeline adds up TP, TN, FP and FN over all five folds and all five seeds, and computes each metric once from the sums. In the inter-subject setting, each participant is tested once. Precision and recall come from the test windows. The  $F_1$ -score comes from the test windows ( $F_1$ -test) and from the training windows ( $F_1$ -train). A high  $F_1$ -train with a low  $F_1$ -test points to overfitting, a fit to the training windows that fails on new windows.

The Wilcoxon signed-rank test compares two models across the 25 participants. Each participant gives one pair of scores, for example the  $F_1$ -test of the EI model and the  $F_1$ -test of the coin-flip baseline. The test ranks the 25 differences by size and asks whether the positive and the negative differences balance out. Demšar recommends the Wilcoxon signed-rank test for comparing two classifiers over several data sets, here the 25 participants, because the test assumes no normal distribution and gives outliers less weight [64]. Demšar also treats the data sets as independent [64]. In the intra-subject setting, the scores of each participant come from the data of that participant alone. In the inter-subject setting, the models for two participants share 23 training participants, so the scores of different participants are not fully independent.

Two comparisons are tested: the EI model against the coin-flip baseline, and each top-feature model against the EI model. Each comparison runs in the intra-subject setting and in the inter-subject setting, for eight metrics:  $F_1$ -train,  $F_1$ -test, precision and recall, each with **STAY** and with **SKIP** as the positive class. The test is two-sided at  $\alpha = 0.05$ , so the test reacts to a model that is better and to a model that is worse. A model counts as better only when  $p < 0.05$  and the mean score of the model over the participants lies above the mean score of the baseline. The result tables mark the better values in green.

The eight result tables hold 64 tests. The  $p$ -values are not corrected for multiple comparisons. At  $\alpha = 0.05$ , about 3 of 64 tests would reach  $p < 0.05$  by chance, even without any real difference between the models. A single significant result therefore serves as exploratory screening. A correction for multiple comparisons is left to future research (Section 5.2).

# Chapter 4

## Results and Discussion

The present chapter reports the six contributions of the present thesis (Section 1.7), one section each:

1. **The experiment** (Section 4.1; Methods Sections 3.1–3.3). Twenty-five participants browsed TikTok while a consumer-grade headband recorded the EEG. The experiment ran in a partly controlled setting that encouraged naturalistic use. The dataset and the experimental protocol form the basis of every later analysis.
2. **The reusable pipeline** (Section 4.2; Methods Sections 3.5–3.8). The full pipeline runs from recording to results on a laptop, with public code. Other studies may reuse the full pipeline.
3. **Swipe bursts** (Section 4.3; Methods Section 3.6.5). Swipes often cluster into bursts. The burst analysis answers none of the three research questions, and is therefore a secondary result.
4. **Engagement Index performance** (Section 4.4; Methods Sections 3.7–3.8). Cohen’s  $d$  compares how well the Engagement Index (EI) and the 236 features separate STAY from SKIP (Section 4.4.1). Logistic regression models then test the EI against the coin-flip baseline, and the three top features against the EI (Sections 4.4.2 and 4.4.3).
5. **Discussion** (Section 4.5). The Discussion interprets the results for neuroscience, psychology, behaviour and application.
6. **A proposed definition of engagement** (Section 4.6). Informed by the results, Section 4.6 proposes a general definition of engagement, four reporting requirements, a hypothesis for field measurement, and two

findings that would show the proposed definition wrong. The present data do not test the proposed definition.

Section 4.4 answers the first two research questions, separability and benchmark comparison (Section 1.5). Section 4.6 answers the third research question, definition.

## Findings Reported in the Methods Chapter

Chapter 3 also reports findings that describe the dataset. Each dataset finding sits next to the step the dataset finding checks: for example, the sampling rate of the recording files (Section 3.6.1), the Bluetooth dropouts (Figure 3.11), the share of samples flagged by the artifact check (Figure 3.12), and the inter-swipe intervals (Figure 3.13). Dataset findings answer no research question. The inter-swipe intervals return as a result in Section 4.3.

The sensitivity analysis also sits in the Methods chapter (Figure 3.10). The sensitivity analysis is an exploratory check that points to future research. The labeling parameters were fixed before the sensitivity analysis (Section 3.4.1), and the sensitivity analysis changed none of the labeling parameters.

## 4.1 Lessons from Running the Experiment

The experiment worked in a partly controlled setting (Section 3.3). The setting still encouraged naturalistic use: self-paced viewing of an algorithmic feed, with a swipe possible at any moment (Section 2.1). No session had a major interruption. Recruitment through email and university mailing lists yielded 28 participants. Most participants were academically oriented (Section 3.1). The author observed that many participants showed interest in the research question.

Guided Access, an iPhone setting, helped in both the baseline phase and the browsing phase. Guided Access kept the iPhone in the open app. Guided Access also ended each phase after a set time: 2 minutes for the baseline phase, and 25 minutes for the browsing phase.

The briefing, meaning the instructions the author gave participants during the session, was hard to keep the same across sessions only from mental notes. After the first few sessions they were written down to standardize the process even better. The briefing explained the scrolling task, the free TikTok browsing of the browsing phase. The briefing covered the “A” keypress at each swipe, the start and the end of the scrolling task, and what to do

when the recording paused due to a Bluetooth dropout. After a Bluetooth dropout, participants were asked to simply keep scrolling. The author would pause or end the experiment only if a participant felt unwell, which did not happen. The wording of the briefing varied between sessions. The effect of the varying wording on the data was not measured. Future replications would benefit from a fixed briefing script and a fixed session flow.

## 4.2 The Reusable Pipeline

The full pipeline is all code of the present thesis, from recording to results: the reproducible recording pipeline (Section 3.5), the reproducible preprocessing pipeline (Section 3.6), feature engineering (Section 3.7) and the result synthesis pipeline (Section 3.8). After several months of development, the full pipeline ran from recording to results.

The main technical problem was the Bluetooth dropout. The EEG during each Bluetooth dropout was lost (Section 3.2). The reproducible recording pipeline handled each Bluetooth dropout: the reproducible recording pipeline reconnected and continued in a new recording file. The reproducible preprocessing pipeline processed each recording file on its own, so the EEG before and after each Bluetooth dropout stayed usable for the analysis (Section 3.6.1). No Bluetooth dropout caused a major session failure. The loss of EEG during each Bluetooth dropout leaves room for improvement.

During development, much of the code became obsolete. The code shrank from 37,189 to 11,311 lines: 69.6% of the lines were temporary, diagnostic, or replaced by better versions. Future projects could archive or delete unused scripts at regular steps. Less code is easier to read, maintain, reuse and extend.

The full pipeline needs a Muse headband, a laptop and open-source Python libraries. The full pipeline uses no data from TikTok: keypresses mark the swipes (Section 3.2). The full pipeline is designed for reuse. The “A” key can mark any action of interest, and one configuration file sets the frequency bands, the electrodes and the analysis steps. The author tested the reuse of the full pipeline successfully. The code is public at [https://github.com/coffeeandcoffee/eeg\\_tiktok\\_pipeline](https://github.com/coffeeandcoffee/eeg_tiktok_pipeline), so each step can be rerun. Other studies that record consumer-grade EEG and mark actions with keypresses may reuse the full pipeline.

### 4.3 Behavioral Swipe-Burst Phenomenon

A burst is several rapid swipes in a row (Section 3.6.5). Each swipe gets a  $\pm 3.0$  s burst window. Swipes less than 6.0 s apart have overlapping burst windows and merge into one burst block. The *block size* is the number of swipes in one burst block. A burst block of size 1 is a single swipe with no other swipe within 6.0 s before or after the swipe.

The burst analysis counts “A” keypresses, not swipes. Every count below therefore relies on each participant following the instructions: one keypress at every swipe to the next video, and no other action in the app (Section 3.3). Participants reported keypress errors for about 1% of keypresses (Section 3.1).

Of all 3,338 swipes, 1,344 (40.3%) stand alone in a burst block of size 1. The other 59.7% of swipes lie in bursts. Counted by burst block instead of by swipe, 1,344 of 1,994 burst blocks (67.4%) have size 1. Burst blocks of size 1 are therefore the most common burst blocks, and the largest burst block holds 23 swipes (Figure 4.2).

Participants differ widely (Figure 4.1). Per participant, the share of burst blocks of size 1 runs from 32.5% (P17) to 100% (P5), with a mean of 69.1% and a standard deviation of 16.2 percentage points. The present thesis calls participants with many large burst blocks *fast scrollers*, for example P17. The present thesis calls participants with mostly burst blocks of size 1 *slow scrollers*, for example P15. The two labels name the two ends of one range, not two separate groups. Each participant was recorded in one 25-minute session only, so the present data cannot show whether a participant scrolls the same way on another day.

The burst window changes every share above. With a  $\pm 5.0$  s burst window, swipes less than 10.0 s apart merge (Figure 4.3). Only 17.0% of swipes then stand alone, and 48.9% of burst blocks have size 1. Per participant, the share of burst blocks of size 1 then has a mean of 46.9% and a standard deviation of 20.8 percentage points. The largest burst block then holds 43 swipes. The author chose the  $\pm 3.0$  s burst window without a source (Section 3.6.5), so every share in the present section depends on an unvalidated choice.

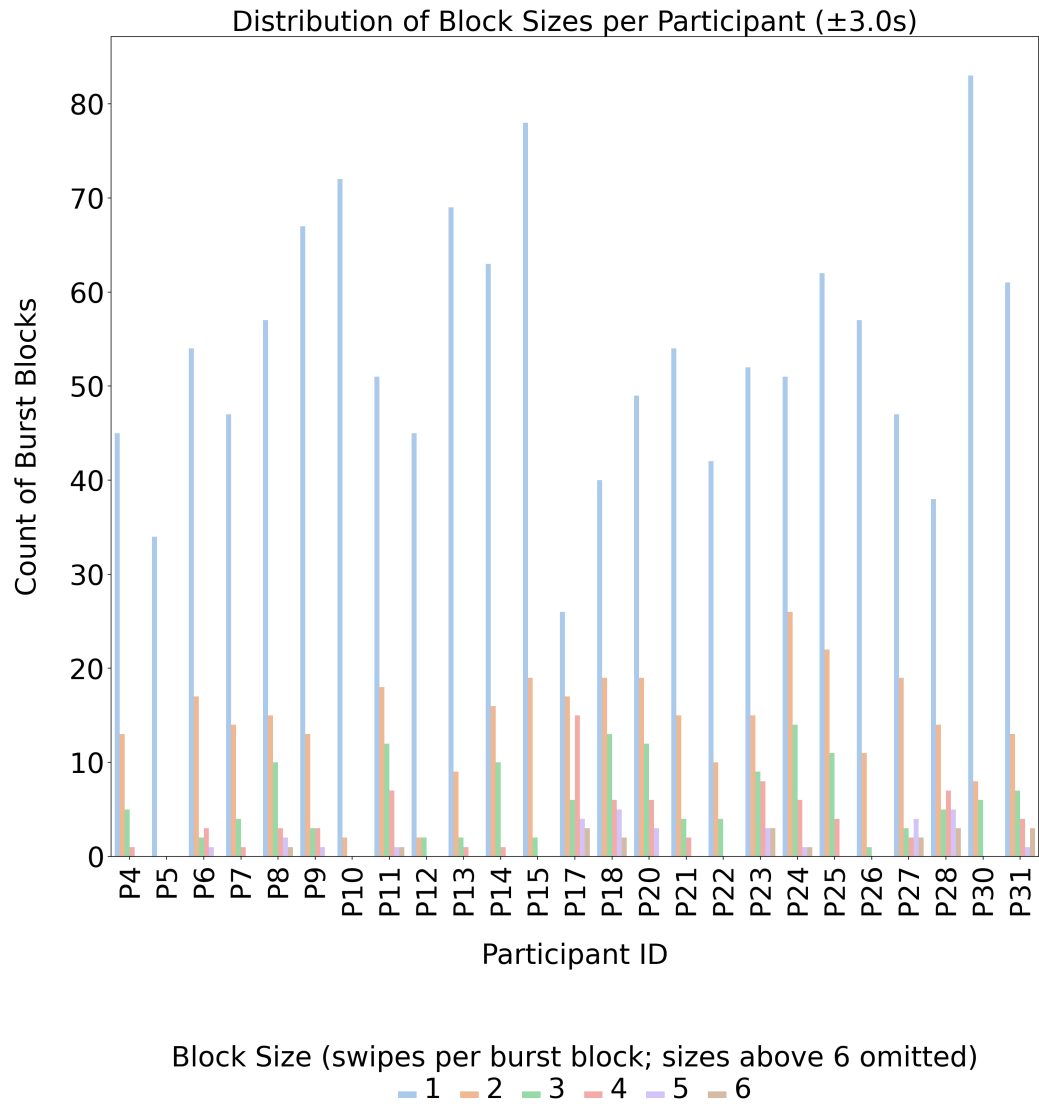


Figure 4.1: Burst blocks per participant, by block size ( $\pm 3.0$  s burst window). Block sizes above 6 are not shown. Fast scrollers, such as P17, have many large burst blocks. Slow scrollers, such as P15, have mostly burst blocks of size 1.

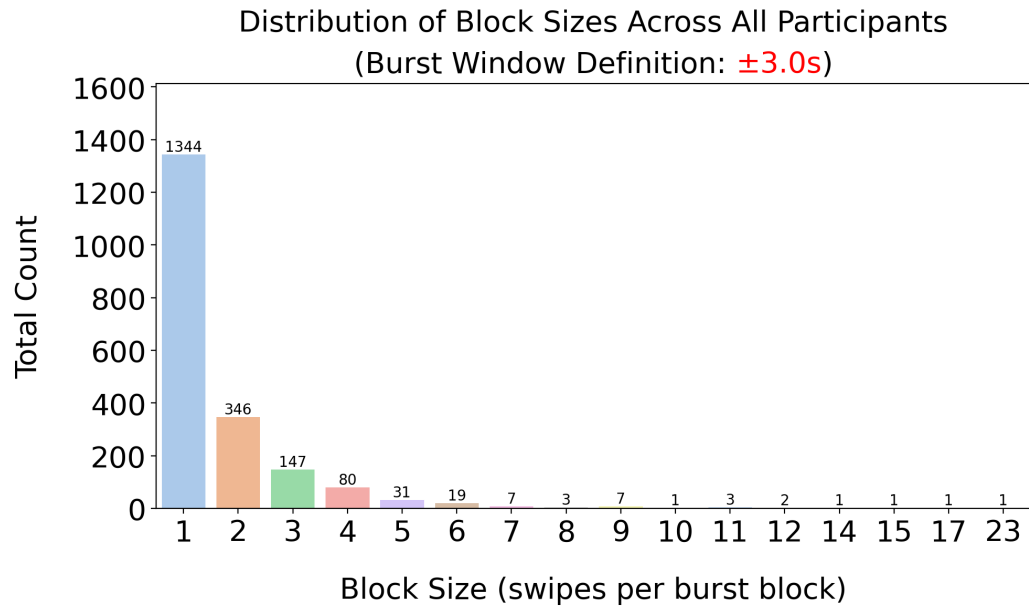


Figure 4.2: Burst blocks of all participants, by block size ( $\pm 3.0$  s burst window).

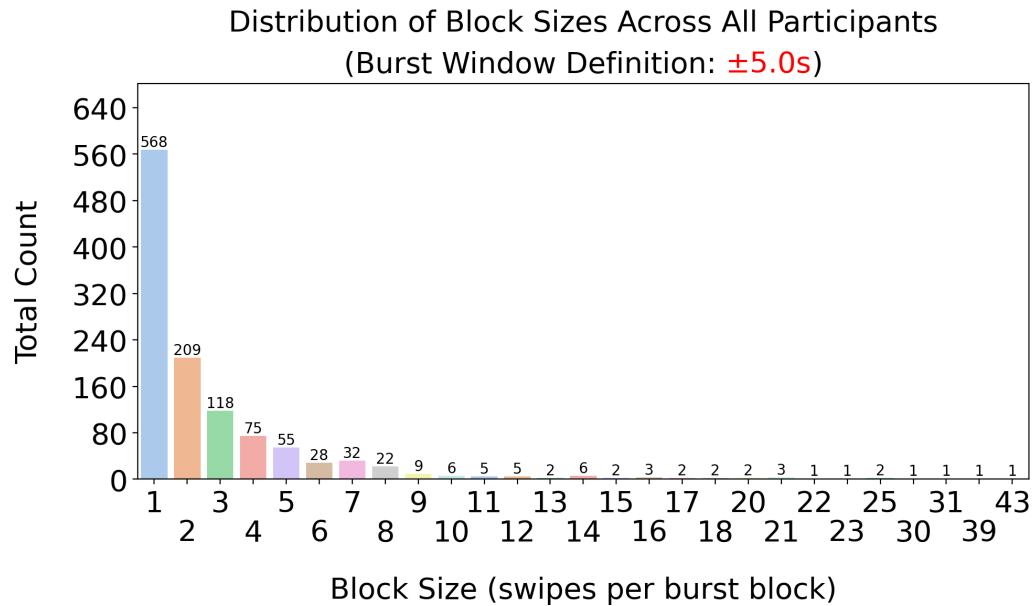


Figure 4.3: Burst blocks of all participants, by block size ( $\pm 5.0$  s burst window).

## 4.4 Engagement Index Performance

### 4.4.1 Class Separation of Single Features

The present section compares the class separation of the Engagement Index (EI) with the class separation of the 236 features. Class separation is measured as  $|d|$ , the size of Cohen’s  $d$  (Section 3.8.2). The EI reaches  $|d| = 0.022$ . All  $|d|$  values in the present section lie under 0.2, the value Cohen gives for a small effect [42], come from one dataset and carry no statistical test.

Figure 4.4 shows the mean  $|d|$  per band. Every band and the raw signal lie above the EI. Alpha lies lowest, close to the EI. High gamma lies highest, about 0.03 above the EI. Only four signals exceed 0.04: the raw signal, delta, low gamma and high gamma. The raw signal ranks second, behind high gamma.

Across the bands,  $|d|$  does not rise or fall steadily with frequency.  $|d|$  is high at delta, dips to the lowest value at alpha, peaks at high gamma and falls again at very high gamma. Section 4.5 discusses delta.

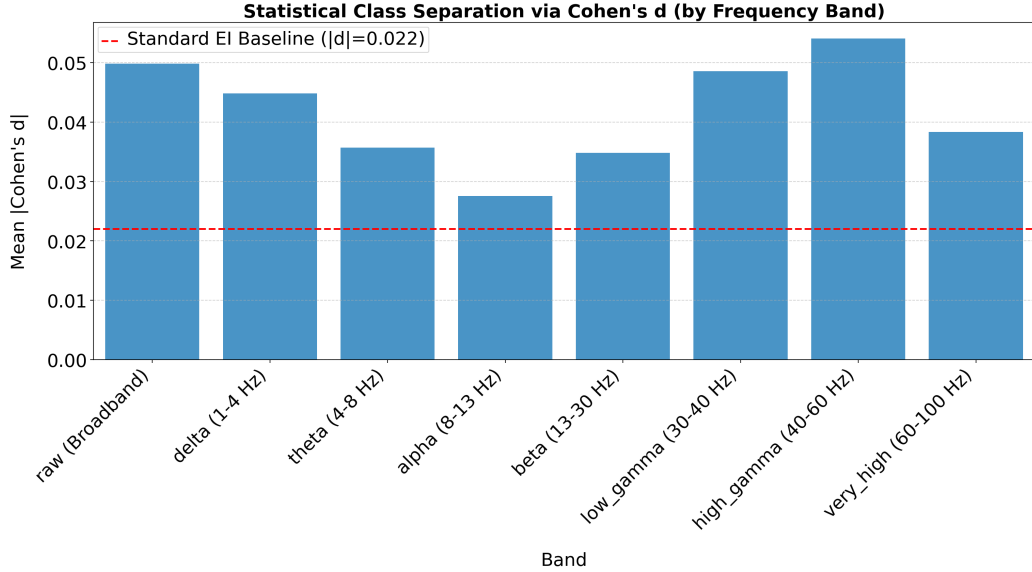


Figure 4.4: Mean  $|d|$  by frequency band, aggregated across electrodes and statistics. The dotted line marks the EI ( $|d| = 0.022$ ); error bars are omitted.

Figure 4.5 shows the mean  $|d|$  per electrode. All four electrodes lie above the EI. TP9 and AF8 lie just under 0.01 above the EI. TP10 lies about 0.024 above the EI, and AF7 about 0.034 above. AF7 sits on the left forehead, and

TP10 sits behind the right ear (Section 3.2).

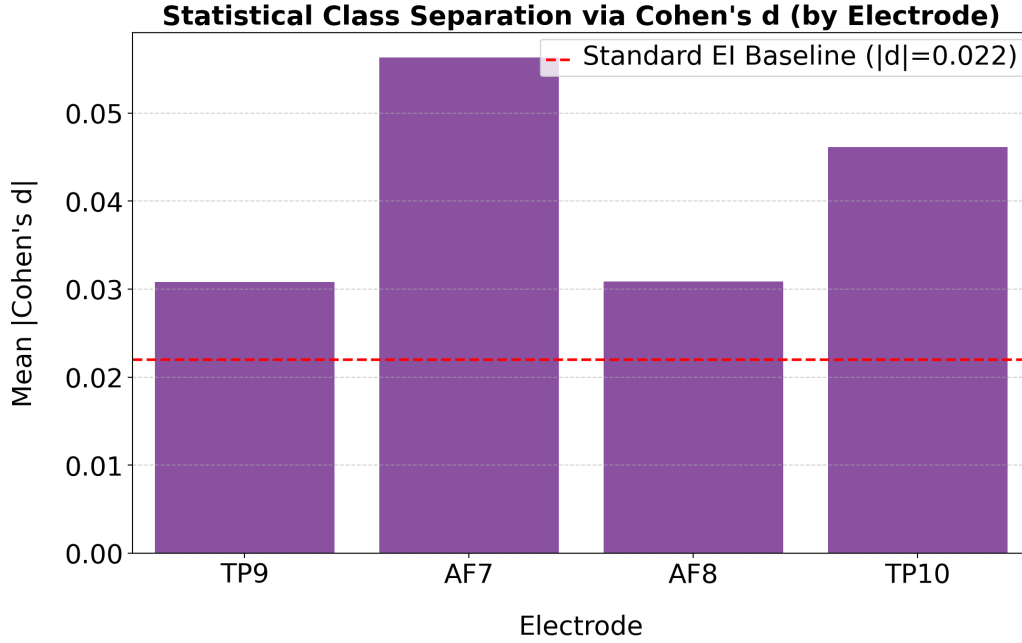


Figure 4.5: Mean  $|d|$  by electrode, aggregated across signal types and statistics. The dotted line marks the EI ( $|d| = 0.022$ ); error bars are omitted.

Figure 4.6 shows the mean  $|d|$  per statistic. Six of the nine statistics lie at least 0.01 above the EI: Peak Frequency, Hjorth complexity, standard deviation, mean, Hjorth mobility and Hjorth activity. Macro Frequency lies about level with the EI. Slope and Relative Power lie below the EI. Peak Frequency reaches nearly 0.07, the highest mean  $|d|$  of any band, electrode or statistic.

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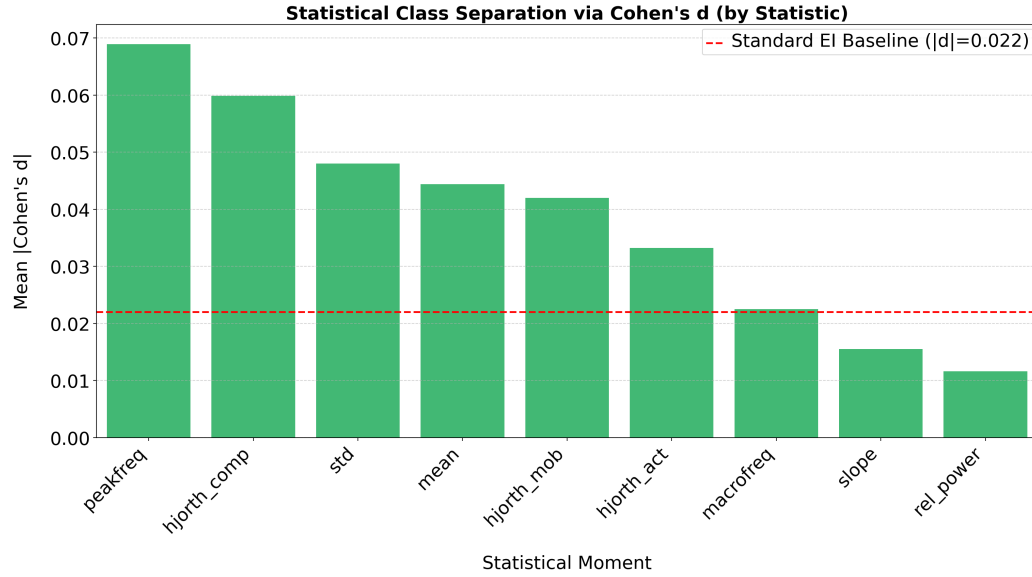


Figure 4.6: Mean  $|d|$  by statistic, aggregated across electrodes and signal types. The dotted line marks the EI ( $|d| = 0.022$ ); error bars are omitted.

The three figures above show means over many features. A single feature fixes one electrode, one band or the raw signal, and one statistic. Table 4.1 lists the 30 single features with the largest  $|d|$ . The strongest single feature is TP10 raw standard deviation, with  $|d| = 0.196$ , about nine times the  $|d|$  of the EI.

| Rank | Feature Name               | Cohen's $d$ |
|------|----------------------------|-------------|
| 1    | TP10 raw std               | 0.196       |
| 2    | AF7 high gamma mean        | 0.168       |
| 3    | AF7 high gamma hjorth mob  | 0.145       |
| 4    | AF7 delta hjorth comp      | 0.141       |
| 5    | AF8 delta peakfreq         | 0.140       |
| 6    | AF7 high gamma peakfreq    | 0.139       |
| 7    | TP9 delta hjorth comp      | 0.137       |
| 8    | TP9 delta peakfreq         | 0.136       |
| 9    | TP10 delta hjorth comp     | 0.131       |
| 10   | TP10 delta peakfreq        | 0.130       |
| 11   | AF7 theta hjorth comp      | 0.129       |
| 12   | AF7 raw std                | 0.129       |
| 13   | AF7 theta peakfreq         | 0.124       |
| 14   | AF7 low gamma hjorth comp  | 0.124       |
| 15   | AF7 delta peakfreq         | 0.123       |
| 16   | TP10 low gamma hjorth act  | 0.119       |
| 17   | AF8 high gamma mean        | 0.119       |
| 18   | AF7 low gamma mean         | 0.118       |
| 19   | AF7 beta mean              | 0.115       |
| 20   | AF8 delta hjorth comp      | 0.115       |
| 21   | TP10 low gamma std         | 0.114       |
| 22   | TP10 high gamma hjorth mob | 0.110       |
| 23   | AF7 very high mean         | 0.109       |
| 24   | TP10 beta hjorth act       | 0.108       |
| 25   | TP9 theta peakfreq         | 0.103       |
| 26   | TP9 high gamma std         | 0.103       |
| 27   | TP10 beta std              | 0.100       |
| 28   | AF8 theta peakfreq         | 0.100       |
| 29   | AF7 beta hjorth comp       | 0.100       |
| 30   | TP9 high gamma hjorth act  | 0.098       |

Table 4.1: Top 30 Features by Separation (|Cohen's  $d$ |).

Figure 4.7 shows the mean  $|d|$  for each band at each electrode. Within the raw signal, TP10 reaches the highest mean  $|d|$ , followed by AF7. TP10 raw standard deviation, the strongest single feature, belongs to the raw signal at TP10. At TP10, the raw signal lies above every single band.

At AF7,  $|d|$  rises from alpha through beta to low gamma.  $|d|$  stays about level from low gamma to high gamma, at about 0.076, and falls slightly at very high gamma. The confidence intervals are wide and overlap.

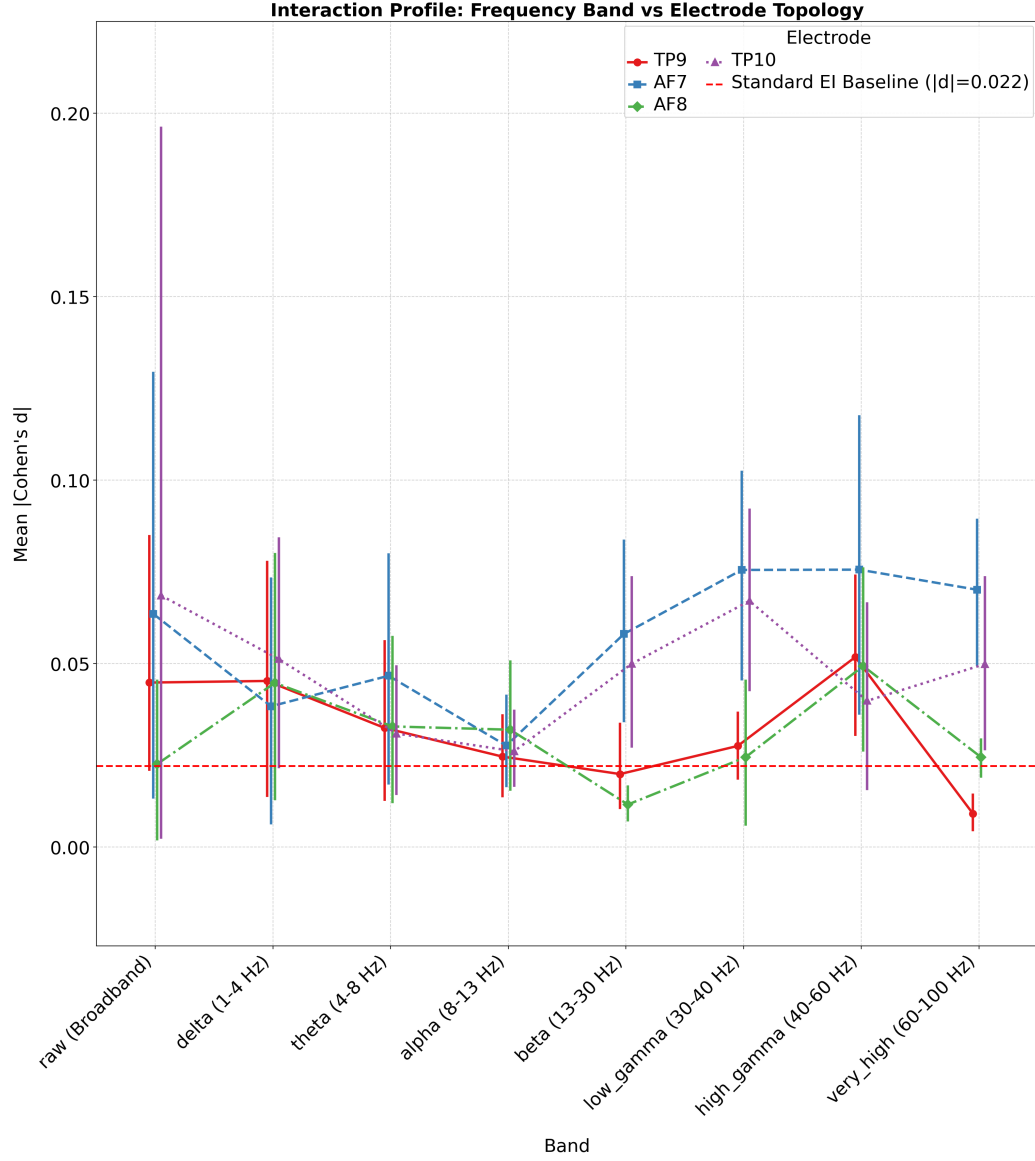


Figure 4.7: Mean  $|d|$  for each frequency-band  $\times$  electrode combination (band–electrode separation profile). Error bars represent bootstrapped 95% confidence intervals.

Five picks lead to the top features (Section 3.8.2). Figure 4.8 shows the result. TP10 raw standard deviation is the overall top feature #1 ( $|d| = 0.196$ ). AF7 high-gamma mean fills three picks: the overall top feature #2, the top feature of the strongest band (high gamma) and the top feature of the strongest electrode (AF7), with  $|d| = 0.168$ . AF8 delta peak frequency is

the top feature of the strongest statistic, Peak Frequency, with  $|d| = 0.140$ . Section 4.5 discusses the possible origins of the three top features.

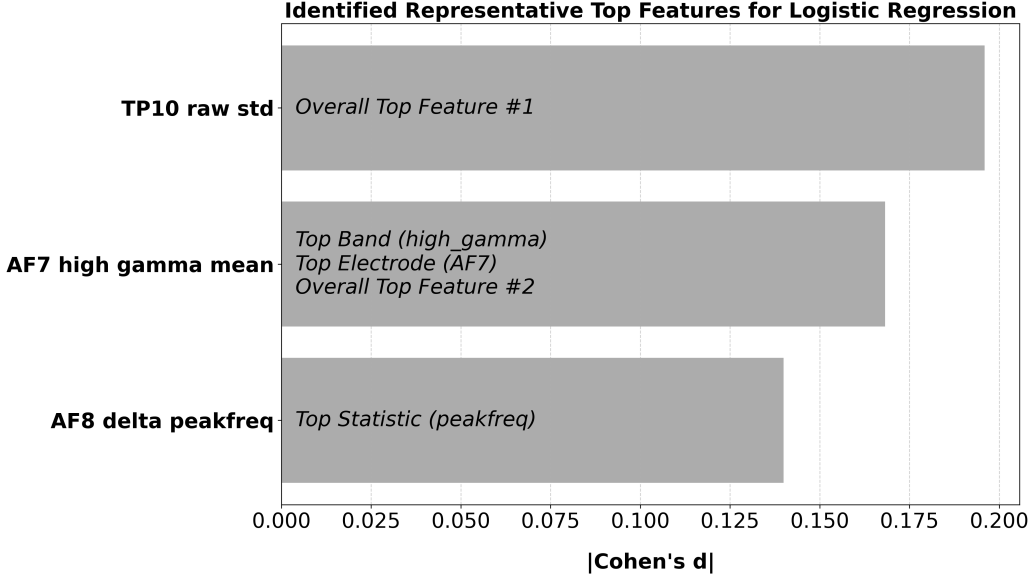


Figure 4.8: The three top features carried forward to the top-feature models. The five picks: overall top feature #1 and #2, and the top feature of the strongest band, of the strongest electrode and of the strongest statistic.

#### 4.4.2 EI Model versus Coin-Flip Baseline

The EI model and the coin-flip baseline predict **STAY** or **SKIP** for the same balanced test windows (Sections 3.8.3 and 3.8.4). The  $F_1$ -score is the primary metric. A green value in the tables is significantly higher than the baseline ( $p < 0.05$ ). The  $p$ -values are uncorrected across 64 tests, so a green value is an exploratory finding (Section 3.8.4).

In the intra-subject setting (Table 4.2), the EI model never beats the coin-flip baseline on the test windows. On four test metrics, the EI model scores significantly lower than the coin-flip baseline:  $F_1$ -test, precision and recall for **STAY**, and precision for **SKIP**. Every value of both models lies between 45% and 55%.

In the inter-subject setting (Table 4.3), the EI model scores significantly higher than the coin-flip baseline on two **STAY** test metrics:  $F_1$ -test (62.5%) and recall (89.2%). The EI model scores significantly lower on all four **SKIP** metrics, down to 9.6% **SKIP** recall. **STAY** precision stays at 49.4%, the level of chance. The EI model calls about 90% of all windows **STAY**, whether a

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window is **STAY** or **SKIP**: the EI model is one-sided (Section 3.8.4). The high **STAY** recall reflects the one-sided calling and does not show that the EI model tells **STAY** from **SKIP**. On balanced test windows, a model that calls every window **STAY** would reach a **STAY**  $F_1$ -score of about 67%, above the 62.5% of the EI model.

**Performance within each participant**

(intra-subject setting: each model trained and tested on one participant)

| <b>Metric</b> |              | <b>Baseline</b> | <b>Engagement Index</b> | <b><math>\Delta</math> [pp]</b>             | <b>Significance test</b>         |
|---------------|--------------|-----------------|-------------------------|---------------------------------------------|----------------------------------|
| <b>Note</b>   |              | coin flip, ~50% | logistic regression     |                                             | Engagement Index<br>vs. baseline |
| <b>STAY</b>   | $F_1$ -train | 50.6%           | <b>53.8%</b>            | <b><math>\blacktriangle +3.2</math></b>     | significantly higher             |
|               | $F_1$ -test  | 52.4%           | <b>50.9%</b>            | <b><math>\blacktriangledown -1.5</math></b> | significantly lower              |
|               | Precision    | 50.1%           | <b>49.1%</b>            | <b><math>\blacktriangledown -1.0</math></b> | significantly lower              |
|               | Recall       | 55.0%           | <b>52.9%</b>            | <b><math>\blacktriangledown -2.1</math></b> | significantly lower              |
| <b>SKIP</b>   | $F_1$ -train | 49.4%           | 49.6%                   | +0.2                                        | not significant                  |
|               | $F_1$ -test  | 47.5%           | 46.9%                   | -0.6                                        | not significant                  |
|               | Precision    | 50.1%           | <b>48.9%</b>            | <b><math>\blacktriangledown -1.2</math></b> | significantly lower              |
|               | Recall       | 45.2%           | 45.1%                   | -0.1                                        | not significant                  |

Table 4.2: EI model versus coin-flip baseline, intra-subject setting. Performance is the mean over the 25 participants.  $\Delta$ : compared model minus the coin-flip baseline, in percentage points (pp). Significance test: two-sided Wilcoxon signed-rank test against the coin-flip baseline ( $\alpha = 0.05$ ,  $n = 25$ , uncorrected). Green cells: significantly higher than the coin-flip baseline; red values: significantly lower; gray row: not significant.

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**Performance across participants**

(inter-subject setting: each model trained on 24 participants, tested on the 25th)

| Metric |                       | Baseline        | Engagement Index    | $\Delta$ [pp] | Significance test             |
|--------|-----------------------|-----------------|---------------------|---------------|-------------------------------|
| Note   |                       | coin flip, ~50% | logistic regression |               | Engagement Index vs. baseline |
| STAY   | F <sub>1</sub> -train | 50.3%           | 62.0%               | ▲ +11.7       | significantly higher          |
|        | F <sub>1</sub> -test  | 50.8%           | 62.5%               | ▲ +11.7       | significantly higher          |
|        | Precision             | 49.9%           | 49.4%               | −0.5          | not significant               |
|        | Recall                | 51.7%           | 89.2%               | ▲ +37.5       | significantly higher          |
| SKIP   | F <sub>1</sub> -train | 49.6%           | 20.6%               | ▼ −29.0       | significantly lower           |
|        | F <sub>1</sub> -test  | 48.9%           | 10.7%               | ▼ −38.2       | significantly lower           |
|        | Precision             | 49.9%           | 23.5%               | ▼ −26.4       | significantly lower           |
|        | Recall                | 48.0%           | 9.6%                | ▼ −38.4       | significantly lower           |

Table 4.3: EI model versus coin-flip baseline, inter-subject setting. Performance is the mean over the 25 participants.  $\Delta$ : compared model minus the coin-flip baseline, in percentage points (pp). Significance test: two-sided Wilcoxon signed-rank test against the coin-flip baseline ( $\alpha = 0.05$ ,  $n = 25$ , uncorrected). Green cells: significantly higher than the coin-flip baseline; red values: significantly lower; gray row: not significant.

Figure 4.9 shows the same pattern per participant. Against the coin-flip baseline, the EI model scores a higher STAY F<sub>1</sub>-test in 23 of 25 participants, and a lower SKIP F<sub>1</sub>-test in 23 of 25 participants.

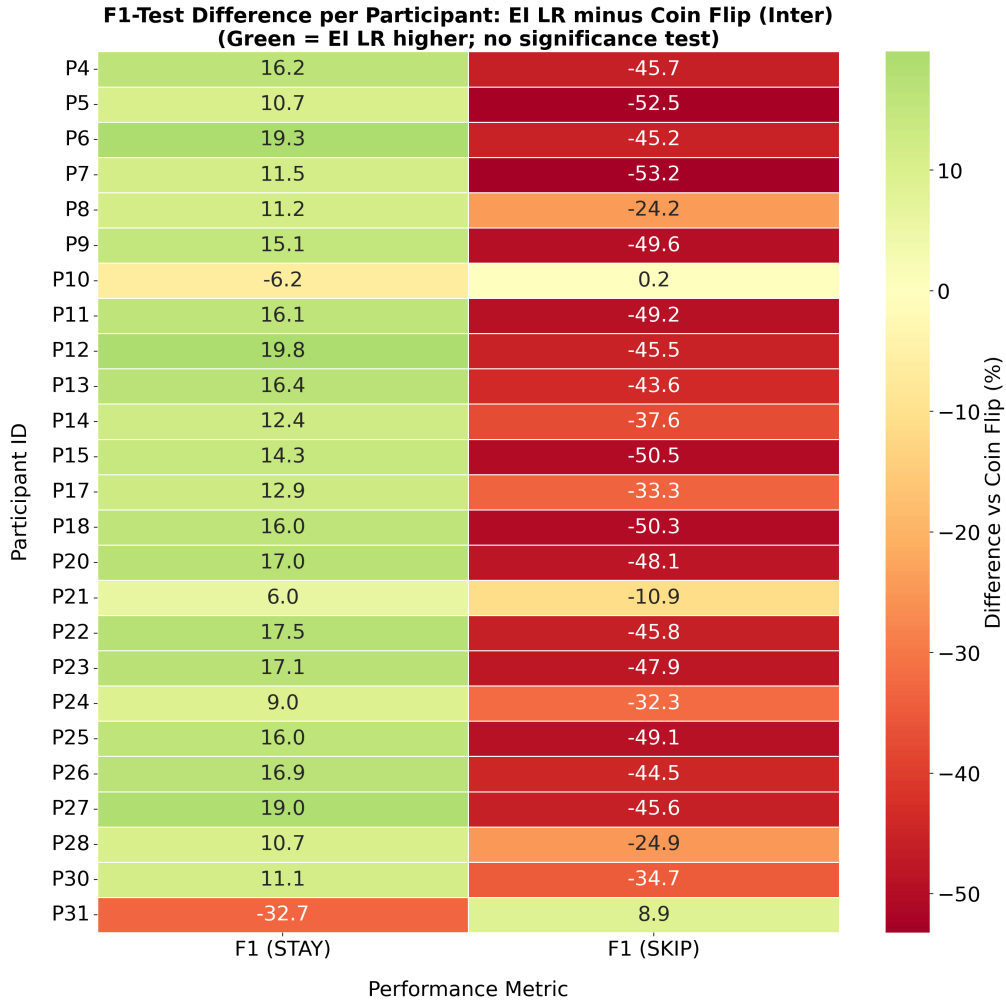


Figure 4.9: Inter-subject setting: difference in  $F_1$ -test between the EI model and the coin-flip baseline, per participant, with **STAY** and with **SKIP** as the positive class. Green: the EI model scores higher. The map shows differences and runs no significance test.

#### 4.4.3 Top-Feature Models

Each top-feature model receives one top feature as the only input (Section 3.8.3). Each top-feature model is compared with the EI model on the same test windows.

### Intra-subject Setting

TP10 raw standard deviation (Table 4.4) beats the EI model on no test metric. **SKIP**  $F_1$ -train rises (52.6% against 49.6%), yet **SKIP**  $F_1$ -test does not (46.8% against 46.9%). A model with one input is unlikely to overfit, so the gain on the training windows alone may come from TP10 raw standard deviation changing between the time blocks of the cross-validation, which requires further validation. On five metrics, TP10 raw standard deviation scores significantly lower than the EI model: all four **STAY** metrics and **SKIP** precision.

#### Performance within each participant

(intra-subject setting: each model trained and tested on one participant)

| Metric |              | Engagement Index    | TP10 raw standard deviation      | $\Delta$ [pp] | Significance test                |
|--------|--------------|---------------------|----------------------------------|---------------|----------------------------------|
| Note   |              | logistic regression | top feature, logistic regression |               | top feature vs. Engagement Index |
| STAY   | $F_1$ -train | 53.8%               | 52.4%                            | ▼ -1.4        | significantly lower              |
|        | $F_1$ -test  | 50.9%               | 49.3%                            | ▼ -1.6        | significantly lower              |
|        | Precision    | 49.1%               | 48.2%                            | ▼ -0.9        | significantly lower              |
|        | Recall       | 52.9%               | 50.4%                            | ▼ -2.5        | significantly lower              |
| SKIP   | $F_1$ -train | 49.6%               | 52.6%                            | ▲ +3.0        | significantly higher             |
|        | $F_1$ -test  | 46.9%               | 46.8%                            | -0.1          | not significant                  |
|        | Precision    | 48.9%               | 47.9%                            | ▼ -1.0        | significantly lower              |
|        | Recall       | 45.1%               | 45.7%                            | +0.6          | not significant                  |

Table 4.4: Top-feature model (TP10 raw standard deviation) versus EI model, intra-subject setting. Both models are logistic regressions with one input. Performance is the mean over the 25 participants.  $\Delta$ : compared model minus the EI model, in percentage points (pp). Significance test: two-sided Wilcoxon signed-rank test against the EI model ( $\alpha = 0.05$ ,  $n = 25$ , uncorrected). Green cells: significantly higher than the EI model; red values: significantly lower; gray row: not significant.

AF7 high-gamma mean (Table 4.5) beats the EI model on two test metrics: **STAY** precision (49.9% against 49.1%) and **SKIP** precision (49.8% against 48.9%). Both gains lie below one percentage point, and both precision values stay below 50%. AF7 high-gamma mean scores significantly lower than the EI model on no test metric.

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**Performance within each participant**

(intra-subject setting: each model trained and tested on one participant)

| Metric |                       | Engagement Index    | AF7 high-gamma mean              | $\Delta$ [pp] | Significance test                |
|--------|-----------------------|---------------------|----------------------------------|---------------|----------------------------------|
| Note   |                       | logistic regression | top feature, logistic regression |               | top feature vs. Engagement Index |
| STAY   | F <sub>1</sub> -train | 53.8%               | 52.6%                            | ▼ -1.2        | significantly lower              |
|        | F <sub>1</sub> -test  | 50.9%               | 52.1%                            | +1.2          | not significant                  |
|        | Precision             | 49.1%               | 49.9%                            | ▲ +0.8        | significantly higher             |
|        | Recall                | 52.9%               | 54.6%                            | +1.7          | not significant                  |
| SKIP   | F <sub>1</sub> -train | 49.6%               | 52.9%                            | ▲ +3.3        | significantly higher             |
|        | F <sub>1</sub> -test  | 46.9%               | 47.2%                            | +0.3          | not significant                  |
|        | Precision             | 48.9%               | 49.8%                            | ▲ +0.9        | significantly higher             |
|        | Recall                | 45.1%               | 45.0%                            | -0.1          | not significant                  |

Table 4.5: Top-feature model (AF7 high-gamma mean) versus EI model, intra-subject setting. Both models are logistic regressions with one input. Performance is the mean over the 25 participants.  $\Delta$ : compared model minus the EI model, in percentage points (pp). Significance test: two-sided Wilcoxon signed-rank test against the EI model ( $\alpha = 0.05$ ,  $n = 25$ , uncorrected). Green cells: significantly higher than the EI model; red values: significantly lower; gray row: not significant.

AF8 delta peak frequency (Table 4.6) beats the EI model on four test metrics: STAY F<sub>1</sub>-test (52.7% against 50.9%), STAY precision (49.9% against 49.1%), STAY recall (58.2% against 52.9%) and SKIP precision (50.0% against 48.9%). Both precision values stay at or below 50%. AF8 delta peak frequency scores significantly lower on SKIP recall (41.7% against 45.1%), and SKIP F<sub>1</sub>-test does not improve (44.5% against 46.9%). AF8 delta peak frequency therefore calls STAY more often than the EI model, without better SKIP detection.

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**Performance within each participant**

(intra-subject setting: each model trained and tested on one participant)

|      | Metric                | Engagement Index    | AF8 delta peak frequency         | $\Delta$ [pp] | Significance test                |
|------|-----------------------|---------------------|----------------------------------|---------------|----------------------------------|
|      | Note                  | logistic regression | top feature, logistic regression |               | top feature vs. Engagement Index |
| STAY | F <sub>1</sub> -train | 53.8%               | 55.4%                            | ▲ +1.6        | significantly higher             |
|      | F <sub>1</sub> -test  | 50.9%               | 52.7%                            | ▲ +1.8        | significantly higher             |
|      | Precision             | 49.1%               | 49.9%                            | ▲ +0.8        | significantly higher             |
|      | Recall                | 52.9%               | 58.2%                            | ▲ +5.3        | significantly higher             |
| SKIP | F <sub>1</sub> -train | 49.6%               | 45.4%                            | ▼ -4.2        | significantly lower              |
|      | F <sub>1</sub> -test  | 46.9%               | 44.5%                            | -2.4          | not significant                  |
|      | Precision             | 48.9%               | 50.0%                            | ▲ +1.1        | significantly higher             |
|      | Recall                | 45.1%               | 41.7%                            | ▼ -3.4        | significantly lower              |

Table 4.6: Top-feature model (AF8 delta peak frequency) versus EI model, intra-subject setting. Both models are logistic regressions with one input. Performance is the mean over the 25 participants.  $\Delta$ : compared model minus the EI model, in percentage points (pp). Significance test: two-sided Wilcoxon signed-rank test against the EI model ( $\alpha = 0.05$ ,  $n = 25$ , uncorrected). Green cells: significantly higher than the EI model; red values: significantly lower; gray row: not significant.

**Inter-subject Setting**

The three top-feature models share one pattern (Tables 4.7, 4.8 and 4.9). Each top-feature model scores significantly higher than the EI model on all four SKIP metrics. Each top-feature model scores significantly lower on STAY F<sub>1</sub>-train, STAY F<sub>1</sub>-test and STAY recall. AF8 delta peak frequency also scores significantly higher on STAY precision (50.9% against 49.4%).

SKIP F<sub>1</sub>-test rises from 10.7% for the EI model to 44.3% for TP10 raw standard deviation, 35.5% for AF7 high-gamma mean and 35.5% for AF8 delta peak frequency. All three values stay below the chance level of about 50% (Section 3.8.4). STAY F<sub>1</sub>-test falls from 62.5% for the EI model to 46.0%, 45.5% and 59.3%.

SKIP recall shows how often each model calls SKIP windows SKIP, and 100% minus STAY recall shows how often each model calls STAY windows SKIP. TP10 raw standard deviation calls about 51% of the SKIP windows and about 49% of the STAY windows SKIP, AF7 high-gamma mean about 43% and 42%, and AF8 delta peak frequency about 30% and 27%. Each

top-feature model therefore calls **SKIP** about as often for **STAY** windows as for **SKIP** windows. The top-feature models are more balanced than the one-sided EI model, although AF8 delta peak frequency still calls about 72% of all windows **STAY**. In the inter-subject setting, none of the four models separates **STAY** from **SKIP** clearly better than chance.

#### Performance across participants

(inter-subject setting: each model trained on 24 participants, tested on the 25th)

|      | Metric                | Engagement Index    | TP10 raw standard deviation      | $\Delta$ [pp] | Significance test                |
|------|-----------------------|---------------------|----------------------------------|---------------|----------------------------------|
|      | Note                  | logistic regression | top feature, logistic regression |               | top feature vs. Engagement Index |
| STAY | F <sub>1</sub> -train | 62.0%               | 49.6%                            | ▼ -12.4       | significantly lower              |
|      | F <sub>1</sub> -test  | 62.5%               | 46.0%                            | ▼ -16.5       | significantly lower              |
|      | Precision             | 49.4%               | 47.5%                            | -1.9          | not significant                  |
|      | Recall                | 89.2%               | 51.0%                            | ▼ -38.2       | significantly lower              |
| SKIP | F <sub>1</sub> -train | 20.6%               | 52.4%                            | ▲ +31.8       | significantly higher             |
|      | F <sub>1</sub> -test  | 10.7%               | 44.3%                            | ▲ +33.6       | significantly higher             |
|      | Precision             | 23.5%               | 43.0%                            | ▲ +19.5       | significantly higher             |
|      | Recall                | 9.6%                | 50.8%                            | ▲ +41.2       | significantly higher             |

Table 4.7: Top-feature model (TP10 raw standard deviation) versus EI model, inter-subject setting. Both models are logistic regressions with one input. Performance is the mean over the 25 participants.  $\Delta$ : compared model minus the EI model, in percentage points (pp). Significance test: two-sided Wilcoxon signed-rank test against the EI model ( $\alpha = 0.05$ ,  $n = 25$ , uncorrected). Green cells: significantly higher than the EI model; red values: significantly lower; gray row: not significant.

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**Performance across participants**

(inter-subject setting: each model trained on 24 participants, tested on the 25th)

| Metric |                       | Engagement Index    | AF7 high-gamma mean              | $\Delta$ [pp] | Significance test                |
|--------|-----------------------|---------------------|----------------------------------|---------------|----------------------------------|
| Note   |                       | logistic regression | top feature, logistic regression |               | top feature vs. Engagement Index |
| STAY   | F <sub>1</sub> -train | 62.0%               | 54.3%                            | ▼ -7.7        | significantly lower              |
|        | F <sub>1</sub> -test  | 62.5%               | 45.5%                            | ▼ -17.0       | significantly lower              |
|        | Precision             | 49.4%               | 44.3%                            | -5.1          | not significant                  |
|        | Recall                | 89.2%               | 58.1%                            | ▼ -31.1       | significantly lower              |
| SKIP   | F <sub>1</sub> -train | 20.6%               | 45.5%                            | ▲ +24.9       | significantly higher             |
|        | F <sub>1</sub> -test  | 10.7%               | 35.5%                            | ▲ +24.8       | significantly higher             |
|        | Precision             | 23.5%               | 52.4%                            | ▲ +28.9       | significantly higher             |
|        | Recall                | 9.6%                | 42.7%                            | ▲ +33.1       | significantly higher             |

Table 4.8: Top-feature model (AF7 high-gamma mean) versus EI model, inter-subject setting. Both models are logistic regressions with one input. Performance is the mean over the 25 participants.  $\Delta$ : compared model minus the EI model, in percentage points (pp). Significance test: two-sided Wilcoxon signed-rank test against the EI model ( $\alpha = 0.05$ ,  $n = 25$ , uncorrected). Green cells: significantly higher than the EI model; red values: significantly lower; gray row: not significant.

**Performance across participants**

(inter-subject setting: each model trained on 24 participants, tested on the 25th)

|      | Metric                | Engagement Index    | AF8 delta peak frequency         | $\Delta$ [pp] | Significance test                |
|------|-----------------------|---------------------|----------------------------------|---------------|----------------------------------|
|      | Note                  | logistic regression | top feature, logistic regression |               | top feature vs. Engagement Index |
| STAY | F <sub>1</sub> -train | 62.0%               | 59.8%                            | ▼ -2.2        | significantly lower              |
|      | F <sub>1</sub> -test  | 62.5%               | 59.3%                            | ▼ -3.2        | significantly lower              |
|      | Precision             | 49.4%               | 50.9%                            | ▲ +1.5        | significantly higher             |
|      | Recall                | 89.2%               | 73.1%                            | ▼ -16.1       | significantly lower              |
| SKIP | F <sub>1</sub> -train | 20.6%               | 37.4%                            | ▲ +16.8       | significantly higher             |
|      | F <sub>1</sub> -test  | 10.7%               | 35.5%                            | ▲ +24.8       | significantly higher             |
|      | Precision             | 23.5%               | 53.3%                            | ▲ +29.8       | significantly higher             |
|      | Recall                | 9.6%                | 29.7%                            | ▲ +20.1       | significantly higher             |

Table 4.9: Top-feature model (AF8 delta peak frequency) versus EI model, inter-subject setting. Both models are logistic regressions with one input. Performance is the mean over the 25 participants.  $\Delta$ : compared model minus the EI model, in percentage points (pp). Significance test: two-sided Wilcoxon signed-rank test against the EI model ( $\alpha = 0.05$ ,  $n = 25$ , uncorrected). Green cells: significantly higher than the EI model; red values: significantly lower; gray row: not significant.

## 4.5 Discussion

The present section asks what the results mean. Three results carry the discussion. First, every band, every electrode and the raw signal separated STAY from SKIP better than the Engagement Index (EI), yet every  $|d|$  stayed below 0.2 (Section 4.4.1). Second, no model predicted STAY or SKIP clearly better than chance, within or across participants (Sections 4.4.2 and 4.4.3). Third, swipes often came in bursts (Section 4.3). The results come from one dataset of 25 participants, and the origin of every feature remains open. Every reading below is therefore a possibility for future tests.

The small  $|d|$  explains why prediction stayed near chance. A  $|d|$  below 0.2 means that the STAY windows and the SKIP windows overlap almost completely. Body height gives an everyday example. Imagine two football teams. The players of team A are on average about 1.5 cm taller than the players of team B. The two averages differ, yet the heights of the two teams overlap almost completely: many players of team B are taller than

many players of team A. Someone who sees only the height of one player can therefore barely guess the team of that player. The windows of the present thesis behave the same way. TP10 raw standard deviation differs on average between **STAY** windows and **SKIP** windows ( $|d| = 0.196$ ), yet a model that sees one window can barely guess the label of that window.

### 4.5.1 Neuroscience

**Why the EI separated so little.** The EI separated **STAY** from **SKIP** least of all tested signals ( $|d| = 0.022$ ). Across participants, the EI model was one-sided and called about 90% of all windows **STAY** (Section 4.4.2). Within participants, the EI model scored near chance. The weak EI result fits one earlier result: in a controlled learning task, Apicella et al. found the accuracy of the EI “not optimal”, and note that the EI is mainly used in non-predictive applications [25]. Three explanations remain open. First, the present thesis tested an adapted EI. The adapted EI uses the variance of the z-scored envelope, the four Muse sites and beta at 13–30 Hz (Section 3.7). Pope et al. used band power at Cz, Pz, P3 and P4, with beta at 13–22 Hz [4]. Part of the weak result may therefore belong to the adaptation. Second, Pope et al. chose the EI for a different job. In the task of Pope et al., a computer switched a tracking task between manual and automated operation within seconds, so every change in task demand came from outside the participant [4]. A swipe comes from inside the participant, without any cue. Third, the EI averages each band over all four sites. Alpha often changes locally: at one moment, alpha can rise at one site and fall at another [65]. An average over two forehead sites and two sites behind the ears may blur such local changes.

**What the EI may track.** The EI has separated whole tasks well. Nuamah et al. recognised five mental tasks from the EI in 10-second trials, with 88.67% accuracy where guessing gives 20% (Section 2.4) [24]. Ovi et al. averaged the EI over each whole task of each participant, and the EI separated a cognitive task from an affective task with a large effect ( $d = -1.03$ ) [66]. In the present thesis, the EI separated 1-second windows before a swipe from sustained viewing only slightly. Taken together, the EI may track which kind of task a person performs over many seconds, rather than the moment before a self-paced action. Section 4.6 returns to the role of the time interval.

**Where the three top features may come from.** Each top feature may reflect brain activity, muscle activity, powerline-noise, or the person being recorded. The present thesis cannot tell these sources apart (Section 2.11). EMG and EEG share frequencies, like the two singers of Section 2.11. Most scalp EEG above 20 Hz is EMG, even in a relaxed person, and the share of EMG is largest at the lower edge of the scalp [51]. Muscle

activity also changes with mental effort [53]. A change in muscle tension before a swipe could therefore separate STAY from SKIP just as a change in brain activity could.

*TP10 raw standard deviation* ( $|d| = 0.196$ ) measures how much the raw signal behind the right ear spreads within one window. TP10 sits next to the small muscles around the ear and near the jaw [53]. The raw signal keeps the 50 Hz powerline-noise, because no notch filter was applied (Section 2.13). The raw signal is also not z-scored (Section 3.6.2). Differences in signal strength between participants, and between the two headband units (Section 3.2), therefore stay in TP10 raw standard deviation. In one EEG study, a broadband signal identified individual people better than any single band [66]. TP10 raw standard deviation may therefore partly reflect which participant, and which headband, is recorded.

*AF7 high-gamma mean* ( $|d| = 0.168$ ) measures the strength of 40–60 Hz on the left forehead. A brain reading exists: attention to a stimulus goes with stronger and faster gamma synchronisation [67]. The evidence for the brain reading comes mainly from recordings inside the visual cortex of monkeys [67], far from a forehead electrode. Two other sources are at least as plausible. AF7 lies near the frontalis, the forehead muscle (Section 2.11), and Pope et al. themselves treated power at 42–100 Hz as EMG [4]. The 50 Hz powerline-noise also lies in the middle of high gamma (Section 2.13). McMenamin et al. advise extreme caution for beta and gamma findings near scalp muscles, and ask at minimum for a scalp map of where each effect lies [53]. Four electrodes cannot give such a map.

*AF8 delta peak frequency* ( $|d| = 0.140$ ) counts how often per second the strength of the delta band peaks on the right forehead (Section 3.7.4). A brain reading exists here too. Reviews link delta to attention, decision making and salience detection [54, 55], and in one EEG study, delta separated two tasks best of all bands [66]. TikTok, the app used by the participants of the present thesis, has a Chinese version called Douyin. In adolescents browsing Douyin, the change in delta during browsing predicted an addiction score [56]. The cited works study mainly the strength of delta, and none labels the seconds before a self-paced swipe (Section 2.12). The link to AF8 delta peak frequency is therefore loose. Low frequencies are not free of muscle either: near a tensed muscle, EMG reaches every band, delta included [49]. AF8 also sits above the right eye, where the artifact check looks for blinks (Section 3.6.4). Whether blinks change AF8 delta peak frequency was not tested.

**Left and right.** AF7 and TP10 separated the classes more than AF8 and TP9 (Section 4.4.1). The pattern may point to a difference between the two sides of the brain. Simpler explanations exist as well. Two headband

units were used. The participant swiped with one hand and pressed the “A” key with the other hand (Section 3.2), and the present thesis recorded neither handedness nor the hand holding the phone. The left–right pattern therefore remains a question for future research.

### 4.5.2 Psychology

The present data suggest that leaving a video may differ from a low level of engagement. Pooled over all participants, the seconds before a swipe differed slightly from sustained viewing in many features (Section 4.4.1). The EI, a composite of three bands, captured the difference least of all. The working definition treats the swipe as the latest observable end of engagement (Section 1.2), and the small differences are compatible with a short event before that end. The present data cannot show what the event is: a decision, a loss of interest, the preparation of the swipe, or muscle tension.

The prediction results add no evidence for such an event. Across participants, the top-feature models scored higher than the EI model on all four SKIP metrics (Section 4.4.3). Yet each top-feature model called SKIP about as often for STAY windows as for SKIP windows. The higher SKIP scores therefore reflect less one-sided calling than the EI model. Within participants, every model stayed near chance (Sections 4.4.2 and 4.4.3). The only sign of a difference before the swipe remains the small pooled  $|d|$ .

Two kinds of swipe may mix in the SKIP class. Some swipes may follow a deliberate judgement of the video. The forehead sites lie over the prefrontal cortex, which is held to be particularly important for top-down control, the guiding of thought and action by internal goals [68]. Other swipes may happen on autopilot. Users of a social media feed describe getting “lost” in the feed [2], and Miller and Cohen note that the prefrontal cortex is not critical for simple, automatic behaviours [68]. A mix of the two kinds of swipe would blur any single signal. The present thesis did not record which kind of swipe occurred. The burst analysis gives the closest behavioural hint (Section 4.3): a swipe after long viewing may more often follow a deliberate judgement, and a swipe inside a burst may more often happen on autopilot. The hint remains untested.

Participants differed widely. Burst tendency varied strongly between participants (Section 4.3), and no model transferred to new participants better than chance. In one fMRI study, prefrontal and other higher-order areas did not synchronise across five people watching the same film, which Hasson et al. link to individual variations [61]. Cross-subject EEG classification is a recognised challenge even in controlled laboratory tasks [25, 22]. EMG also differs strongly between people [53]. One difference to Hasson et al. matters:

the five people of Hasson et al. watched the same film, whereas each participant of the present thesis saw a different feed. Part of the weak transfer may therefore come from different videos as well as from different brains. Within participants, the models also stayed near chance, so the present data cannot separate a person-specific signal from a missing signal.

### 4.5.3 Behaviour

Swipes often came in bursts: 59.7% of swipes lay in bursts under the  $\pm 3.0$  s burst window, and 83.0% under a  $\pm 5.0$  s burst window (Section 4.3). The swipes of one burst may not be independent decisions. The **SKIP** class therefore mixes swipes after long viewing with swipes a few seconds after the previous swipe. In bursts, 23% of all **SKIP** windows lie within the pre-swipe gap or the post-swipe gap of a neighbouring swipe (Section 3.4.2). The sensitivity analysis points the same way. The largest  $|d|$  rose as the **SKIP** windows moved further from the swipe, and so closer to the previous swipe (Section 3.4.5). Rimbart et al. attribute a similar gain partly to activity after the previous movement [36]. Part of the separation between **STAY** and **SKIP** may therefore belong to the previous swipe rather than to the coming swipe.

Every share and every  $|d|$  above depends on labeling parameters and a burst window set by the author (Sections 3.4.5 and 3.6.5). Future behavioural analyses should test whether a pattern holds across several reasonable values.

### 4.5.4 Application

The present results support no applied system yet. No model predicted a swipe clearly better than chance, and muscle activity, powerline-noise and differences between participants may drive the top features. The applied value lies in three narrower results. The full pipeline records consumer-grade EEG with a time mark for every swipe, runs from recording to results with public code, and uses no data from TikTok (Section 4.2). Features with a known meaning, such as the strength of one band at one electrode, can be compared one by one with the EI. And the EI, the standard EEG measure of engagement, separated the seconds before a swipe from sustained viewing least of all tested signals. The recommendations below rest on these three results and on the reviewed literature. Each recommendation states whether the recommendation rests on the present results or on the reviewed literature.

### Recommendations for Policy Makers

First, **report more than watch time**. Social media designers are tasked with maximising time on site [2]. Long viewing can reflect absorbed attention or zoned-out scrolling: in two daily-life studies, time spent in a feed was not significantly associated with dissociation, and more social media use than usual went with fewer flow reports [2, 3]. Knierim et al. advise platforms to measure the quality of the experience rather than time on the platform alone [3]. Transparency reports and audits could therefore require engagement measures checked against recall or behaviour change. The recommendation rests on the reviewed literature. The present EEG results did not measure dissociation, flow or recall.

Second, **test interface features that interrupt scrolling**. On a custom Twitter app, custom lists and reading-history labels went with lower self-reported dissociation [2]. The evidence comes from one study of text posts, with daily averages that cannot show cause and effect. Replication on short-form video platforms should come before regulation. The recommendation rests on the reviewed literature.

Third, **treat engagement optimisation as auditable**. In the present thesis, one researcher with a consumer headband and a laptop measured small average EEG differences before a swipe. Platforms likely hold far richer behavioural data. The present thesis has no access to platform data, so the point is a reasonable extrapolation. Regulators could require platforms to disclose which behavioural target a feed optimises, such as continued viewing, because long viewing can also reflect zoned-out scrolling [2].

### Recommendations for Digital Global Public Health Research

For digital public health research on disease prevention and evidence-based health communication, three extensions follow.

First, **pre-test health messages**. The full pipeline needs a consumer headband, a laptop and open-source software (Section 4.2). Small pre-tests of health messages, such as two versions of a vaccination message, therefore become more affordable for public institutions. EEG studies already compare advertisements [17]. The full pipeline took several months of development (Section 4.2), so the full pipeline lowers the barrier to entry but remains a research tool. A pre-test would also need a validated signal, which the present thesis does not yet provide. The recommendation rests on the present results.

Second, **report when viewers leave a health message**. Studies of health messages on TikTok mostly count views, likes and comments [1, 30]. In one content analysis, institutional accounts received fewer likes, comments

and shares than personal creators, in a comparison without a statistical test that rests on two public-health videos [1]. Campaigns could also report churn, the share of viewers who leave during each quarter of a video, as Salminen et al. did for advertisements [23]. Churn comes from TikTok analytics and needs no EEG. The present thesis adds the reason to look at the moment of leaving: in the working definition, the swipe is the latest observable end of engagement (Section 1.2).

Third, **expect viewers to differ**. Precision public health tailors prevention to groups with different needs. In the present thesis, participants differed widely in burst tendency (Section 4.3). One session per participant cannot show whether burst tendency stays stable across days. If burst tendency proves stable, health messages could be tailored to groups of viewers, and one message design for all viewers would reach some viewers poorly. The recommendation rests on the present results and remains a hypothesis.

### Recommendations for Health Message Design

(i) **Win interest early**. Most viewers left TikTok advertisements within the first quarter of the video [23]. The present thesis placed the SKIP windows 5 to 2 s before each swipe by argument (Section 3.4.1), and the seconds before a swipe differed only slightly from sustained viewing. Where in the opening seconds of a video such a difference is strongest, and which content features relate to the difference, remains for future research. The recommendation rests mainly on the reviewed literature.

(ii) **Judge messages by recall and behaviour change**. A message watched to the end while zoned out may be remembered worse than a shorter message that holds attention [2, 32]. Evaluation should therefore weight recall and behaviour change above the share of viewers who watch to the end. The recommendation rests on the reviewed literature.

(iii) **Expect bursts**. Swipes cluster into bursts, and burst tendency varies sharply between participants (Section 4.3). By the definition of a burst, a video inside a burst is viewed for less than 6 s. Whether a message inside a burst is remembered worse was not tested. Burst-aware timing of messages therefore remains a question for future research. Until then, campaigns may lower the risk that a single message falls into a burst by spreading several short messages over time, each with a different opening hook. The recommendation rests on the present results, and the spreading of messages is untested.

**Societal Scope: An Explicitly Speculative Outlook**

The following paragraph changes register. The paragraph states a hypothesis, not a result of the present thesis. Knierim et al. found that more social media use than usual went with fewer flow experiences, and name displacement as one possible explanation: scrolling may fill time that active pursuits could otherwise use [3]. Knierim et al. tie flow to such active pursuits, among them learning, creating and conversing [3]. Displacement may therefore also reduce time and energy for other people, including the in-person contact through which relationships form. The present thesis did not test displacement, and can neither support the hypothesis nor show the hypothesis to be wrong. Knierim et al. also note that social media can serve rest [3]. If displacement exists, measures of the moment a viewer leaves, such as the swipe studied here, could help evaluate interventions against displacement, such as interface features that interrupt scrolling [2].

Finally, a dual-use note. A marker that helps health communicators hold attention could equally help commercial or political actors. The present thesis therefore argues for open research under ethics oversight (Section 3.1), paired with the third recommendation for policy makers.

**4.6 Implication: Engagement Definition**

The sections above concerned one behaviour on one platform: the swipe. The present section steps outside the swipe. The present section proposes a general definition of engagement, informed by the results of the present thesis and by the definitions reviewed in Section 2.4. This definition is by no means complete nor validated. It's mere purpose is to start a discussion about a general definition of engagement crossing multiple academic fields which, as far as the author's knowledge allows to see, seems under-discussed. The intended continuation of this thought is, that later work in other settings can use, test and most importantly correct the proposed definition. The proposed definition is a proposal about the construct, and the present data do not test the proposed definition. Section 4.6.5 states starting points of what would show the proposed definition to be wrong.

**4.6.1 Four Findings**

Four findings motivate a more general definition of Engagement.

**Staying is no proof of engagement.** Although longer interaction is often equated with engagement, the evidence does not support this. Long viewing

can reflect either absorbed attention or zoned-out scrolling, and sustained attention fluctuates while a person remains in a task (Section 2.6). The present thesis did not measure attention during **STAY** viewing, so this conclusion rests on the reviewed literature (Section 3.4.2). The finding therefore argues against defining engagement solely in terms of time spent.

**The swipe is a datable end.** A keypress timestamps every swipe (Section 3.2). Although EEG before a swipe differed only slightly from sustained viewing, with  $|d|$  at most 0.196 (Section 4.4.1), the datable end arises from the behaviour itself: the swipe. Some swipes may occur on autopilot, after engagement has already faded (Section 4.5), but the swipe still marks the latest possible end of engagement, if not the exact end. This finding argues against a definition based merely on a declining level of engagement and instead proposes a precise moment of definite disengagement.

**No single signal defines engagement.** Pooled across participants, many single features separated **STAY** from **SKIP** better than the EI (Section 4.4.1). Yet when used as the sole input to a model, the same features performed near chance both within and across participants (Section 4.4.3). The present data cannot distinguish a person-specific signal from a missing signal (Section 4.5). The finding therefore argues against definitions that rely on a narrow signal, such as AF7 high-gamma mean, the EI ratio, likes, or interaction duration, and proposes more holistic metrics, with outcome-oriented behaviour-change metrics being the most desirable.

**A number means little without an interval.** The largest  $|d|$  increased with longer pre-swipe gaps, **SKIP** periods, and window lengths (Section 3.4.5). The EI separated whole tasks over many seconds in other studies, yet separated 1-second windows before a swipe only slightly (Section 4.5). Measured over different intervals, the same type of signal yields different values. This finding therefore argues against reporting engagement without specifying the measurement interval, reinforcing a holistic approach that considers the environment in which the interval and measurement are embedded.

#### 4.6.2 The Proposed Definition

The proposed definition inherits the limits of the four findings. The effects are small, come from one dataset of 25 demographically narrow participants, and may partly reflect muscle activity or powerline-noise (Section 4.5). The

proposed definition is therefore a first version, open to correction. The proposed definition keeps the working definition of Section 1.2 and replaces the four domain-bound parts: the viewer becomes a person, the content on screen becomes an object, the next video becomes the available alternatives, and the swipe becomes an observable exit.

**Proposed definition (Engagement).** *Engagement is a person’s attention to a particular object, held against available alternatives, and ending at the latest at an observable exit.*

**Attention** rules out dissociated viewing, in which a person scrolls a feed without really paying attention to the scrolling [2]. Time spent, used as the only metric, counts the dissociated viewer and the attentive viewer the same way (Staying is no proof of engagement).

**To a particular object** requires engagement to be engagement with something, such as a message, a task, a person or a piece of work. The marketing literature likewise names the object of engagement, such as a brand or a community [10]. “An engaged user” without an object is not yet a measurable claim.

**Held against available alternatives** means that the person could turn to something else at any moment. In the infinite feed, the next video is always one thumb movement away.

**Ending at the latest at an observable exit** gives engagement an end that can be dated. The exit is the latest possible end. Engagement may fade earlier without any visible sign, as in dissociated viewing or in a swipe on autopilot. The swipe is a datable end. O’Brien and Toms likewise describe disengagement as ceasing to be engaged, and also as stopping use [11].

**What the proposed definition inherits.** From O’Brien and Toms: engagement is a process of four stages, from the point of engagement through sustained engagement and disengagement to re-engagement [11].

**Where the proposed definition departs.** Nuamah et al., citing earlier work, define task engagement as “effortful striving towards task goals” [24]. The computer-science and HCI literatures often treat engagement as flow [9], a state of deep, effortless absorption [3]. The proposed definition deliberately leaves out the word *effortful*, so that flow can count as engagement. The word *attention* already keeps dissociated viewing out, so the proposed definition does not become too narrow.

**What the proposed definition drops.** Duration becomes an indicator. Duration covers two things here: the period over which a person uses an intervention, part of the definition of Perski et al. [9], and the watch time of one video. Counts of use, the most common measure in social-media engagement research [10], become indicators as well. Every measuring instrument drops out: no index, scale, rating, formula or telemetry (No single signal defines engagement). An EEG feature, a self-report item, a linguistic marker and a behavioural trace can therefore all serve as indicators. None of the indicators is engagement itself.

**What the proposed definition adds.** To the author’s knowledge, none of the definitions reviewed in Section 2.4 ends engagement at an observable exit. The interval enters as a reporting requirement (Section 4.6.3).

### 4.6.3 Implications for Measurement

Three implications follow from the proposed definition.

**Duration is one indicator.** Duration may be reported as one indicator. Duration should not serve as the yardstick against which other measures are validated.

**The exit is a primary measurement target.** An instrument should be judged by how well the instrument detects the change toward the exit, as well as by how well the instrument scores the middle of viewing. In the present thesis, the adapted EI separated the seconds before a swipe from sustained viewing least of all tested signals (Section 4.4.1). The result concerns the adapted EI, and says nothing yet about engagement itself.

**Indicators are plural and are checked against each other.** Yardley et al. recommend multiple measures of engagement and call for methods to triangulate the measures [6]. O’Brien and Toms planned to check self-report against behavioural and physiological measures [11].

The finding *A number means little without an interval* and the three implications lead to four reporting requirements. An engagement measurement states:

- the **object** engaged with;
- the **interval** over which engagement is claimed;
- the **indicator** used;
- the **exit criterion**: what counts as an exit, observed directly rather than inferred from time spent alone.

#### 4.6.4 A Hypothesis for Field Measurement

EEG is time-precise: EEG records 256 times per second during viewing, unlike a questionnaire filled in afterwards. Yet EEG is hard to use at scale. The EEG pipeline of the present thesis needed a headband, a baseline recording for each participant, a keypress at every swipe, and a quiet room. Yardley et al. rate body-route measures as often intrusive, expensive and hard to scale (Section 2.5). Even in a partly controlled setting, no model of the present thesis separated SKIP windows from STAY windows clearly better than chance (Sections 4.4.2 and 4.4.3).

The proposed definition names no measuring instrument, and so leaves room for cheaper instruments. No route measures engagement directly, so no route serves as ground truth. Witnesses offer an analogy. No single witness sees everything. When independent witnesses agree on the time of an event, the account gains credibility. When witnesses disagree, the disagreement adds information. In the analogy, the ask route, the log route and the body route are the witnesses, and the exit is the event.

The present thesis therefore states the following hypothesis. The hypothesis awaits testing in follow-up research.

*Where EEG cannot be used at scale, engagement as proposed in Section 4.6.2 may be measurable with cheaper field measures: short self-report questionnaires repeated over time (the ask route), and behavioural or linguistic traces in everyday activity and communication (the log route). A field measure gains support when the field measure agrees with indicators from other routes at the exit. The proposed definition implies two checks: the field measure should react at the exit, and the field measure should do more than count time.*

The hypothesis faces a known obstacle. In the one study Perski et al. cite on the question, body-route measures and ask-route measures of engagement agreed only weakly [9].

The present data cannot test the hypothesis. The present thesis offers only a narrow first step: the keypress dated each exit, and the EEG differed from sustained viewing only slightly before the exit. The present thesis recorded no ask-route measure of engagement, no second setting and no later outcome. In other settings, such as a work setting, EEG is hard to use, and which routes can date an exit there remains open.

### 4.6.5 What Would Show the Proposal Wrong

Two findings would show the proposed definition to be wrong, or in need of a change, and existing methods can already look for both.

First, an indicator that marks the change toward the exit in new people, without calibration to each person. Keeping every signal out of the proposed definition would then be needless caution, and a definition that names the indicator would serve better. Even the best result of Apicella et al. across participants, 72.8% for two levels of cognitive engagement, first removed the average signal of each participant [25].

Second, dissociated viewing that produces the same recall and the same behaviour change as attentive viewing. The word *attention* would then add nothing.

### 4.6.6 Scope of the Proposal

Nothing in the proposed definition is specific to short-form video. TikTok served as a test case with a convenient property: a frequent, voluntary, time-stamped exit, about 134 swipes per participant in 25 minutes (Section 3.1). The proposed definition is meant also for settings where exits are rarer, slower and harder to see: a learner abandoning a course, a patient dropping out of a digital behaviour-change intervention [9], or a colleague withdrawing from shared work. In such settings, one caution applies: stopping use can also mean that a person no longer needs the intervention [6].

Under the proposed definition, engagement with a health message in a feed and engagement with shared work in a distributed team can be treated as one construct with different exits. In the feed, the exit is a thumb movement and is logged. In remote work and online collaboration, the exit of engagement is slower and quieter, and may show only in what a person stops contributing to. In remote work, hours logged, messages sent or meetings attended can most easily stand in for engagement unnoticed. The proposed definition asks for an exit criterion in addition.

# Chapter 5

## Conclusion and Future Research

### 5.1 Conclusion

The present thesis asked three questions. Can EEG show the seconds before a viewer decides to swipe away? Is the Engagement Index (EI),  $\beta/(\alpha + \theta)$ , the standard EEG measure of engagement, the right instrument for the seconds before a swipe? And which general definition of engagement could serve other fields as a starting point? Twenty-five participants browsed a TikTok “For You” feed, reset before each participant, freely and alone for 25 minutes. A four-channel Muse S headband recorded the EEG, and a keypress marked every swipe. Each 1.0 s window gave 236 features. Cohen’s  $d$  ranked the 236 features and the EI. Logistic regression then tested the EI and the three top features, in the intra-subject setting and in the inter-subject setting. The effects are small, and the answers are narrow.

**Separability.** On average, the seconds before a swipe (SKIP) differed slightly from sustained viewing (STAY). The strongest single feature, TP10 raw standard deviation, reached  $|d| = 0.196$ . Every  $|d|$  stayed below 0.2, the value Cohen gives for a small effect, and no statistical test was run on  $|d|$ . No model predicted STAY or SKIP clearly better than chance. The data therefore suggest a small average difference. The data do not show that the small average difference can predict a single swipe.

**Benchmark comparison.** Every band, every electrode and the raw signal separated STAY from SKIP better than the EI. The EI reached  $|d| = 0.022$ , and TP10 raw standard deviation about nine times as much. In the intra-subject setting, the EI model scored near chance. In the inter-subject setting, the EI model was one-sided: the EI model called about 90% of all windows STAY. STAY recall reached 89.2%, SKIP recall only 9.6%, and STAY precision stayed at 49.4%, the level of chance. Each top-feature model

detected **SKIP** significantly better than the **EI** model in the inter-subject setting, on all four **SKIP** metrics. Yet **SKIP**  $F_1$ -test reached at most 44.3%, below the chance level of about 50%. Each top-feature model called **SKIP** about as often for **STAY** windows as for **SKIP** windows. The top-feature models were therefore less one-sided than the **EI** model, without separating **STAY** from **SKIP** better than chance. In the intra-subject setting, TP10 raw standard deviation beat the **EI** model on no test metric. AF7 high-gamma mean and AF8 delta peak frequency beat the **EI** model on a few test metrics, among the **SKIP** metrics only on **SKIP** precision, and every model stayed near chance. The present thesis has therefore found no better index. In the present data, the **EI** separated the seconds before a swipe from sustained viewing least of all tested signals.

**Parameter sensitivity.** The largest  $|d|$  of any single feature rose with every longer pre-swipe gap, every longer **SKIP** period and every longer window length. Across the grid of pre-swipe gap and **SKIP** period, the largest  $|d|$  ran from 0.158 to 0.236. Every  $|d|$  of the present thesis therefore depends on the labeling parameters. The default pre-swipe gap rests on the literature on movement preparation, and the default window length on the literature on window length. To the author’s knowledge, no study gives a value for the **SKIP** period before a self-paced action. The default values were fixed before the sensitivity analysis.

**Bursts.** Swipes often came in bursts. Under the  $\pm 3.0$  s burst window, 40.3% of swipes stood alone, and 59.7% lay in bursts. Under a  $\pm 5.0$  s burst window, only 17.0% of swipes stood alone. Burst tendency differed widely between participants. The author set the burst window without a source.

**Limitations.** The present thesis has the following limits:

- **Muscle activity.** All three top features may partly reflect EMG (Section 2.11). TP10 lies at the lower edge of the scalp, where the share of EMG is largest, even at rest. AF7 and AF8 lie near the forehead muscle. Near a tensed muscle, EMG reaches every band, delta included. Common cleaning methods such as ICA reduce EMG without removing EMG completely.
- **Powerline-noise.** AF7 high-gamma mean (40–60 Hz) and TP10 raw standard deviation both contain the 50 Hz powerline-noise. No reported analysis used a notch filter (Section 2.13).
- **Suspected noise in the top features.** A rough check after the main analysis raised a suspicion that powerline-noise makes up a large part of the raw signal, and leaks into other bands, such as delta at AF8. The same check suggested that noise alone may produce Peak Frequency

5.1 CONCLUSION

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and Macro Frequency values similar to those of the recordings. The separation of some top features may therefore partly reflect noise. The suspicion awaits a proper test.

- **No artifact removal.** The artifact check flagged blinks and EMG, and removed nothing (Section 3.6.4).
- **Participants and headbands.** Two headband units were used. Differences between the two units cannot be separated from differences between participants. TP10 raw standard deviation is not z-scored, so TP10 raw standard deviation may partly reflect which participant and which headband was recorded (Section 4.5).
- **Sample.** The sample was demographically narrow and academically oriented, and each participant was recorded once. Three analysed participants reported an ADHD or attention-related diagnosis, and one reported taking psychoactive medication. ADHD and psychoactive medication were no exclusion criteria (Section 3.1). About half of the analysed participants received a 20-euro voucher. The effect of the voucher is beyond the scope of the present thesis.
- **Keypresses.** Participants reported keypress errors for about 1% of keypresses (Section 3.1). Unreported keypress errors cannot be ruled out. The delay between each swipe and its “A” keypress was not measured.
- **SKIP windows in bursts.** In bursts, 23% of the SKIP windows lie within the pre-swipe gap (2.0 s) or the post-swipe gap (0.5 s) of a neighbouring swipe. These SKIP windows may contain preparation or movement of the neighbouring swipe, and were kept (Section 3.4.2).
- **Statistics.** The  $p$ -values are not corrected for multiple comparisons, and serve as screening only (Section 3.8.4).
- **Sensitivity analysis.** The sensitivity analysis was a rough check. The sensitivity analysis crossed the pre-swipe gap and the SKIP period without a statistical test, varied the window length alone at a fixed 1.0 s stride, and pooled all participants. Whether the effect of the window length depends on the pre-swipe gap or the SKIP period, or differs between participants, is untested (Section 3.4.5).

Part of what looks like a brain signal may be jaw or forehead muscle activity. Measuring how much comes from the brain and how much from muscles is

the most important next step. The findings can be trusted only after the share of brain activity and the share of muscle activity are measured.

**Method.** The present thesis also leaves a working method. Naturalistic use of an algorithmic feed, with no video list chosen by the researcher, no fixation and no trial structure, was recorded on consumer hardware, with a keypress marking every swipe. The full pipeline ran from recording to results with open-source code, and used no data from TikTok (Section 4.2). The full pipeline lowers the barrier to such studies, and remains a research tool (Section 4.5).

**Definition.** Section 4.6 draws one further conclusion, beyond TikTok. Section 4.6 proposes a definition of engagement that names no measuring instrument, four reporting requirements (object, interval, indicator, exit criterion), and a hypothesis for cheaper field measures where EEG cannot be used at scale. The hypothesis awaits testing.

Social media designers are tasked with maximising time on site [2]. In the literature reviewed here, long viewing can reflect attention held on the content or zoned-out viewing, normative dissociation (Section 2.6). The present thesis has not decoded the swipe. The present thesis offers a small first step toward measuring, independently of the platforms, the moment a viewer leaves.

## 5.2 Future Research

**Replication.** Future research should first replicate the present findings in an independent sample. A replication would ideally use one headband unit, artifact control, a broader and less academically oriented sample, and correction for multiple comparisons. Clear exclusion criteria for ADHD and psychoactive medication, and a fixed briefing script (Section 4.1), would reduce variation between participants and sessions. Webcam video, which the recording pipeline can already record (Section 3.5), could measure the delay between each swipe and its “A” keypress. A key test is whether the three top features, TP10 raw standard deviation, AF7 high-gamma mean and AF8 delta peak frequency, stay informative across sessions, content types and user groups. Whether the 20-euro voucher affected the data also remains to be tested.

**Brain or artifact.** The present results leave open whether the top features reflect brain activity, muscle activity, powerline-noise or the person recorded. Future work should control EMG in all bands, delta included, and not only in high gamma. Useful steps include motion monitoring, ICA, ICA with extra EMG recordings as in Li et al. [52], channel-quality checks and

checks per headband unit. A test without a person attached, as in Pattisapu and Ray [57], could show how much powerline-noise the Muse S picks up, and whether the Muse S shows extra peaks at 14, 36 and 64 Hz. Comparing results before and after conservative cleaning would show how much of the separation remains. Recording handedness and the hand holding the phone would test the difference between left and right electrodes (Section 4.5).

**Theory.** The EI may track which kind of task a person performs over many seconds, rather than the moment before a self-paced action (Section 4.5). Future studies should separate sustained attention, the decision to leave, the preparation of the swipe and the swipe itself. EEG could be combined with behavioural timing, self-report, eye tracking or later content ratings. A study could test whether a video inside a burst is remembered or judged worse (Section 4.3). Analyses aligned to each swipe, second by second, could show whether a signal comes before the decision, during the preparation of the swipe, or after the decision. The proposed definition is a target for such work. The two findings that would show the proposed definition wrong (Section 4.6.5) and the hypothesis for field measurement (Section 4.6.4) await testing.

**Method.** The labeling parameters deserve a study of their own. A full crossing of pre-swipe gap, SKIP period and window length, resolved per participant, could show whether the effects of the three labeling parameters depend on each other and differ between people (Section 3.4.5). The burst window needs the same care. Future analyses should test whether burst results hold across several burst windows, and whether removing bursts changes the separation between STAY and SKIP (Section 3.6.5). Several features need testing or validation: an EI built on the level of band strength, as in Pope et al., the minimum and the maximum of each band, and the two features defined in the present thesis, Peak Frequency and Macro Frequency (Sections 3.7 and 3.7.4). Other transparent models, such as hierarchical models, and burst-aware features or feature selection per person may add value. More complex models can, however, reduce interpretability and raise the risk of overfitting.

**Public health.** Short-form video also carries public communication, education and health promotion. If EEG-based models of engagement become robust and explainable, such models may help to fit the format, timing and length of a message to the attention of the viewer, for example in vaccine communication or mental-health micro-interventions. Such models must avoid manipulation, respect autonomy, and disclose how engagement is inferred and used. Where in the opening seconds of a video the difference before a swipe is strongest, and whether burst-aware timing of messages helps, remain open (Section 4.5).

**Other settings.** Finally, future research should test other contexts, populations and content. The present thesis studied naturalistic use of TikTok. Similar dynamics may appear in other media, in education or in other digital decisions, but need testing in a broader sample. Testing the same feature families in other content domains would show whether the effects are specific to short-form video or reflect engagement more generally.

## Chapter 6

# Plain-Language Summary: Common Questions

The questions that dominate public debate about short-form video are not the questions an EEG study is built to answer. They are, however, the questions on which people actually act, as parents, as teachers, as voters, as regulators, and as communicators trying to reach an audience. This chapter therefore steps out of technical register and answers the most common of those questions in ordinary language, using only what this thesis and the literature it reviews can support. It deliberately revisits ground already covered in Chapters 4 and 5, restated for a reader who has not read them.

Three cautions apply to everything below. This was one study of 25 young adults, mostly students, each recorded once for 25 minutes. It measured brain activity in the seconds around a swipe; it did not measure anyone's health, mood, grades, sleep, or long-term wellbeing, and it followed nobody over time. Every answer here is therefore a *direction of evidence*, not a settled finding, and is written to be revised. Where the honest answer is "this study cannot tell you," that is stated plainly rather than filled in with plausible-sounding inference.

**Is short-form video damaging my brain?** This study cannot tell you, and nobody should cite it as evidence either way. It never looked for damage. It compared two ordinary moments, still watching and about to swipe away, in healthy young adults over 25 minutes. What it did find is that those two moments look slightly different in the brain recording, on average. That is a statement about how attention may work, not about harm. Claims in circulation that scrolling "rewires" or "shrinks" the brain go well beyond what evidence of this kind can carry. *What to do with this:* treat confident claims in both directions with suspicion. The strong version of the harm story and

the dismissive “it’s all moral panic” reply are both running ahead of the data.

**If I watch something for a long time, does that mean I was interested in it?** Probably less often than you would assume, and this is the clearest practical message of the thesis. Platforms, advertisers and much public commentary treat time spent watching as if it measured interest. But long, uninterrupted viewing can mean being absorbed, or being zoned out and later remembering almost nothing [2, 3]. The measurement side of this thesis points the same way: the standard laboratory formula for “engagement” (Section 4.4.2) was far worse at telling these two moments apart than several plain, single measurements of the raw signal, the best of which separated them by close to nine times as much. Even the best single measurement separated the two moments only slightly, so the brain data give no licence to read watch time as interest either. *What to do with this:* stop reading watch time as proof that something worked. For anyone reporting reach, health campaigns especially, a high view count is a statement about distribution, not about whether anything landed. Ask separately whether people acted, returned, or could recall the message.

**Can a cheap headband read my mind, or tell what I am about to do?** No, not in any sense that should worry you today, and the failures are as informative as the successes. Using a consumer headband costing a few hundred euros, the signal that distinguished “about to swipe” from “still watching” showed up only as a small average difference. Prediction stayed close to guessing, both when the system was tuned to one person and when a system trained on a group was applied to a new person (Sections 4.4.2 and 4.4.3). In plain terms: there is at most a faint average difference, not a readable thought. *What to do with this:* do not buy claims that wearable brain sensors can currently detect engagement, attention or intent in a general population. Equally, note the direction of travel: this was four cheap sensors, one short session, one small group. Organisations with far more data and far longer observation are not bound by those limits, which is the real reason to ask now who is permitted to measure attention and under what oversight.

**Why is it so hard to stop scrolling once I start?** This study offers one small, concrete piece of the picture. Swipes do not arrive evenly spaced. They cluster into short bursts, meaning runs of rapid skipping with calmer stretches in between (Section 4.3). About two in five swipes stood alone, and the other three in five came inside a burst. How burst-prone someone was

varied enormously between individuals, far more than any average conveys. The pattern is real; the “it” that happens is not the same for everyone. *What to do with this:* if you want to interrupt your own scrolling, the useful target is the fast run rather than the total minutes, since the two are not the same behaviour. And be sceptical of advice derived from an average user, because on this measure the average user barely exists.

**Should phones be banned in schools, or social media restricted by age?** This thesis has nothing to say about either policy, and it would be misleading to enlist it on either side. It tested no intervention, involved no minors, and measured no outcome that such a policy would target. One indirect observation is worth carrying into that debate, however: the behaviour measured here differed so much from person to person that a single rule calibrated to the average would fit very few individuals well. That is a caution about uniform rules generally, not an argument for or against any particular one. *What to do with this:* demand that whoever proposes a rule states which outcome it is supposed to improve and how that will be checked afterwards. This study cannot supply that evidence, and neither can most of what is currently cited in the debate.

**What should regulators actually require, if not bans?** The most defensible ask suggested by this work is disclosure rather than prohibition. Feeds are tuned to maximise something, most commonly continued watching, which on the evidence reviewed here is not a reliable indicator that the viewer is attending, benefiting, or even remembering. Optimising it is therefore not the same as serving the user, and the gap between the two is currently invisible from the outside. A further practical point: this study was run on off-the-shelf hardware by a single researcher, which means independent measurement of these questions is now cheap enough that no regulator need rely solely on platform-supplied figures. *What to do with this:* require platforms to disclose what their systems optimise for and to open those targets to outside audit; fund small independent replication rather than waiting for internal data to be volunteered.

**I make health or education content. How do I hold attention?** Honestly: this thesis does not yet give you a tool. It shows that the seconds before leaving differ slightly, on average, from staying, a small first step toward one day detecting disengagement as it begins, but the difference is currently too weak, and its origin in brain or muscle too uncertain, to build anything you could deploy. What it does give you is a warning about your

own dashboard. If you are judging content by average watch time, you are optimising against a number that cannot distinguish an absorbed viewer from a dissociated one, and you may be steadily selecting for content that holds eyes without landing a message. *What to do with this:* keep watch time as a distribution check, but judge success on something downstream that you actually care about, such as recall, clicks, appointments or behaviour change, and accept that this is harder and slower to measure. That difficulty is not a reason to fall back on the easier number.

**So what is the single most useful thing to take away?** That the industry-standard way of measuring engagement is weaker than its ubiquity implies, and that better measurement is now within reach of people outside the platforms. This thesis does not show that anyone has decoded the decision to swipe; it shows that the seconds before leaving differ slightly, on average, from staying, that the conventional engagement metric captures this difference least of all tested signals, and that the equipment required to investigate this costs a few hundred euros rather than a laboratory budget. Whether that capability is used to make feeds more compelling or to hold them accountable is not a scientific question, and will not be settled by studies like this one.

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# Appendix A

## Kurzzusammenfassung (German Summary)

**Das Gehirn vor dem Swipe: Kann Consumer-EEG das Engagement der Zuschauenden beim naturalistischen TikTok-Browsing vorher-sagen?**

Kurzvideo-Plattformen wie TikTok gliedern die Aufmerksamkeit ihrer Nutzerinnen und Nutzer in eine schnelle Folge von Mikroentscheidungen: weiterschauen oder wegwischen. Plattformkennzahlen behandeln ununterbrochene Sehdauer häufig als Maß für Engagement, obwohl langes Scrollen ebenso mit passivem, dissoziativem Konsum vereinbar ist wie mit gerichteter Aufmerksamkeit. Die vorliegende Arbeit untersucht, ob die Elektroenzephalografie (EEG) während naturalistischer TikTok-Nutzung die Sekunden vor einem freiwilligen Wischen (SKIP) vom fortgesetzten Schauen (STAY) unterscheiden kann, und ob einzelne EEG-Merkmale dabei den herkömmlichen Engagement Index ( $\beta/(\alpha + \theta)$ ) übertreffen.

**Methode.** 25 Teilnehmende (13 weiblich, 12 männlich; Alter  $M = 24,6$ ,  $SD = 3,0$ ) trugen ein Consumer-EEG-Stirnband mit vier Elektroden (Muse S; AF7, AF8, TP9, TP10). Nach einer zweiminütigen Ruhemessung nutzten sie 25 Minuten lang frei einen zuvor zurückgesetzten TikTok-„For You“-Feed. Jedes Wischen wurde per Tastendruck im EEG markiert. SKIP-Fenster stammen aus den 3 Sekunden, die 2 Sekunden vor jedem Wischen enden; STAY umfasst die übrige ununterbrochene Sehzeit. Aus Fenstern von 1 Sekunde wurden 236 Merkmale über sieben Frequenzbänder, vier Elektroden und das Rohsignal berechnet.

**Ergebnisse.** Gemessen mit Cohens  $d$  trennten alle Frequenzbänder und alle Elektroden die beiden Klassen im Mittel stärker als der Engagement Index. Das stärkste Einzelmerkmal, die Standardabweichung des Rohsignals an TP10, erreichte etwa das Neunfache des Engagement Index; alle Effekte sind jedoch klein. Als logistische Regression lag das Engagement-Index-Modell innerhalb der Teilnehmenden nahe am Zufallsniveau. Über Teilnehmende hinweg ordnete es etwa 90 % aller Fenster STAY zu und erkannte SKIP kaum. Drei Merkmale aus der Rangfolge nach Cohens  $d$  (Standardabweichung des Rohsignals an TP10, High-Gamma-Mittelwert an AF7, Delta-Peak-Frequenz an AF8) erkannten SKIP über Teilnehmende hinweg signifikant besser als das Engagement-Index-Modell, blieben aber unter dem Zufallsniveau. Innerhalb der Teilnehmenden verbesserten die Merkmale an AF7 und AF8 nur die SKIP-Präzision. Die  $p$ -Werte sind nicht für multiples Testen korrigiert. Eine Verhaltensanalyse zeigte zudem, dass Wischbewegungen in schnellen, individuell unterschiedlichen Serien („Bursts“) auftreten.

**Schlussfolgerung.** Der Engagement Index erfasst in diesem Paradigma nur einen begrenzten, einseitigen Teil des EEG-Signals. Spezifischere EEG-Merkmale enthalten zusätzliche, bislang schwache Information über das bevorstehende Wegwischen. Muskelaktivität (EMG) und Netzbrummen (50 Hz) sind als Störquellen nicht ausgeschlossen und erfordern eine artefaktkontrollierte Replikation. Die Arbeit stellt außerdem eine offene, reproduzierbare Aufnahme- und Analysepipeline für Consumer-EEG bereit und schlägt eine Definition von Engagement vor, die an kein Messinstrument gebunden ist.